



SECJ2203: Software Engineering

System Documentation (SD)

KADA's Cooperative Web-Based System

Version 4.0

Date: **17 / 2 / 2025**

Faculty of Computing

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Revision Page

a. Overview

The current version of the System Documentation (SD) covers all the chapters which include Software Requirement Specification, Software Design Document and Software Test Description.

b. Target Audience

The target audience are for those who are involved with the project such as the lecturers from the Software Engineering course, Software Development technology course and Database course. Another target audience is the product owner of the system.

c. Project Team Members

This table below shows the role, task and status for each team member.

Member Name	Role	Task	Status
IMAN ABADI BIN MOHD NIZWAN	Scrum Master	Completed	Coordination tasks
1) MOHAMED ALIF FATHI BIN ABDUL LATIF 2) MUHAMMAD AFIQ DANIAL BIN ROZAIDIE 3) IMAN ABADI BIN MOHD NIZWAN 4) AFIF SHAQIR IRFAN BIN ARQAM	Developer	Completed	Feature integration
ENCIK AHMAD (KADA representative)	Product Owner	Completed	User feedback

d. **Version Control History**

This version control shows all the version history that has been made for this project. This history helps to track back the progress of this project.

Version	Primary Author(s)	Sprint/Iteration	Description of Version	Date Completed
1.0	Team leader 1 (IMAN ABADI BIN MOHD NIZWAN)	Sprint 1	Completed Chapter 1 and 2	28/11/2024
2.0	Team leader 1 (IMAN ABADI BIN MOHD NIZWAN)	Sprint 2	Revised Chapter 1 and 2, Completed Chapter 3-7.	21/12/2024
3.0	Team leader 1 (IMAN ABADI BIN MOHD NIZWAN)	Sprint 3	Revised Chapter 1-7, Completed Chapter 8 and 9	17/2/2025

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1. Introduction

1.1 Purpose

The purpose of this document is to present details of the Web-based Platform for Kemubu Agricultural Development Board or KADA's system. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli. This document is also intended for both KADA's cooperative staff and our developers of the system. This will be proposed and presented to the KADA's cooperative representative.

1.2 Scope

The scope of this project is to propose a system that will be a website that replaces manual systems by using multiple technologies to ensure the smooth-running system for the cooperative. Each of this technology has its own functionality to help the development of this system. The system will have some existing functionality but with additional changes using these technologies will increase efficiency to enhance the working experience because the technologies used will develop a web server with a relational database management system. The system will cover functionalities such as user authentication, membership application, membership application, administrative processes and successful payment notification. The system will not cover scopes that utilize payment gateways for tasks such as making a payment through the system, buying share modal and claiming of member benefits.

1.3 Definitions, Acronyms and Abbreviation

Table 1.1: Definitions, Acronyms and Abbreviation

Acronyms	Description
KADA	Kemubu Agricultural Development Board
HTTPS	Hypertext Transfer Protocol Secure
SRS	Software Requirement Specification
HTML	Hypertext Markup Language

iOS	iOS: iPhone Operating System
SSL	Secure Sockets Layer
TLS	Transport Layer Security
SMTP	Simple Mail Transfer Protocol
RAM	Random Access Memory
AES	Advanced Encryption Standard

1.4 References

1.5 Overview

The current version of the System Documentation (SD) covers all the chapters which include Software Requirement Specification, Software Design Document and Software Test Description.

2. Specific Requirements

2.1 Persona

Personas are intended users of the system and their characteristics, such as their technical expertise, knowledge, and physical abilities. This section provides the description of the user of the system determining their needs and making a story out of the user needs to break down the requirements of the system.

2.1.1 Persona 1: KADA's Member

User Need

- KADA members need a way to apply for a loan and access the dashboard.
- KADA members need a way to apply for a membership

User Stories

- As a KADA member, I want to apply for a loan so I can borrow a large amount of money.
- As a KADA member, I want to log in the system and check my debts in the dashboard.

2.1.2 Persona 2: Clerk

User Need

- Clerk needs a way to manage data contained within the system's database and provide financial statements for members and generate reports.

User Stories

- As a clerk, I want to be able to log into the system so that I can manage the data in the system's database.
- As a clerk, I want to be able to provide financial statements for both members' financial reports.
- As a clerk, I want to be able to generate reports to be used for meetings.

2.1.3 Persona 3 : KADA's Board of Director

User Need

- The Board of Directors needs a way to approve the loan applications, membership applications and member withdrawal applications and access the dashboard.

User Stories

- As a member of the Board of Directors, I want to be able to log in to the system so that I can approve the loan applications, membership applications and member withdrawal application.

2.2 System Features

The KADA's Cooperative web system is a software system that operates on both mobile devices and desktop, including Android, iOS, Windows and Mac operating systems. The system provides a means for KADA employees to apply as a member of the cooperative to enjoy the benefits of the cooperative.

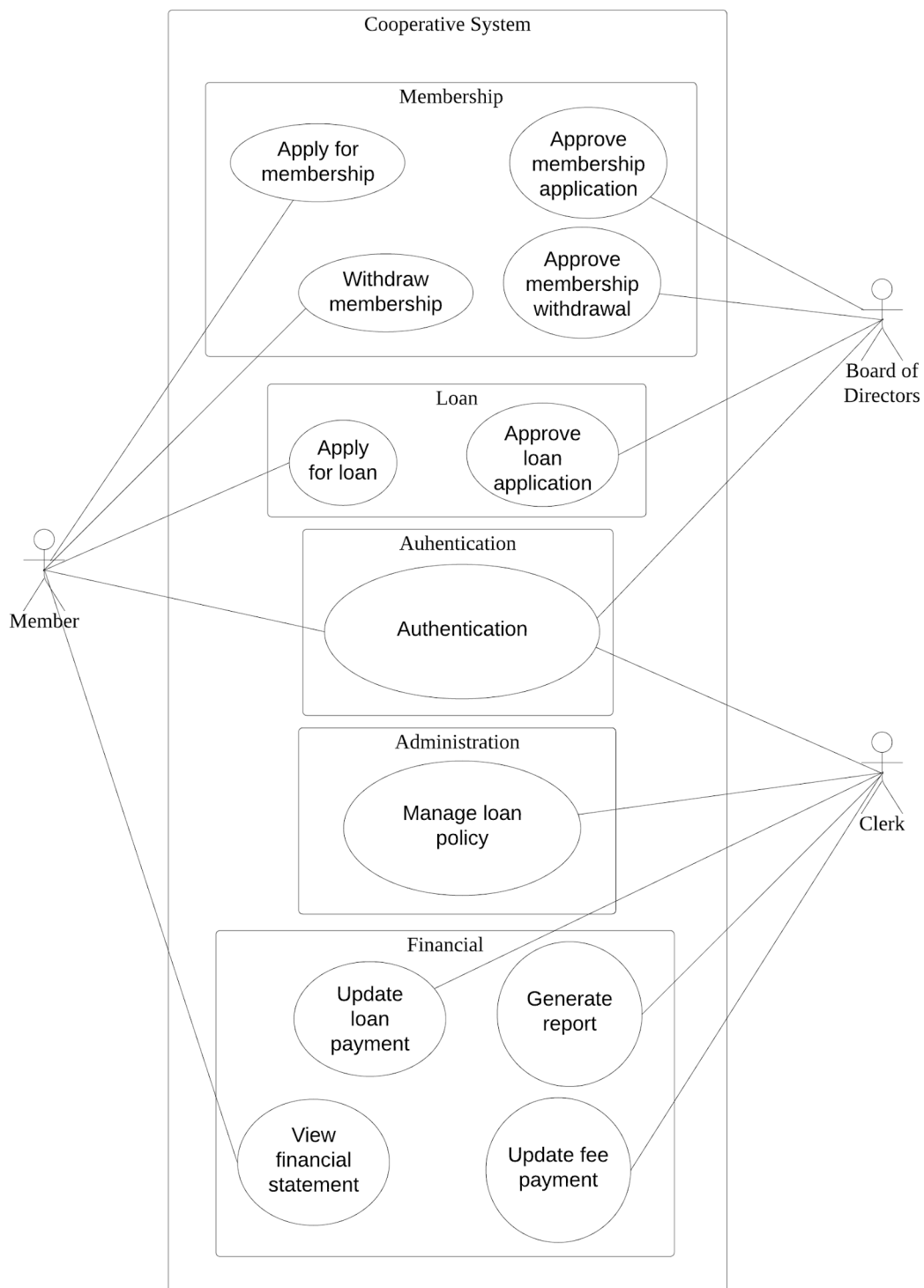


Figure 2.1: Use Case Diagram for KADA's Cooperative Web-Based System

Figure 2.1 shows the use case that is involved within the system. The users involved in the system are clerks, members and board of directors. Each of them have their own respective use cases that determine their boundaries within the system.

Table 2.1: Description of Module and Functions for KADA's Cooperative Web-Based System

Use case	Function	Description
US001	Apply for Membership	This use case allows the board of directors to be able to apply for membership.
US002	Apply for Loan	This use case allows members to be able to apply for a loan.
US003	Withdraw as Membership	This use case allows member to withdraw their position as a member
US004	Approve Membership Withdrawal	This use case allows the board of directors to approve or reject member withdrawal applications.
US005	View Financial Statement	This use case allows members to view their financial statements such as loan payment statements and their cooperative shares.
US006	Update Loan Payment	This use case allows clerk to update loan payment for each members
US007	Update Fee Payment	This use case allows clerk to update fee payment for each members
US008	Approve Membership Application	This use case allows board of director to approve or reject the membership application
US009	Approve Loan Application	This use case allows the board of directors to approve or reject the loan application
US010	Manage Loan Policies	This use case allows the clerk to generate reports so that the statistical report of the cooperative can be summarised.
US011	Generate Report	This use case allows the clerk to generate reports like number of membership, number of loan applications and details and number of member's withdrawal for monthly or yearly
US012	Authentication	This use case allows the web system to distinguish the user type based on their Identification number

2.3 Launch Phase

Launch phases are used to track the development phase of the system. The phases are named as sprints. The sprints are considered successful when all the modules in the sprint are completed. Each has the team members assigned to the modules of the respective sprints.

Table 2.2: Sprints and Assigned Member

Sprint	Team members assigned
Sprint #1 Membership Application, Loan Application, Update Loan Payment, Update Fee Payment and Manage Loan Policies	Iman Abadi Bin Mohd Nizwan Mohamed Alif Fathi Bin Abdul Latif Muhammad Afiq Danial Bin Rozaidie Afif Shaqir Irfan Bin Arqam
Sprint #2 Authentication, View Financial Statement, Loan Approval and Membership Approval	Iman Abadi Bin Mohd Nizwan Mohamed Alif Fathi Bin Abdul Latif Muhammad Afiq Danial Bin Rozaidie
Sprint #3 Generate Report	Iman Abadi Bin Mohd Nizwan
Sprint #4 Member Withdrawal and Approve Membership Withdrawal	Muhammad Afiq Danial Bin Rozaidie Mohamed Alif Fathi Bin Abdul Latif

Table 2.2 shows the module/use cases for each sprints and the members assigned for each sprints. The consecutive sprints can only be done when the previous sprints have been completed.

2.4 User Story Details

User story details clarifies the flow of the use cases from Figure 2.1

2.4.1 US001: Apply for Membership

Table 2.3: User Story Description for Apply for Membership

User story: Apply for Membership	
ID: US001	
User Story Description As an employee of KADA, I want to be a member of KADA to have a permission to apply for personal loan	
Flow of events: 1. The employee accesses the application page. 2. The employee key in all their personal details. 3. The system verified the input and waited for approval from the Board of Directors.	
Alternative flow : 1. If an employee fills in invalid details, the system will prompt an error message and members redirect to the application page. 2. If the applicant was a member but has withdrawn and wants to reapply, the system will fill out all the information based on the past records of the member.	
Postcondition: The employee successfully applied for a membership	
Precondition: Applicant is an employee of KADA	
Exception flow: 1. If the application was already submitted or has been approved then the application cannot be registered again.	

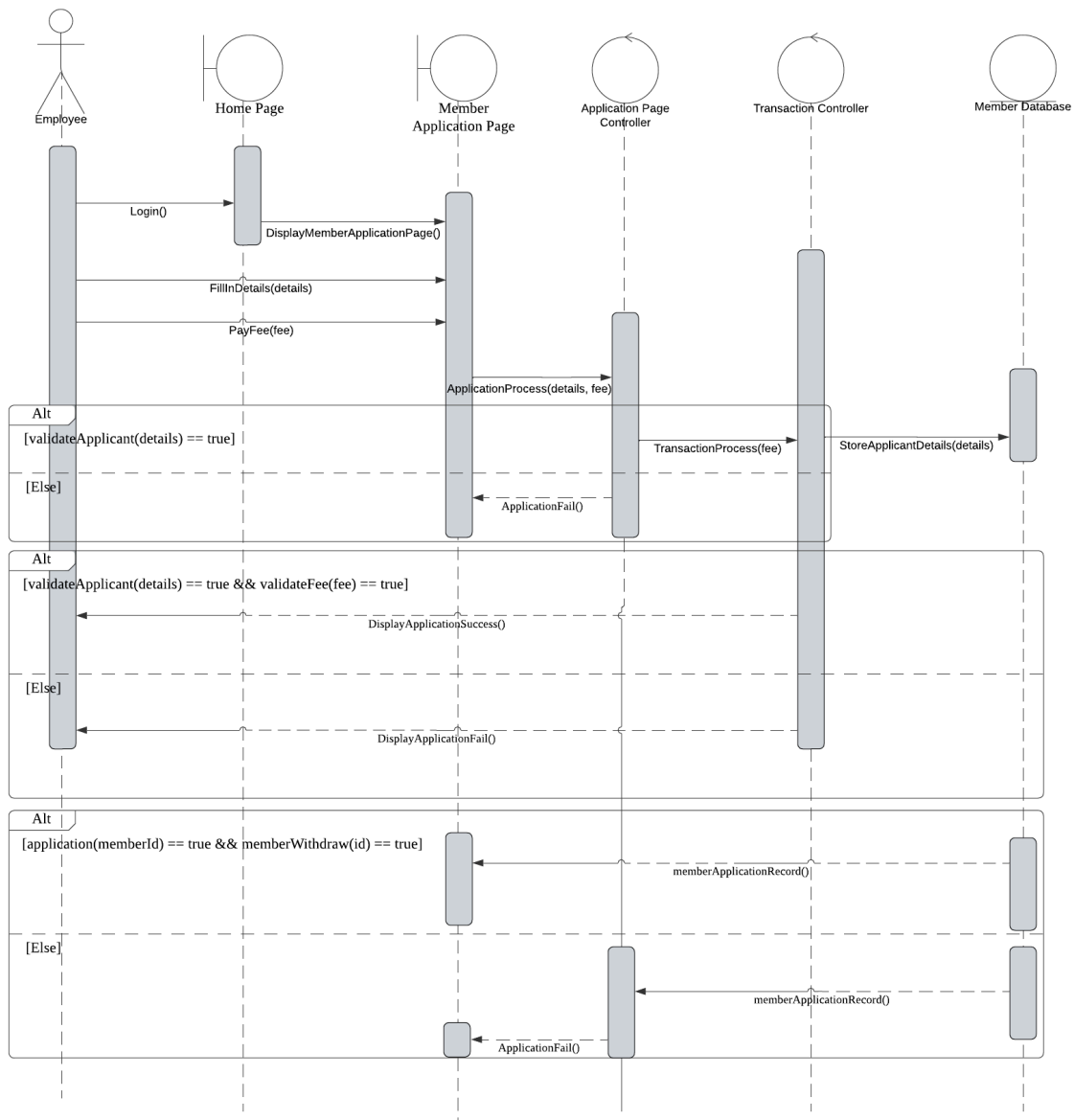


Figure 2.2: Sequence Diagram for Apply for Membership

Figure 2.2 shows the sequence diagram for membership application. The member first has to log into the system and navigate to the membership application page. Then the member has to fill in all the details. The system will validate the application based on the data entered. The member will be notified of whether the application is successful or not. The member cannot apply for another application if they are already known as a member and if their application has not been approved or rejected.

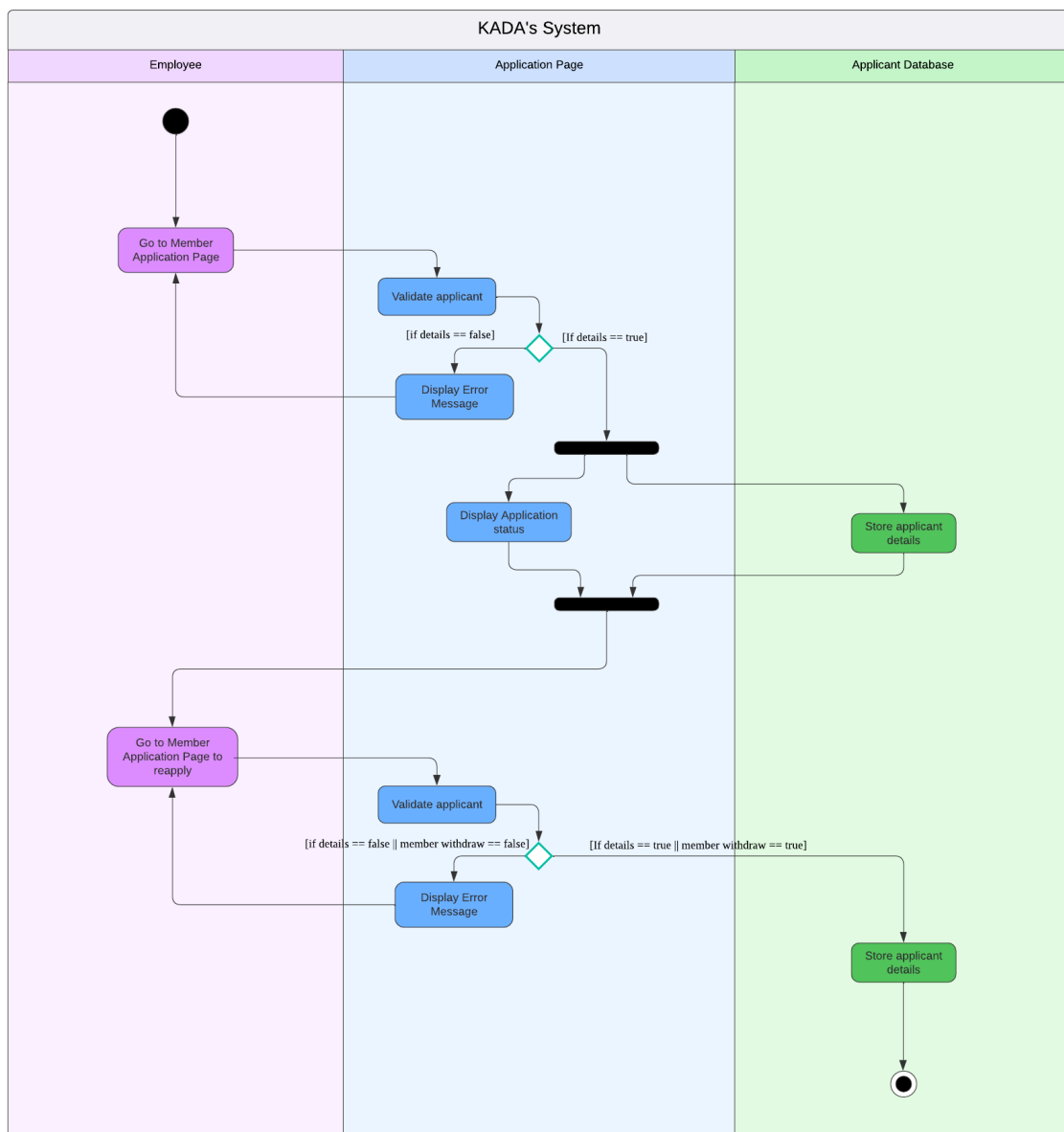


Figure 2.3: Activity Diagram for Apply for Membership

Figure 2.3 shows the activity diagram for membership application. The member first has to log into the system and navigate to the membership application page. Then the member has to fill in all the details. The system will validate the application based on the data entered. The member will be notified of whether the application is successful or not. The member cannot apply for another application if they are already known as a member and if their application has not been approved or rejected.

2.4.2 US002: Apply for Loan

Table 2.4: User Story Description for Apply for Loan

User story: Apply for Loan
ID: US002
User Story Description As a member of KADA, I want to be able to access the loan application page so that I can apply for a loan.
Flow of events: 1. The member accesses the application page. 2. The member key in their details (like amount of loan). 3. The system verified the input and waited for approval from the Board of Directors.
Alternative flow 1: If members don't have modal shares more than or equal to 300, members cannot proceed with the application.
Alternative flow 2: If members fill in invalid details, the system will prompt an error message and members redirect to the application page.
Postcondition: The member successfully applied for a loan.
Precondition: The member is already registered to the KADA membership.

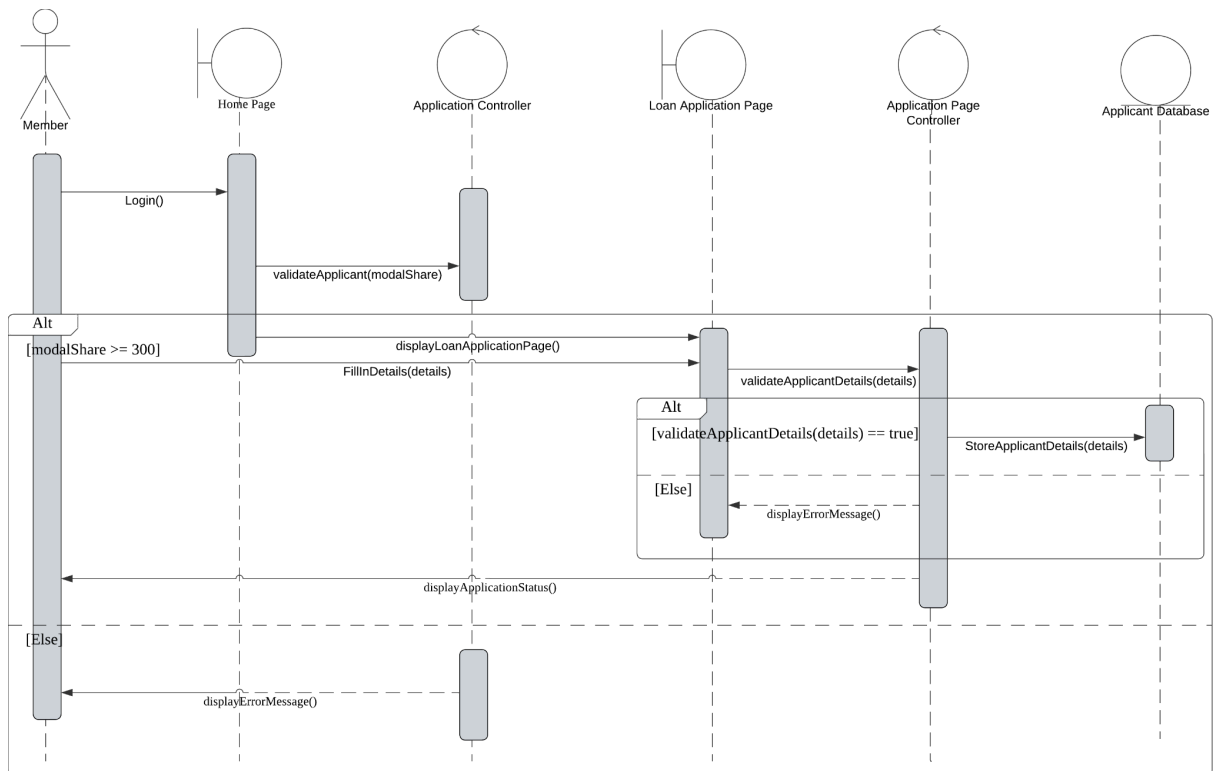


Figure 2.4: Sequence Diagram for Apply for Loan

Figure 2.4 shows the sequence diagram for loan application. The member first has to log into the system and navigate to the loan application page. If the member does not RM 300 of modal share, they are not able to apply for a loan. If they are able to apply for a loan, they have to fill in the details of the application. The system will validate the details entered and will display an error message if the details provided are invalid.

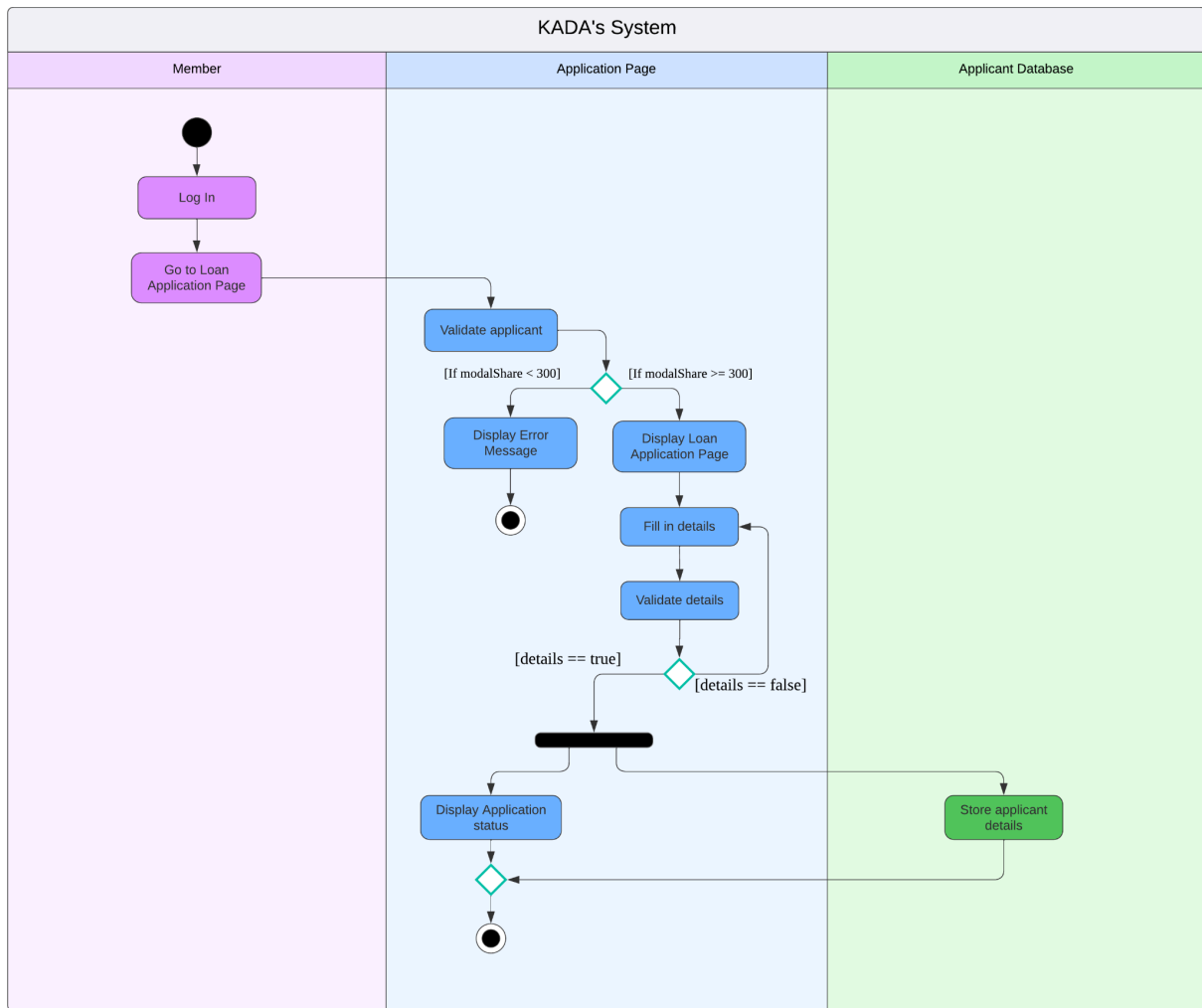


Figure 2.5: Activity Diagram for Apply for Loan

Figure 2.5 shows the activity diagram for loan application. The member first has to log into the system and navigate to the loan application page. If the member does not RM 300 of modal share, they are not able to apply for a loan. If they are able to apply for a loan, they have to fill in the details of the application. The system will validate the details entered and will display an error message if the details provided are invalid.

2.4.3 US003: Withdraw Membership

Table 2.5: User Story Description for Withdraw Membership

User story: Withdraw Membership
ID: US003
User Story Description As a member of KADA, I want to be able to quit the membership.
Flow of events: 1. The member accesses the dashboard. 2. The member key in their reason for membership withdrawal. 3. The system verified the input and waited for approval from the Board of Directors.
Postcondition: The member successfully applied for a withdrawal.
Precondition: The member is already registered to the KADA membership.
Exception flow:

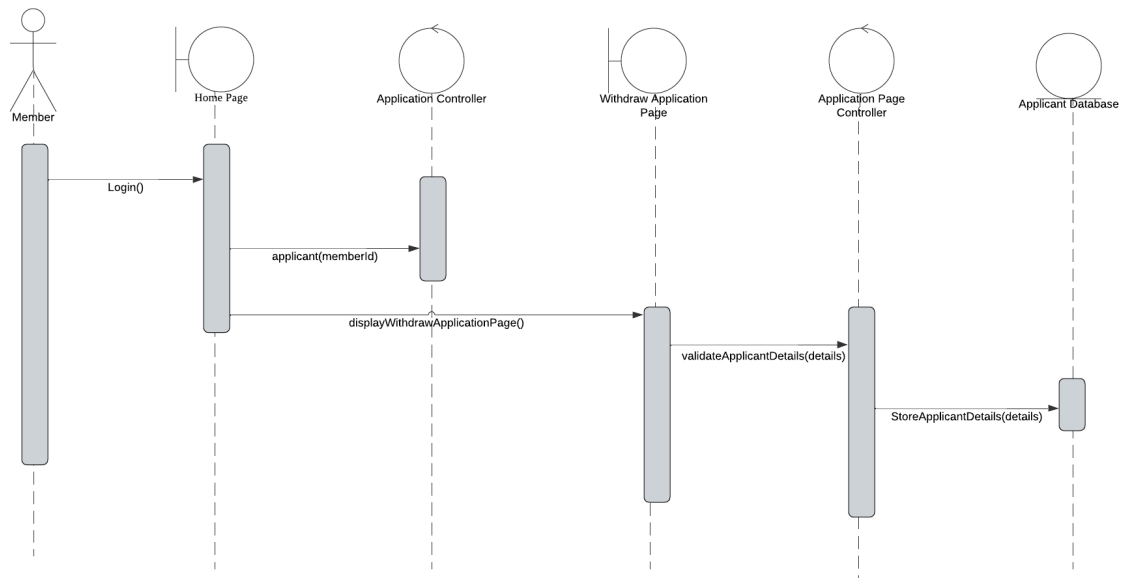


Figure 2.6: Sequence Diagram for Withdraw Membership

Figure 2.6 shows a sequence diagram for withdrawal membership. The member needs to login and be able to access the dashboard. Then, the member needs to key in their reason for the membership withdrawal. The system will verify the input and have to wait for the approval from the Board of Directors.

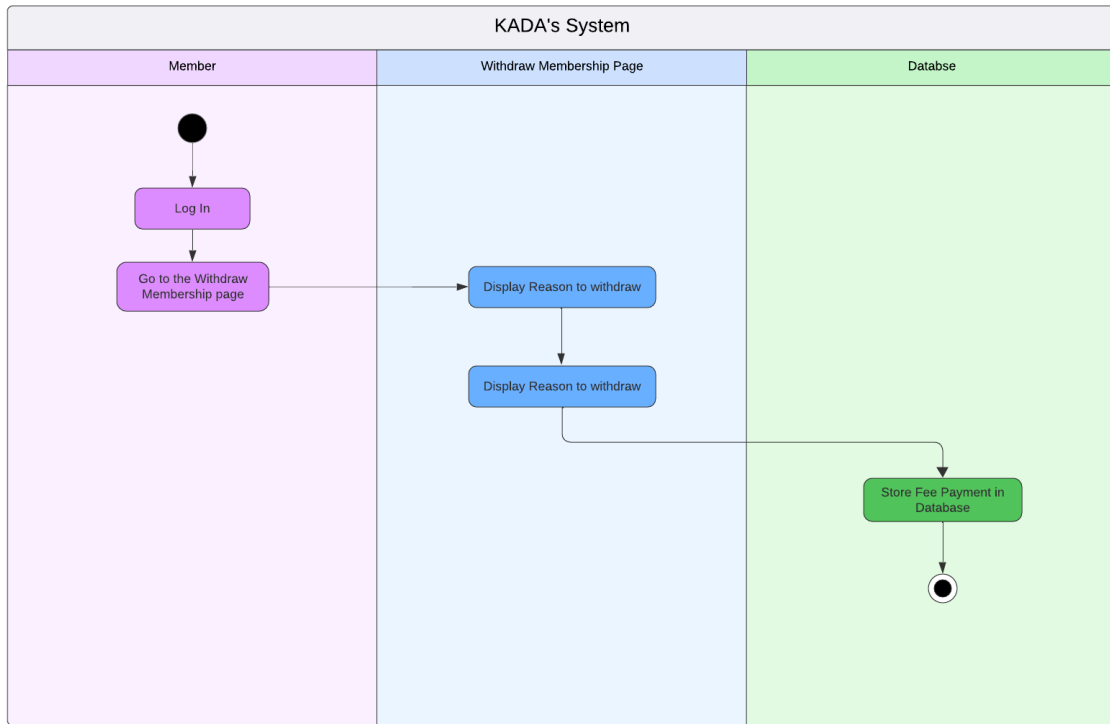


Figure 2.7: Activity Diagram for Withdraw Membership

Figure 2.7 shows an activity diagram for withdrawal membership. First, members log in to the system. Then, the member needs to go to the withdrawal membership page. Then, members key in their reason and the reason then store it in a database for BOD to review for approval.

2.4.4 US004: Membership Withdrawal Approval

Table 2.6: User Story Description for Membership Withdraw Approval

User story: Withdraw Membership
ID: US004
User Story Description As a board of director, I want to approve or reject a member withdrawal application.
Flow of events: 1. The Board of Directors logs into the system. 2. The system displays the Membership Withdrawal Application Page. 3. The Board of Directors views a list of submitted applications 4. The Board of Directors selects an application to review. 5. The Board of Directors decides whether to approve or reject the applications.
Postcondition: 1. The member approved is no longer a member of the cooperative. 2. The member rejected is still a member of the cooperative.
Precondition: 1. The applicant is a member of the cooperative

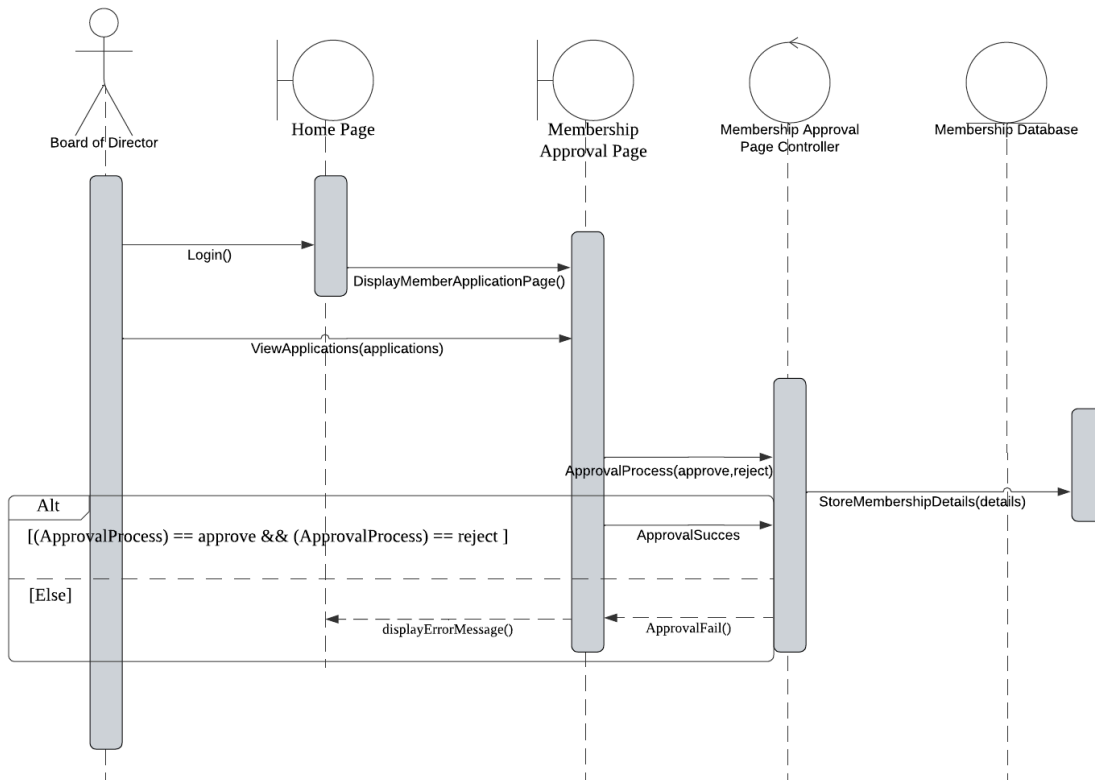


Figure 2.8: Sequence Diagram for Membership Withdrawal Approval

Figure 2.8 shows a sequence diagram for approval withdrawal membership. The Board of Directors need to login and navigate to the membership withdrawal application. The list of applicants will be displayed. The Board of Directors can check the details as well as the reason for them to withdraw. Then, the BoD can decide whether to approve the application or reject the application.

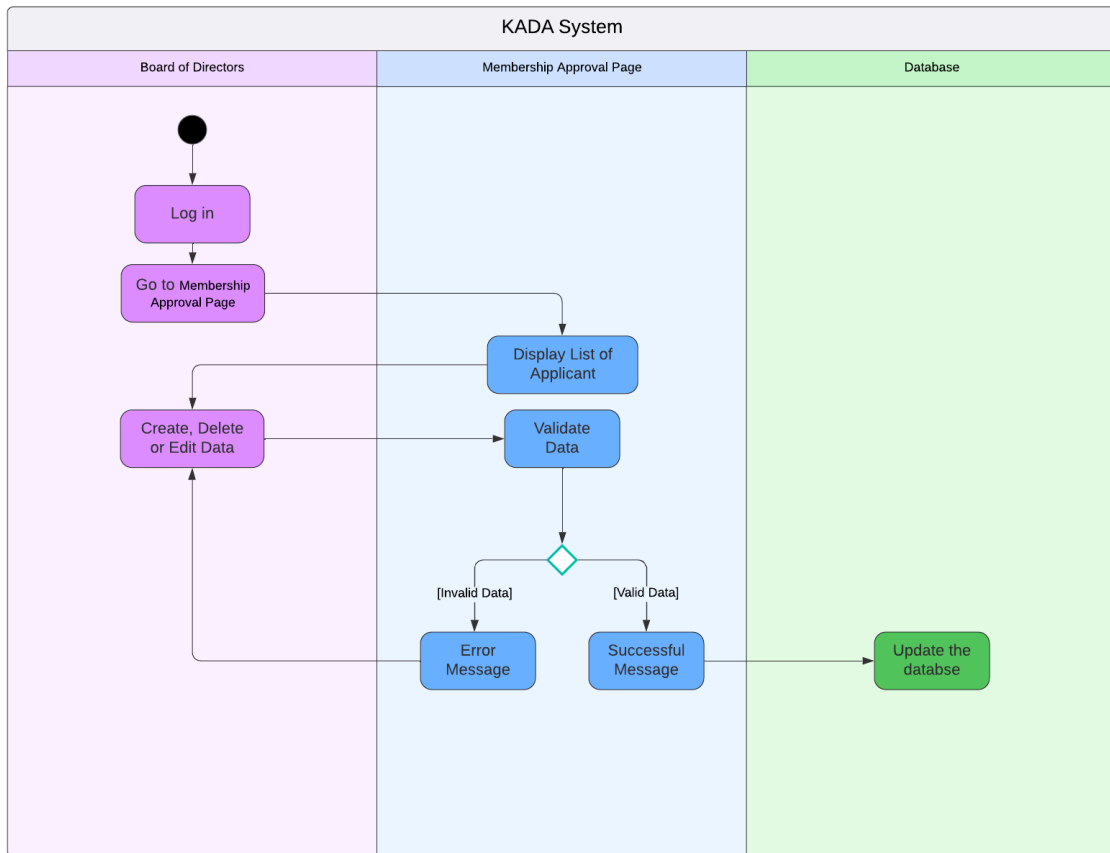


Figure 2.9: Activity Diagram for Membership Withdrawal Approval

Figure 2.9 shows the approval process for the membership withdrawal. First, BOD log in to the system. Then, BOD goes to the membership approval page to display the list of applicants and then BOD can approve the application by updating the status in database table member to inactive.

2.4.5 US005: View Financial Statement

Table 2.7: User Story Description for View Financial Statement

User story: View Financial Statement
ID: US005
User Story Description As a member of KADA, I want to be able to check or view the current financial status.
Flow of events: 1. The member accesses the dashboard to see the financial details in there. 2. The system generates and displays the financial statement in a PDF file. 3. The members can view or print the statement generated by the system.
Precondition: The members already login with their own authenticated username and password.
Postcondition: 1. The members successfully view or print their financial statement

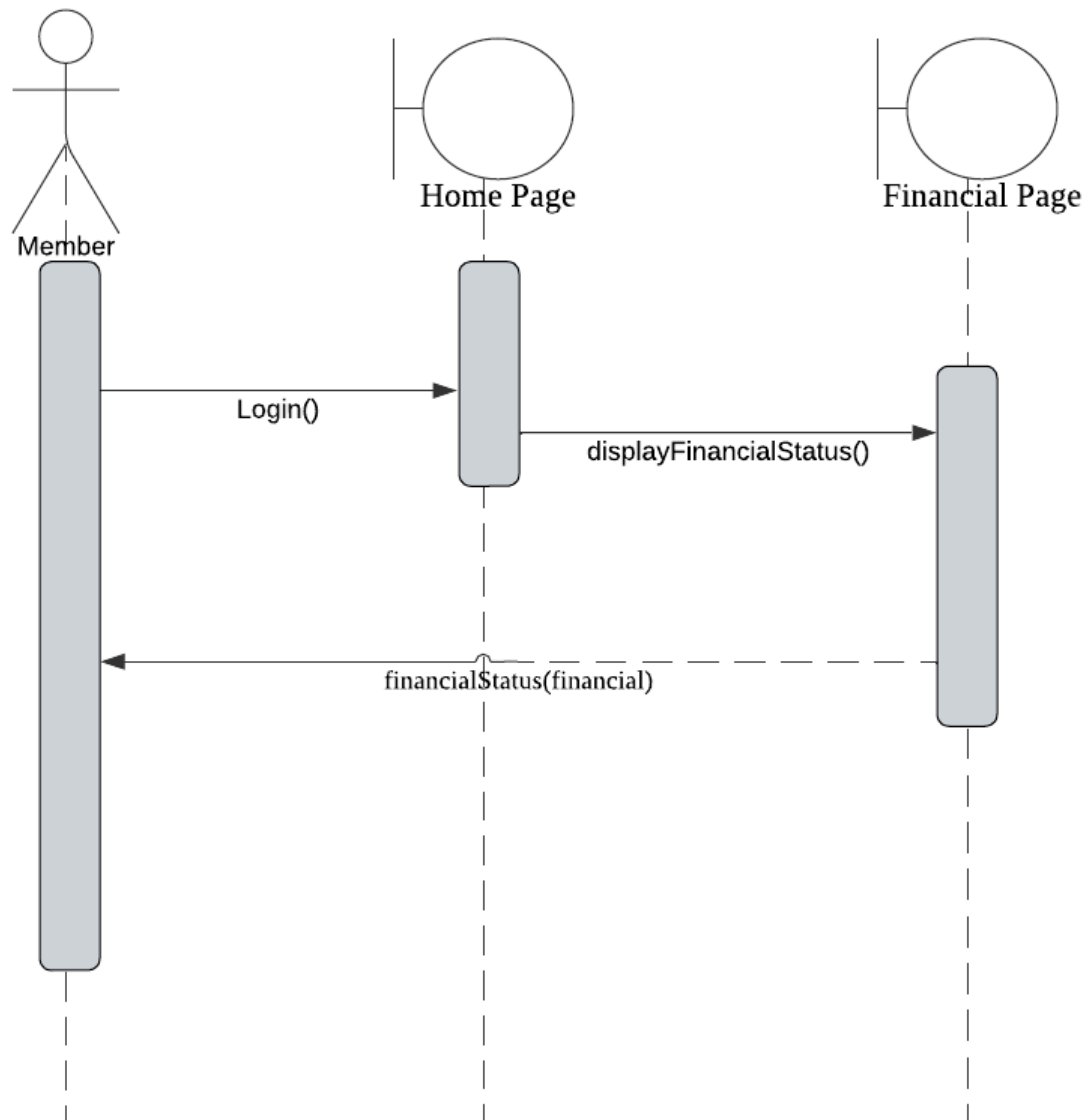


Figure 2.10: Sequence Diagram for View Financial Statement

Figure 2.10 shows the sequence diagram of viewing financial statements. When the member logs into the system, their financial statement will be shown in their dashboard. The dashboard will display their share and loan details. The members can print their statement using their current share and loan details in a pdf file.

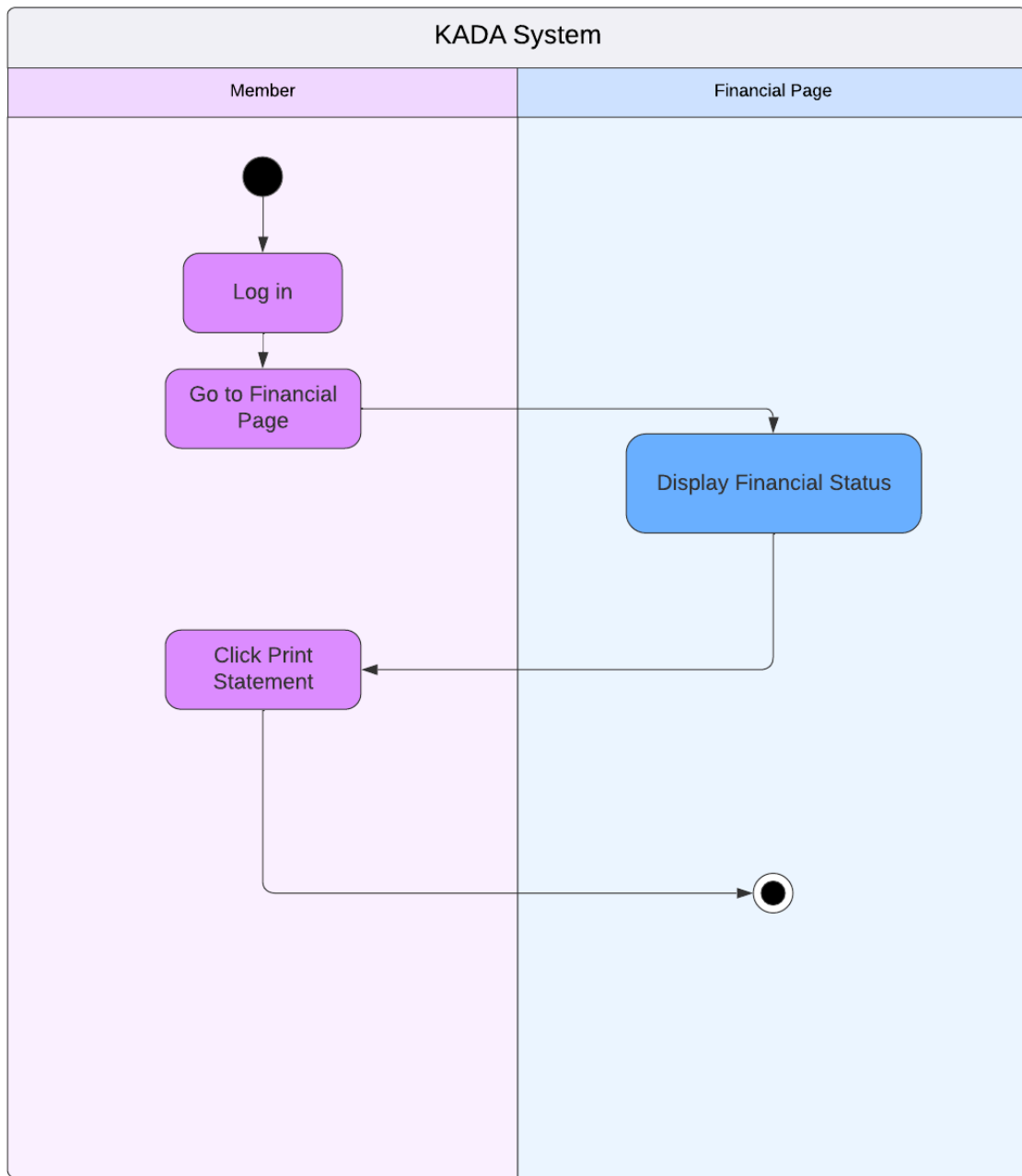


Figure 2.11: Activity Diagram for View Financial Statement

Figure 2.11 shows the activity diagram of viewing financial statements. When the member logs into the system, their financial statement will be shown in their dashboard. The dashboard will display their share and loan details. The members can print their statement using their current share and loan details in a pdf file.

2.4.6 US006: Update Loan Payment

Table 2.8: User Story Description for Update Loan Payment

User story: View Financial Statement
ID: US006
User Story Description As a clerk, I want to update the loan payment of the members that have an ongoing loan.
Flow of events: 1. The clerk accesses the update member loan page. 2. The system displays the members with active loans. 3. The clerk accesses the members loan details. 4. The system displays the loan details such as the monthly payment. 5. The clerk chooses the date of payment. 6. The clerk updates the payment.
Precondition: 1. The members viewed has an active loan 2. The user is logged in as a clerk.
Postcondition: 1. The members' loan payment is recorded and the loan application is updated.

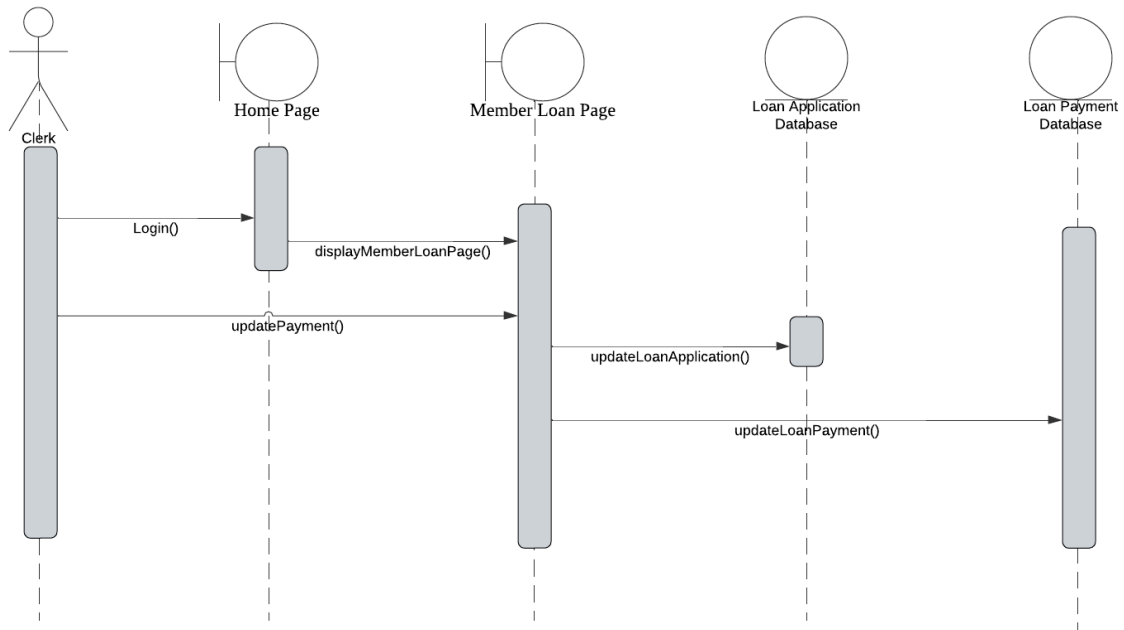


Figure 2.12: Sequence Diagram for Update Loan Payment

Figure 2.12 shows the sequence diagram for updating loan payment. The clerk will navigate to the member loan page and view a list of members with active loans. Then the clerk can pick any of the members that would want to update the payments.

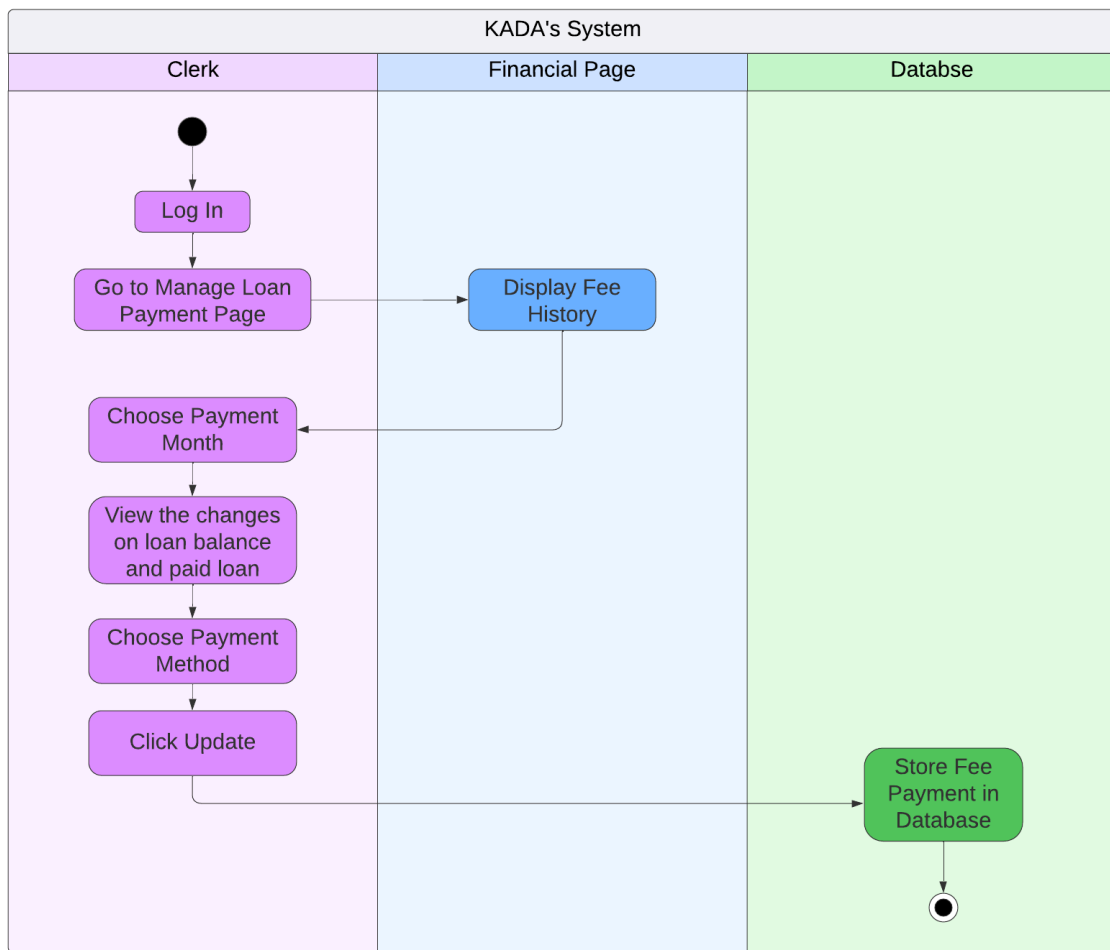


Figure 2.13: Activity Diagram for Update Loan Payment

In figure 2.13, It shows the process of the clerk updating the loan payment. First, the clerk logs in to the system and goes to manage the loan payment page. Then, The Financial Page displays the fee history. Then, the clerk can choose the payment month and view the changes on loan balance and the loan that will be paid. After that, the clerk chooses the payment method and clicks update to store the loan payment data in the database.

2.4.7 US007: Update Fee Payment

Table 2.9: User Story Description for Update Fee Payment

User story: Update Fee Payment
ID: US007
User Story Description As a clerk of KADA, I want to be able to update the fee payment for each member.
Flow of events: 1. The clerk accesses the list member of the update fee payment page. 2. The system displays the list of members. 3. The clerk selects the members and access the update fee payment page. 4. The clerk chooses the monthly date, key in the entry fee and other fees are automatically key in because it is already fixed. 5. The system updates the fee payment.
Precondition: 1. The members viewed are active. 2. The user is logged in as a clerk.
Postcondition: 1. The members' fee payment is recorded and the members' fee details are updated.

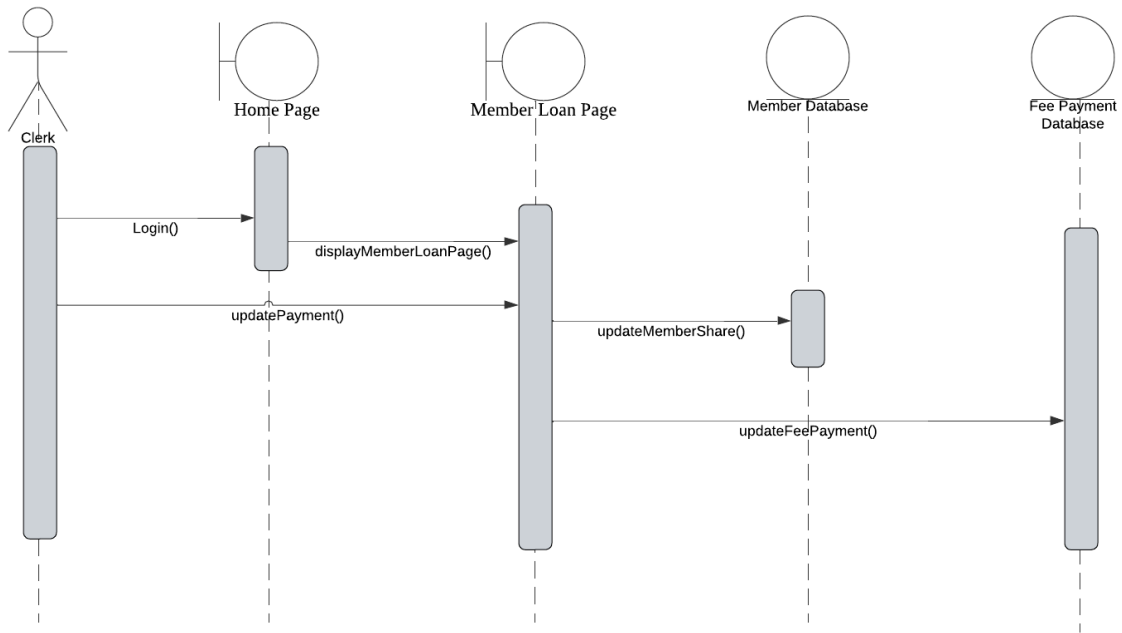


Figure 2.14: Sequence Diagram for Update Fee Payment

Figure 2.14 shows a sequence diagram for update fee payment. Clerk first needs to login into the system and navigate to the member loan page. The system will display the list of the members. The clerk can select the member and also update the fee payment. Then, the system will update the payment details into the member account.

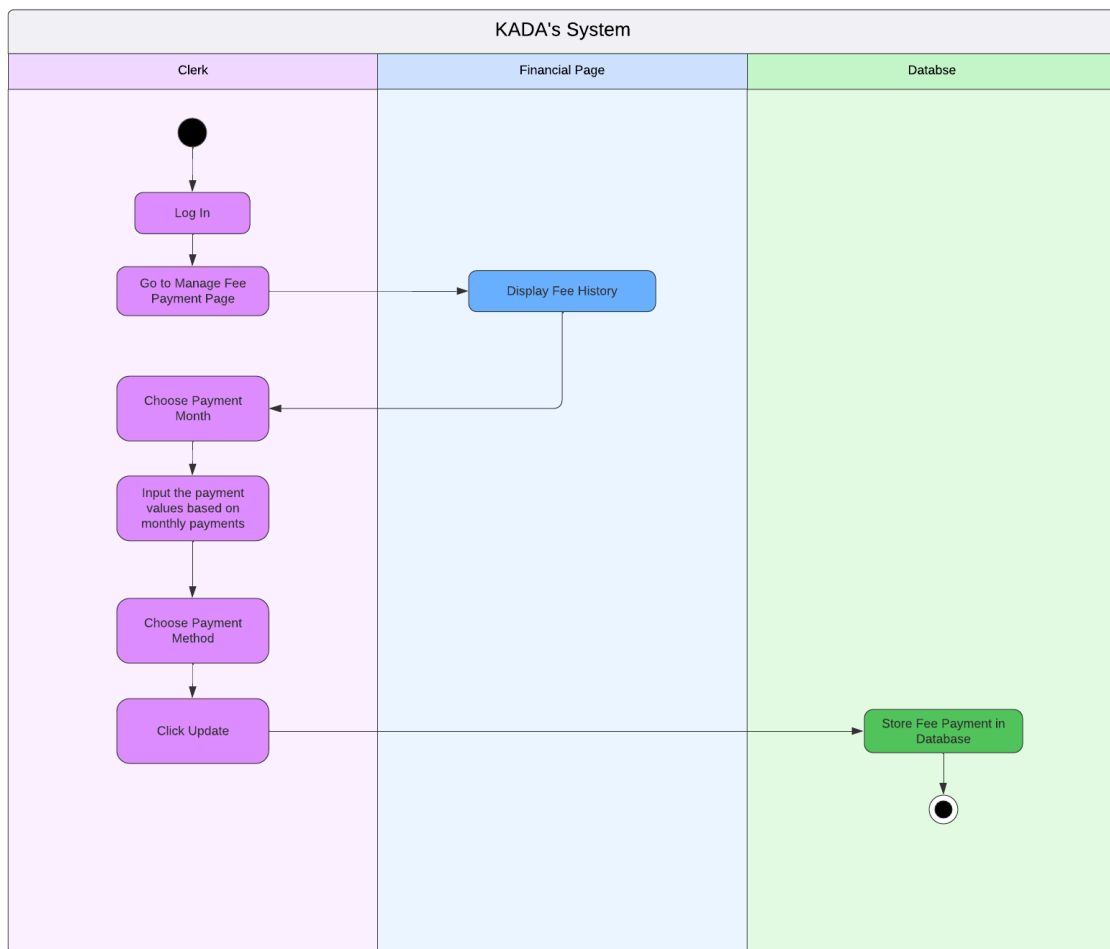


Figure 2.15: Activity Diagram for Update Fee Payment

For figure 2.15, it shows the process of activity diagram for updating fee payment. Clerk first needs to login into the system and navigate to the member fee payment page. The system will display the list of the members. The clerk can select the member and also update the fee payment. Then, the system will update the payment details into the member account.

2.4.8 US008: Approve Membership Application

Table 2.10: User Story Description for Approve Membership Application

User story: Withdraw Membership
ID: US004
User Story Description As a board of director, I want to approve or reject a membership application.
Flow of events: 1. The Board of Directors logs into the system. 2. The system displays the Membership Application Page. 3. The Board of Directors views a list of submitted applications 4. The Board of Directors selects an application to review. 5. The Board of Directors decides whether to approve or reject the applications.
Postcondition: 1. The member approved is a member of the cooperative 2. The member rejected is not a member of the cooperative.

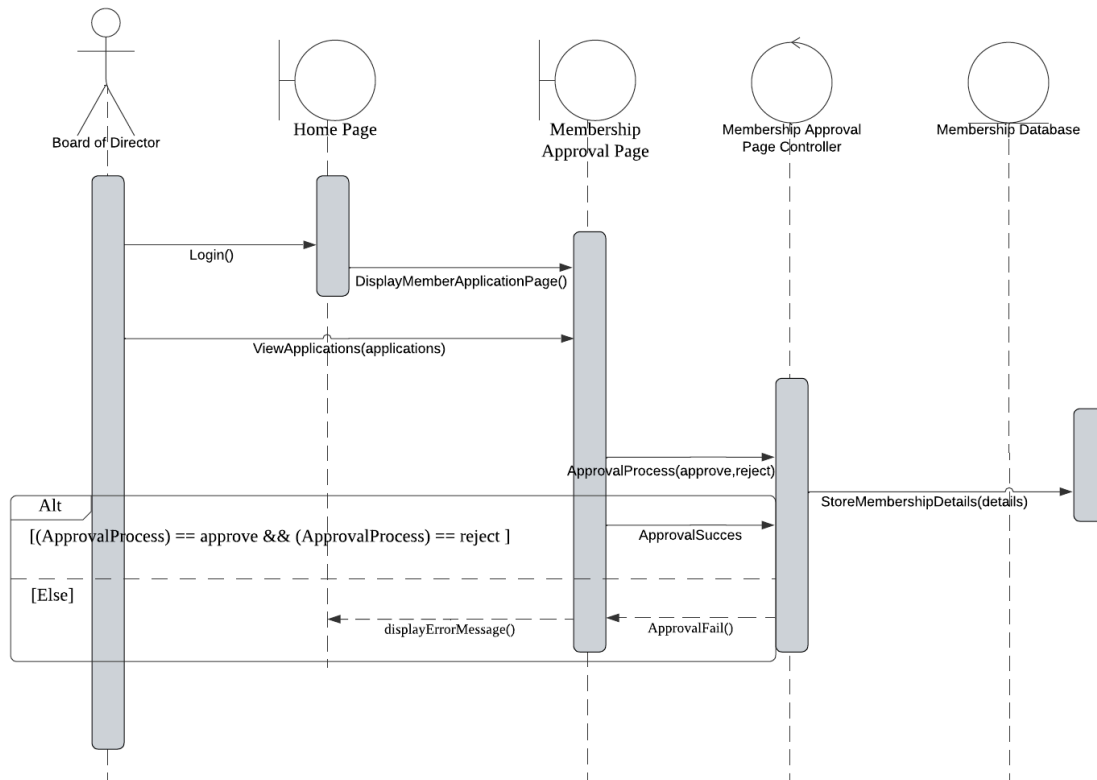


Figure 2.16: Sequence Diagram for Approve Membership Application

Figure 2.16 shows a sequence diagram for Approve Membership Application. The Board of Directors need to login first and the system will display BoD dashboard. BoD will navigate to the list of membership applications. BoD is also able to view and see the details of the applicant and decide to approve or reject the application. If the application is approved then it will store to the database and send the message to the applicant if it is rejected.

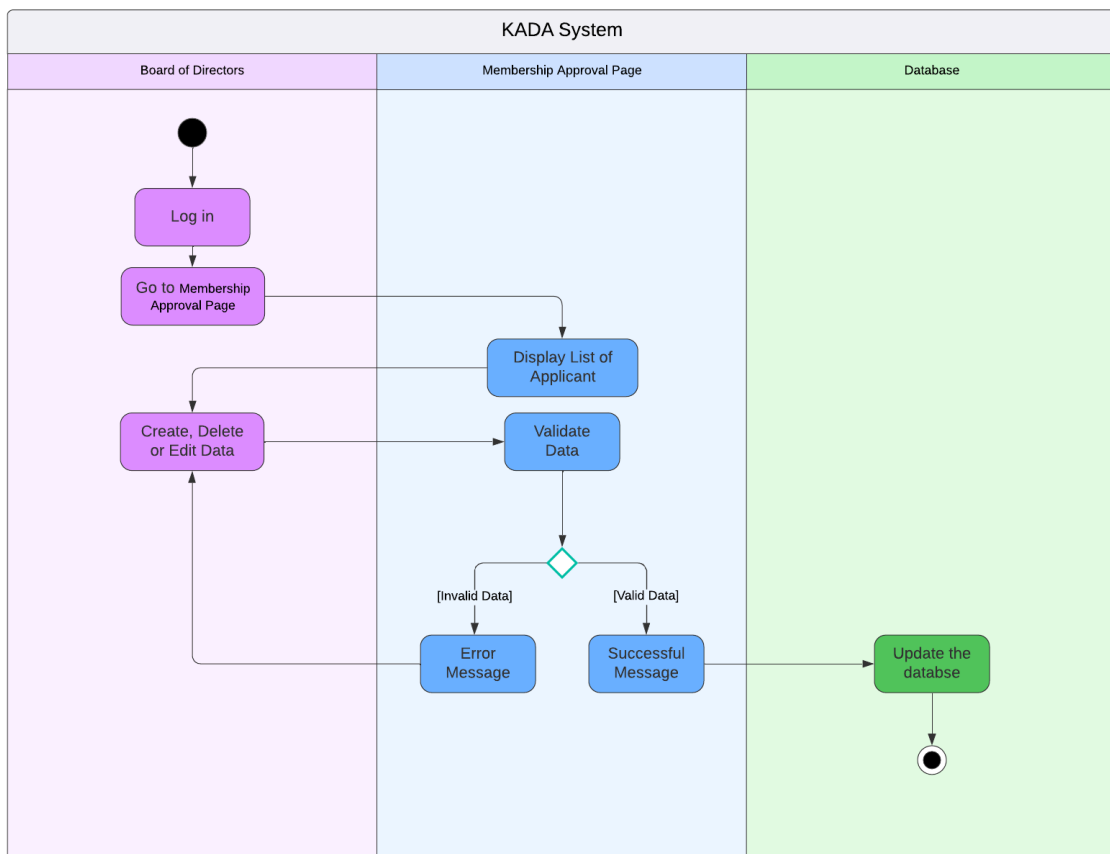


Figure 2.17: Activity Diagram for Approve Membership Application

Figure 2.17 shows an activity diagram for Approve Membership Application. The Board of Directors need to login first and the system will display BoD dashboard. BoD will navigate to the list of membership applications. BoD is also able to view and see the details of the applicant and decide to approve or reject the application. If the application is approved then it will store to the database and send the message to the applicant if it is rejected.

2.4.9 US009: Approve Loan Application

Table 2.11: User Story Description for Approve Loan Application

User story: Approve Loan Application
ID: US009
User Story Description As a Board of Directors, I want to access the loan approval page so that I can approve or reject the list of loan applications.
Flow of events: 1. The Board of Directors logs into the system. 2. The system displays the Loan Application Page. 3. The Board of Directors views a list of submitted applications 4. The Board of Directors selects an application to review. 5. The Board of Directors decides whether to approve or reject the applications.
Alternative flow : 1. If there's an issue during approval for example a database error, the system displays an error message. 2. The Board may retry the process.
Postcondition: The board of director change the status of loan application either approved or rejected
Precondition: 1. The Board of Directors must be logged into the system. 2. The loan application data must be available for review.

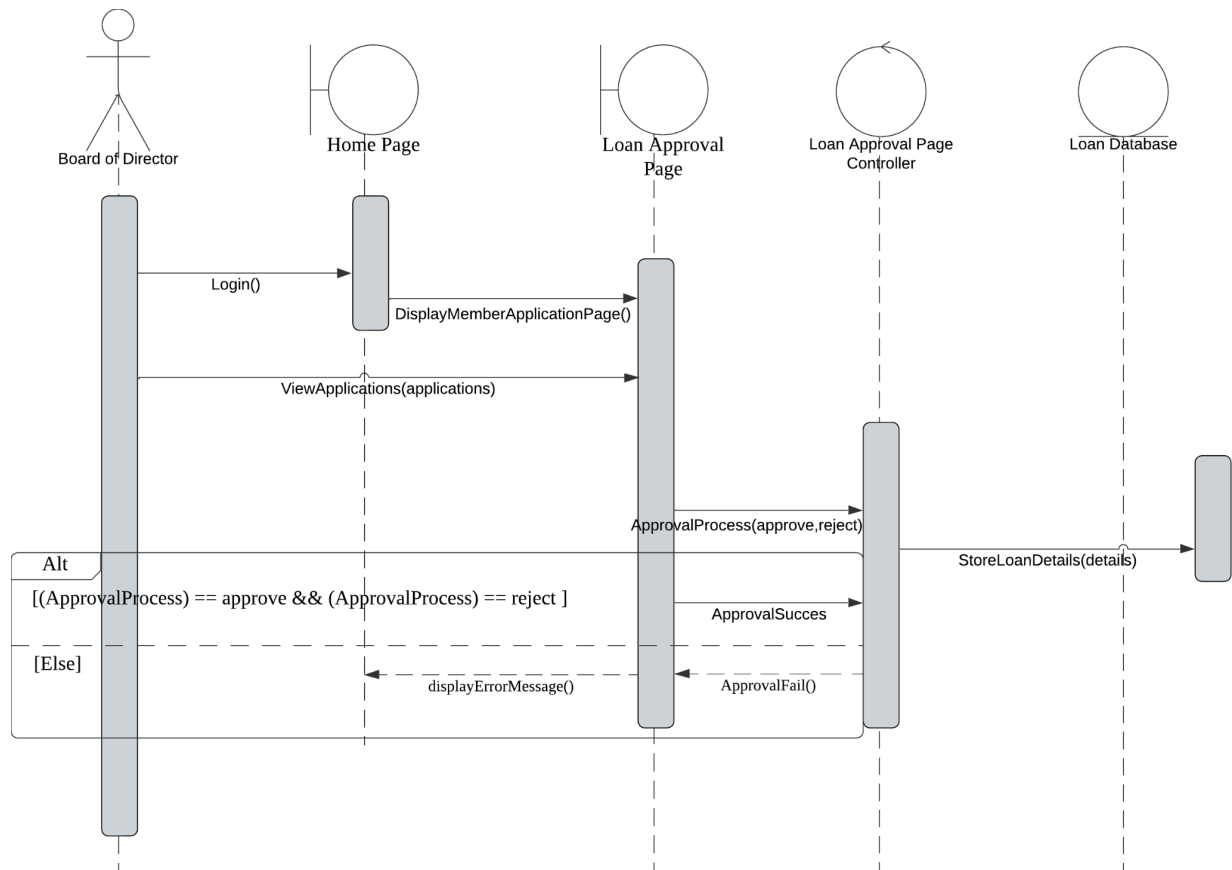


Figure 2.18: Sequence Diagram for Approve Loan Application

Figure 2.18 shows the sequence diagram for approving loan applications. The board of directors view the list of members that applies for a loan. The board of directors view a member that applies for a loan to see their details. They can either approve or reject the loan application.

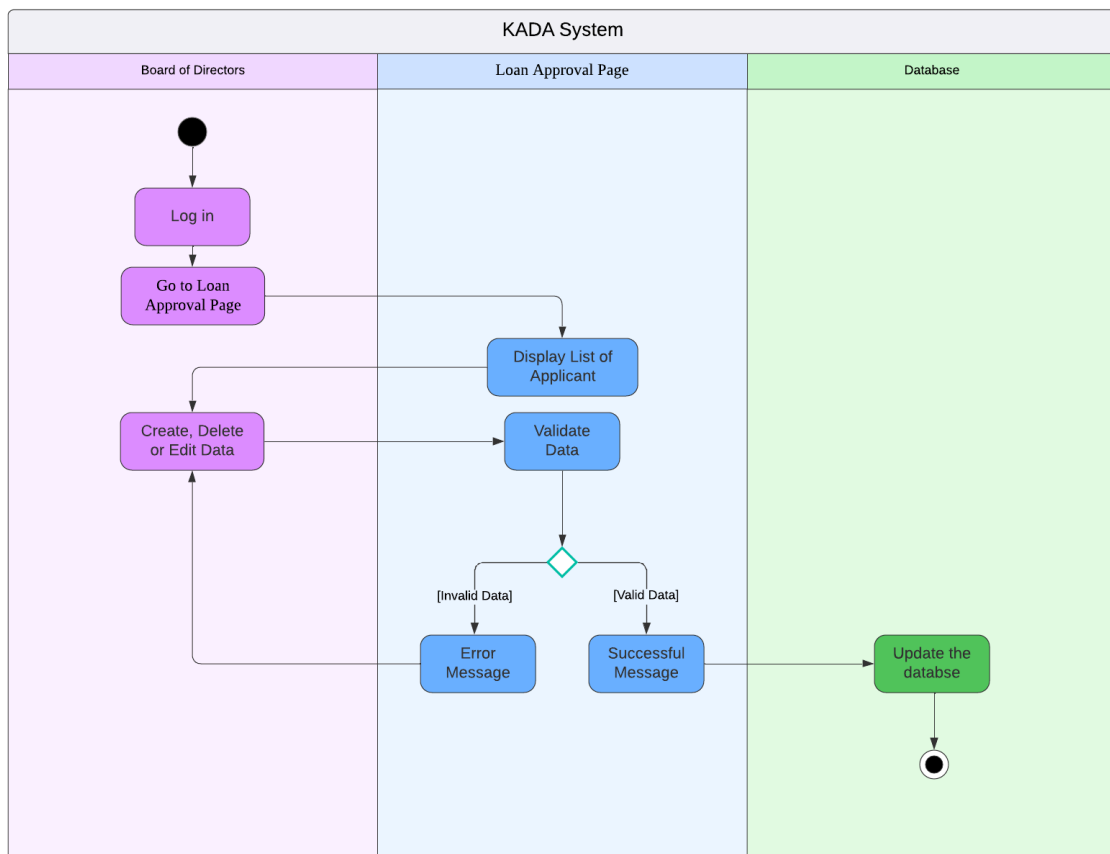


Figure 2.19: Sequence Diagram for Approve Loan Application

Figure 2.19 shows the activity diagram for approving loan applications. The board of directors view the list of members that applies for a loan. The board of directors views a member that applies for a loan to see their details. They can either approve or reject the loan application.

2.4.10 US010: Manage Loan Policies

Table 2.12: User Story Description for Manage Loan Policies

User story: Manage Cooperative Database and Policies
ID: US010
User Story Description: As a clerk, I want to be able to log into the system so that I can manage the cooperative's loan policy.
Flow of events: 1. The clerk navigates to the loan policy page 2. The clerk chooses the type of loan. 3. The clerk updates the details of the loan such as the loan rate and max amount of loan.
Postcondition: The database is updated successfully.
Precondition: The user is logged in as an administrator.

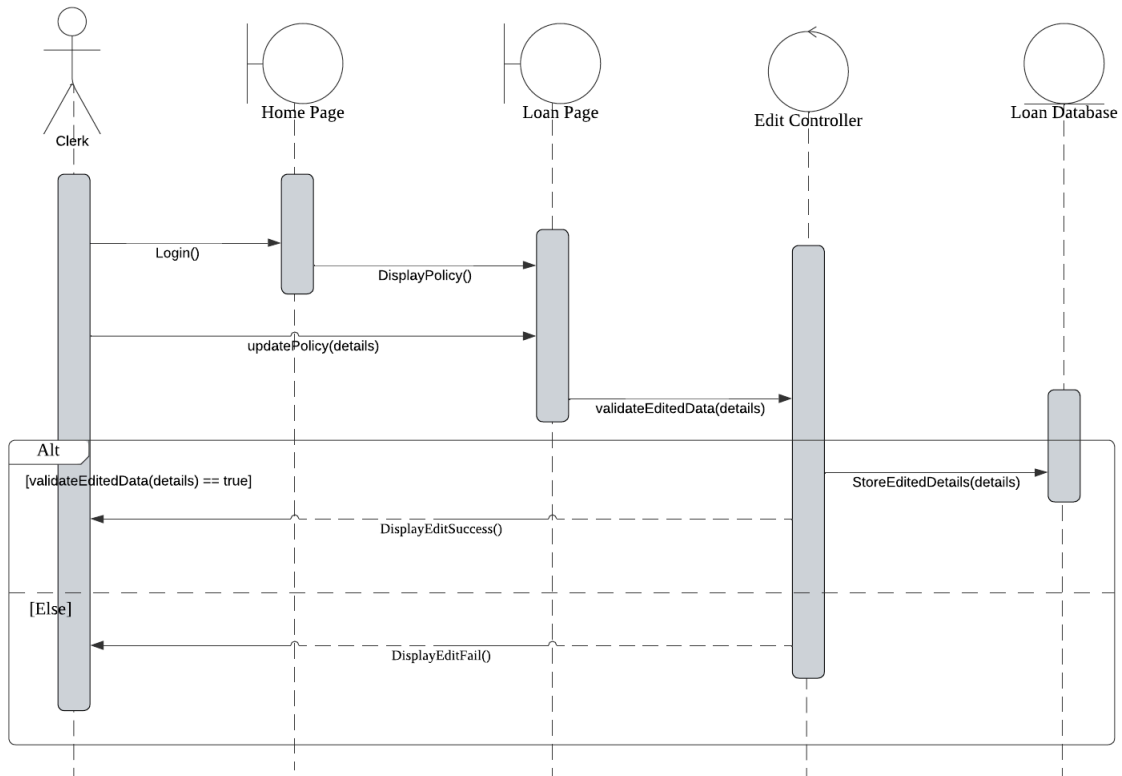


Figure 2.20: Sequence Diagram for Manage Loan Policy

Figure 2.20 shows the sequence diagram for managing loan policy. The user assigned for this task is the clerk. The clerk can update the database policy such as the loan rate, maximum and minimum amount of loan amount as well minimum share modal required to apply for a loan.

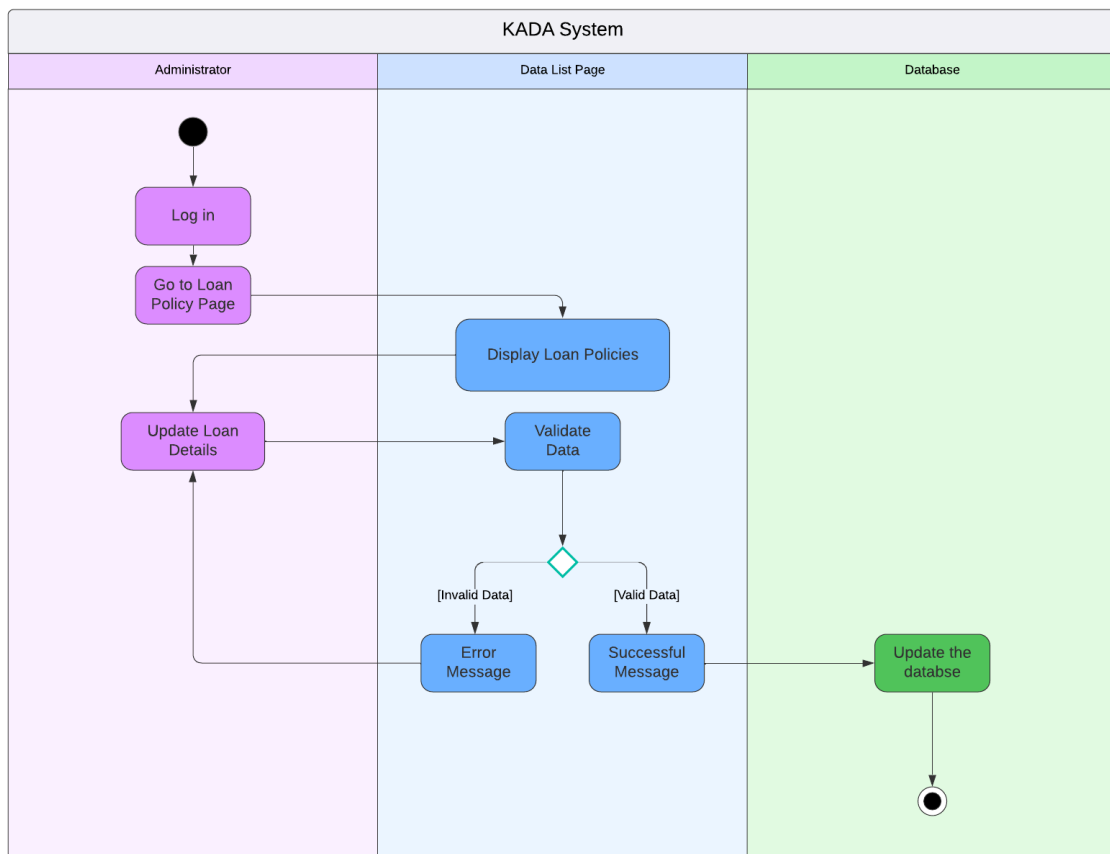


Figure 2.21: Activity Diagram for Manage Loan Policy

Figure 2.21 shows the activity diagram for managing loan policy. The user assigned for this task is the clerk. The clerk can update the database policy such as the loan rate, maximum and minimum amount of loan amount as well minimum share modal required to apply for a loan.

2.4.11 US011: Generate Report

Table 2.13: User Story Description for Generate Report

User story: Generate Report
ID: US011
User Story Description: As a clerk, I want to be able to generate report for the use of the cooperative
Flow of events: 1. The clerk navigates to the generate report page 2. The clerks choose whether to generate a monthly or yearly report. 3. If the clerk chose a monthly report, the clerk input the month and year of the report. 4. If they choose a yearly report, the clerk inputs the year. 5. The clerk clicks submit and a PDF report will be displayed.
Postcondition: A report according to the chosen report type and inputted values
Precondition: The user is logged in as a clerk.

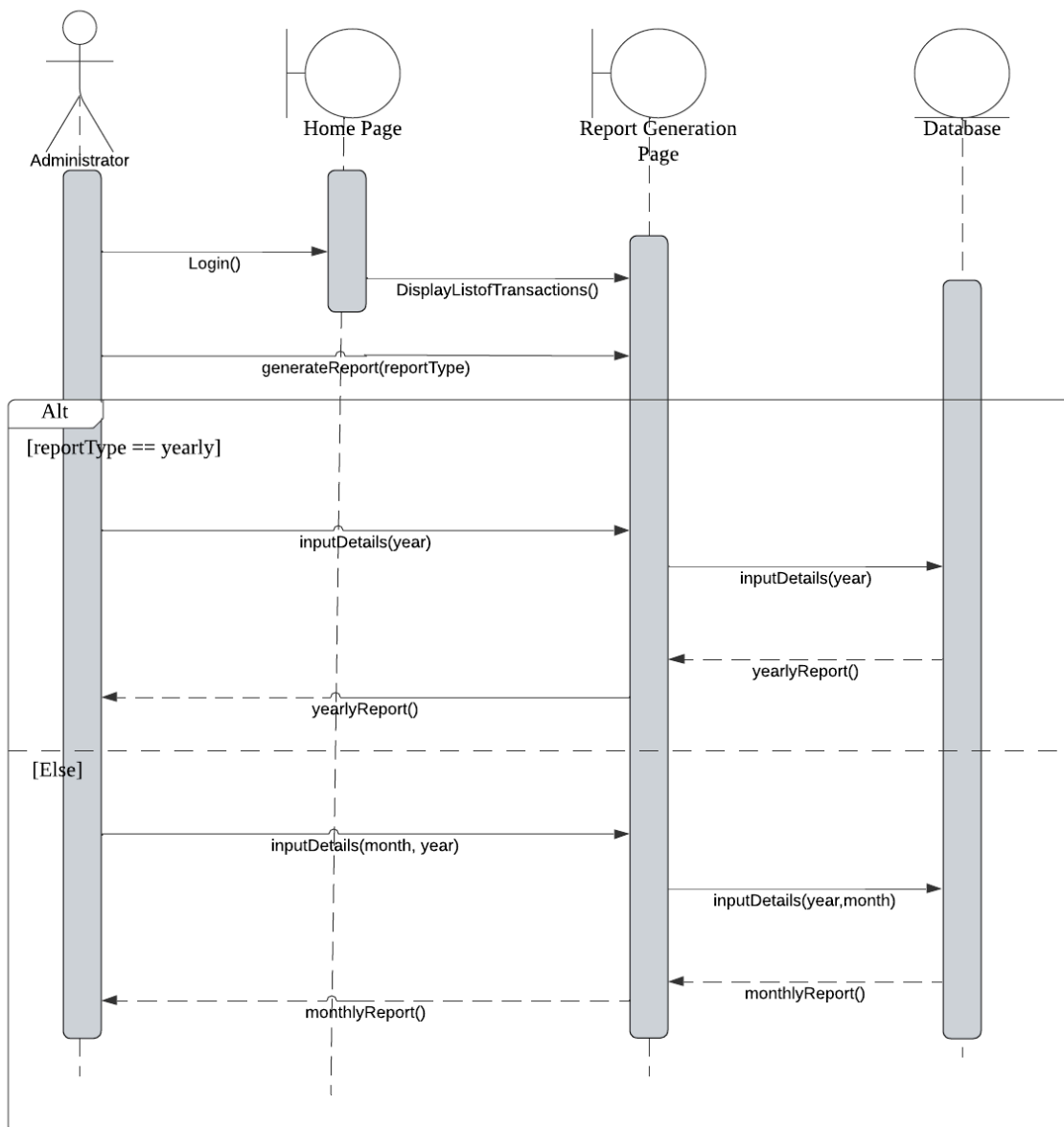


Figure 2.22: Sequence Diagram for Generate Report

Figure 2.22 shows the process for generating reports. This process starts with the clerk login to the system and going to the page. After that, the clerk chooses the report type which is yearly or monthly. After that, The system will retrieve the data from the database and generate the report in a pdf file.

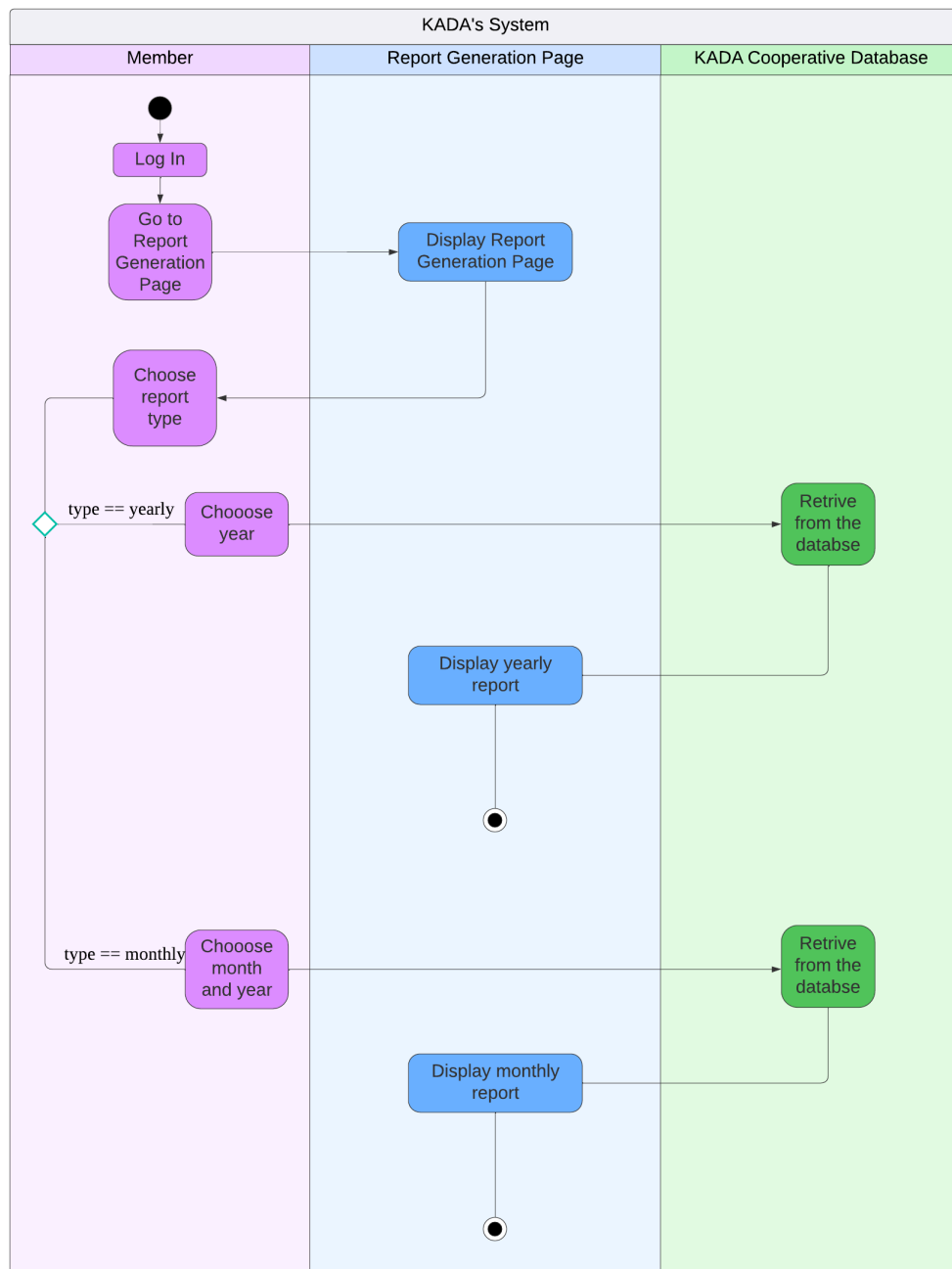


Figure 2.23: Sequence Diagram for Generate Report

Figure 2.23 shows the process for generating reports. This process starts with the clerk login to the system and going to the page. After that, the clerk chooses the report type which is yearly or monthly. After that, The system will retrieve the data from the database and generate the report in a pdf file.

2.4.12 US012: Authentication

Table 2.14: User Story Description for Authentication (Member)

User story: Authentication
ID: US012
User Story Description: As a member, I want to be able to log into the system so that I can access the features in the system.
Flow of events: 1. The members go to the login page. 2. The members key in staff ID and password. 3. The system validates the staff ID and password by checking it in the database.
Postcondition: The member logged in successfully.
Precondition: The member has already registered the membership.

Table 2.15: User Story Description for Authentication (Clerk)

User story: Authentication
ID: US012
User Story Description: As a clerk, I want to be able to log into the system so that I can access the features in the system.
Flow of events: 4. The clerk goes to the login page. 5. The clerk key in staff ID and password. 6. The system validates the staff ID and password by checking it in the database.
Postcondition: The clerk logged in successfully.
Precondition: The clerk has already registered.

Table 2.16: User Story Description for Authentication (Board of Director)

User story: Authentication
ID: US012
User Story Description: As a board of director, I want to be able to log into the system so that I can access the features in the system.
Flow of events: 7. The board of directors goes to the login page. 8. The board of directors key in staff ID and password. 9. The system validates the staff ID and password by checking it in the database.
Postcondition: The board of directors logged in successfully.
Precondition: The board of directors has already registered.

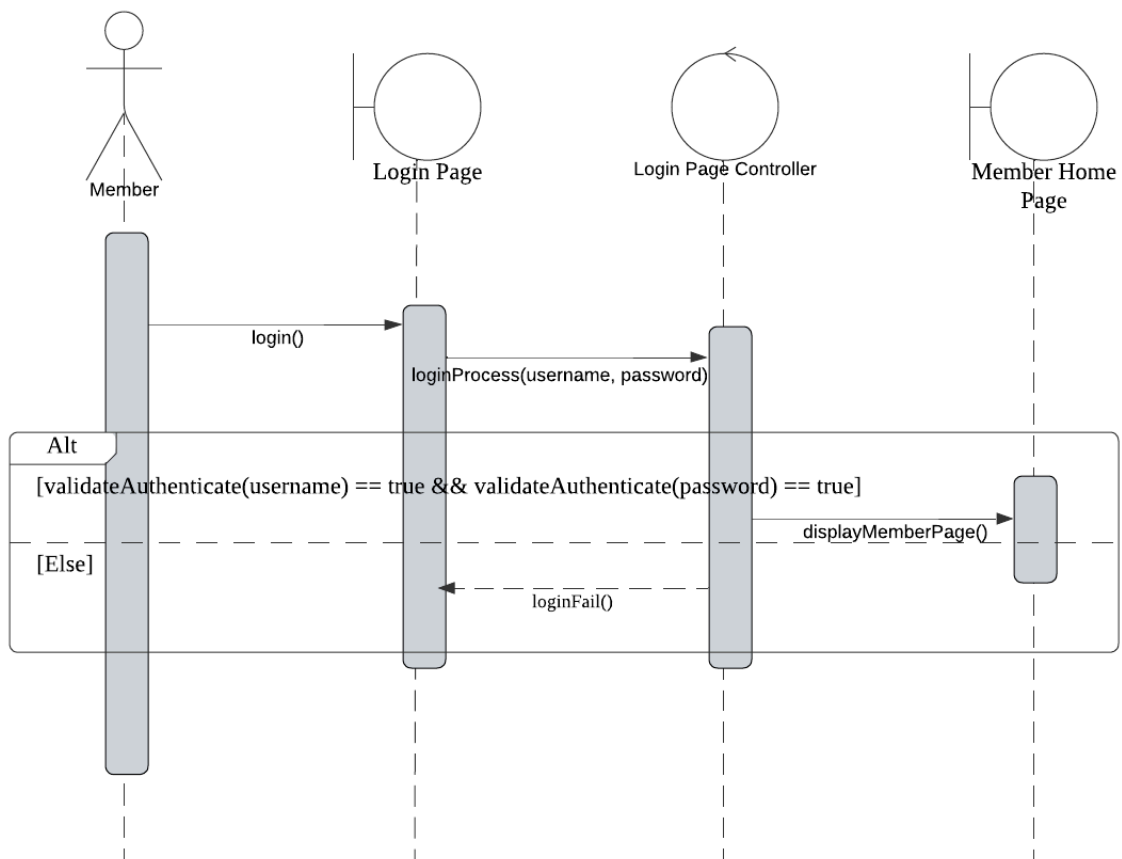


Figure 2.24: Sequence Diagram for Authentication (Member)

Figure 2.24 shows the sequence diagram for authenticating a user that is a member. The user has to input credentials such as staff number and password. If the system detects the user to be a member, they will be directed to the member dashboard page. The system will alert the user if the credentials are invalid.

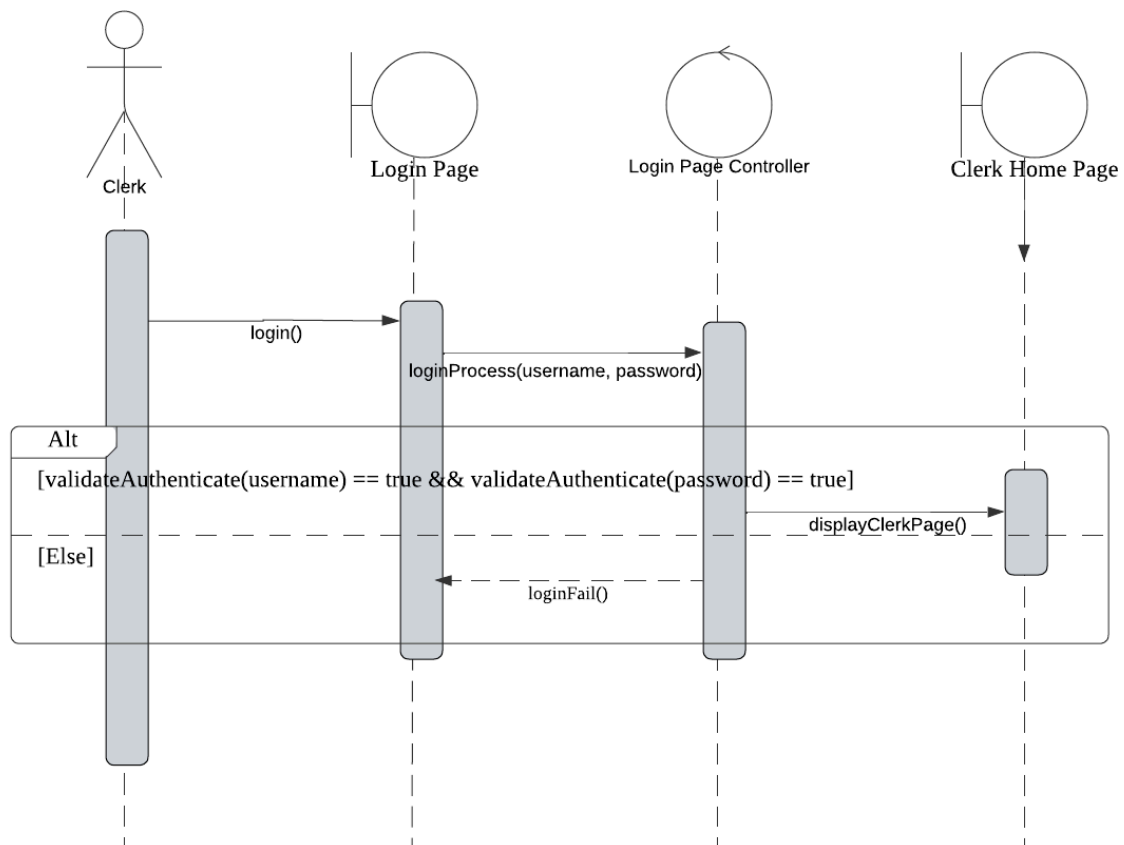


Figure 2.25: Sequence Diagram for Authentication (Clerk)

Figure 2.25 shows the sequence diagram for authenticating a user that is a clerk. The user has to input credentials such as staff number and password. If the system detects the user to be a clerk, they will be directed to the clerk dashboard page. The system will alert the user if the credentials are invalid.

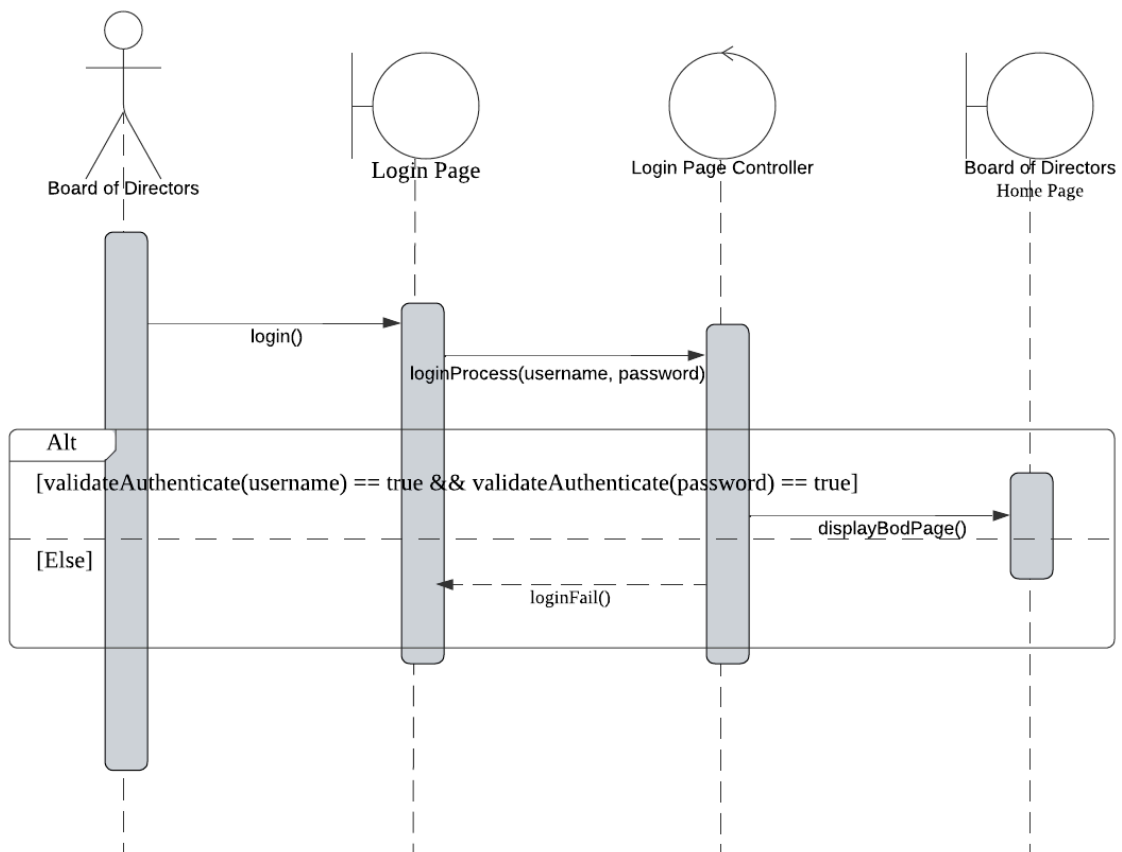


Figure 2.26: Sequence Diagram for Authentication (Board of Directors)

Figure 2.26 shows the sequence diagram for authenticating a user that is a board of director. The user has to input credentials such as staff number and password. If the system detects the user to be a board of director, they will be directed to the board of director dashboard page. The system will alert the user if the credentials are invalid.

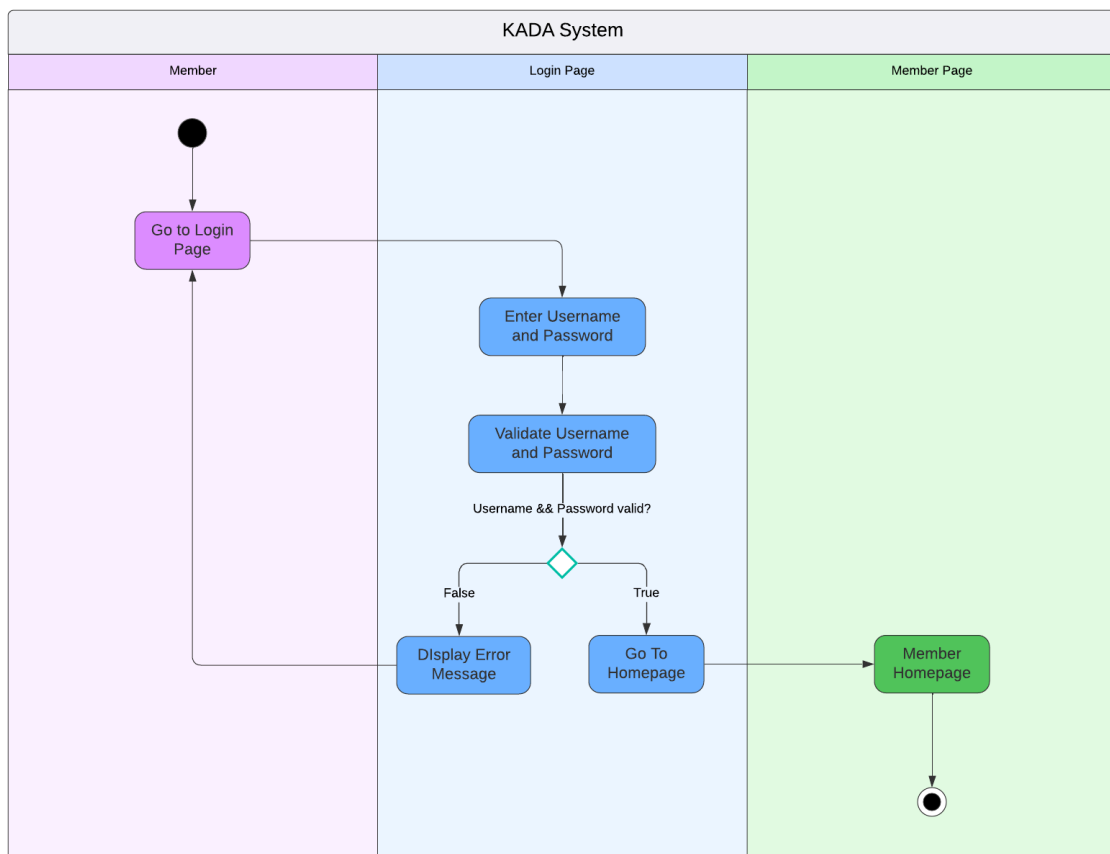


Figure 2.27: Activity Diagram for Authentication (Member)

Figure 2.27 shows the activity diagram for authenticating a user that is a member. The user has to input credentials such as staff number and password. If the system detects the user to be a member, they will be directed to the member dashboard page. The system will alert the user if the credentials are invalid.

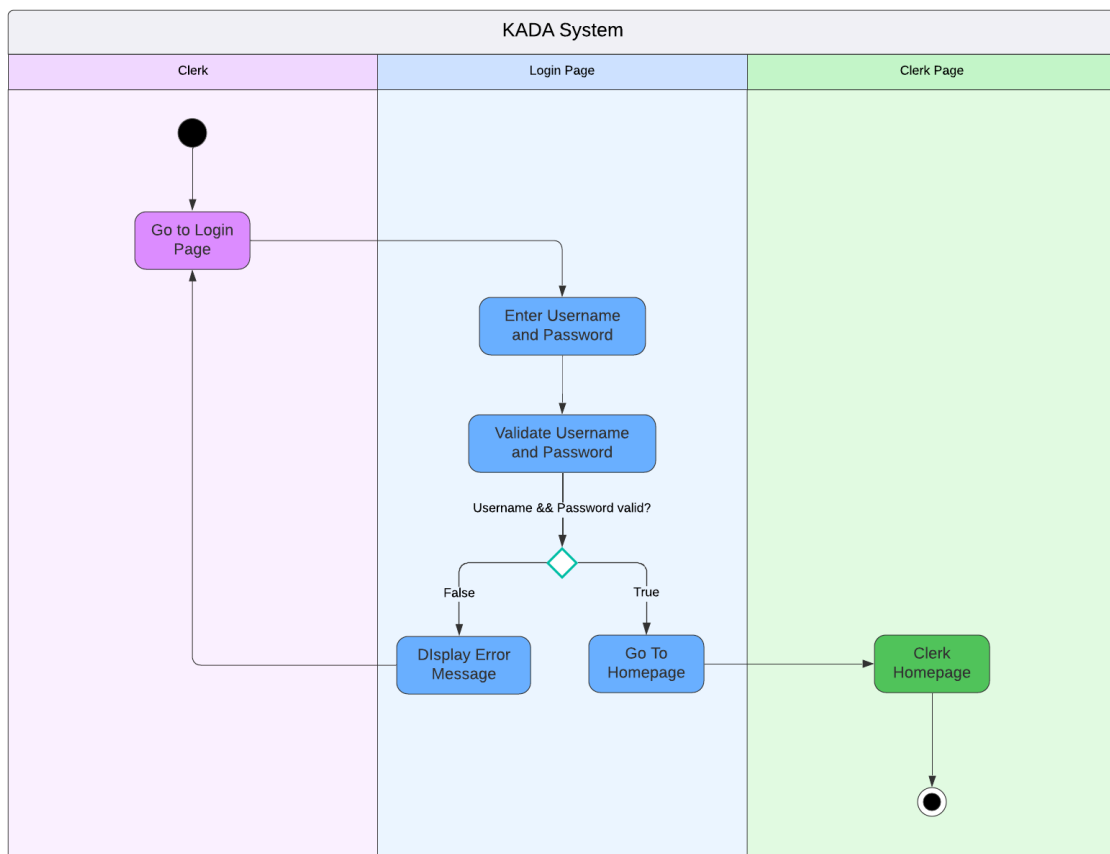


Figure 2.28: Activity Diagram for Authentication (Clerk)

Figure 2.28 shows the activity diagram for authenticating a user that is a clerk. The user has to input credentials such as staff number and password. If the system detects the user to be a clerk, they will be directed to the clerk dashboard page. The system will alert the user if the credentials are invalid.

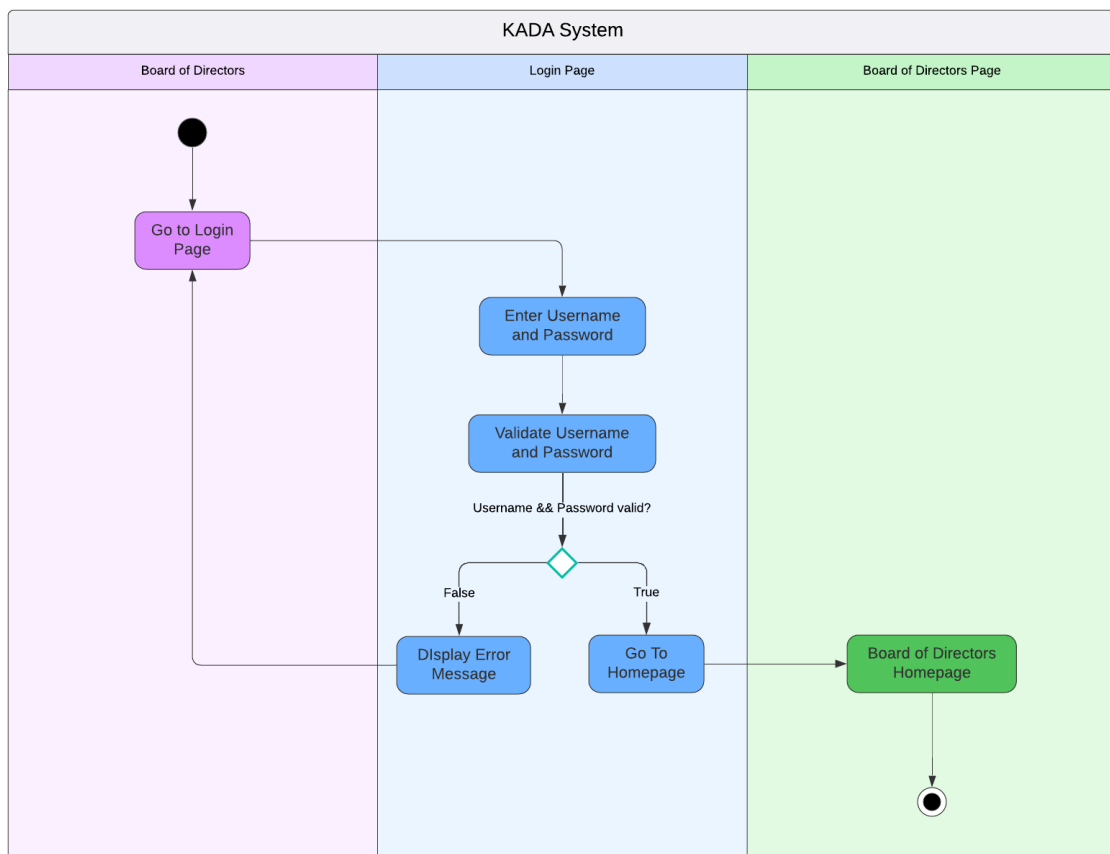


Figure 2.29: Activity Diagram for Authentication (Board of Directors)

Figure 2.29 shows the activity diagram for authenticating a user that is a board of director. The user has to input credentials such as staff number and password. If the system detects the user to be a board of director, they will be directed to the board of director dashboard page. The system will alert the user if the credentials are invalid.

2.5 Performance and Other Requirements

Performance requirements define the system's capability to respond to user requests and handle data in a timely manner. These requirements include the following:

- **Response Time:** The time it takes for the system to respond to a user request.
- **Throughput:** The number of requests the system can handle in a given period of time.
- **Capacity:** The maximum number of users or amount of data that the system can handle.
- **Availability:** The percentage of time the system is operational and accessible to users.

Software system attributes define the overall qualities or characteristics of the software. These attributes are the foundation on which the system is built, and they include the following:

- **Correctness:** The system shall be able to correctly behave as per the system requirements, specifications and features.
- **Maintainability:** The system shall be able to be maintained easily by adding new features and updating existing features with new technologies.
- **Usability:** The system shall be able to be usable towards the users of the system for a good customer experience.
- **Portability:** The system shall be portable where the system can be migrated to other environments with different hardware and operating systems.
- **Compatibility:** The system shall be able to be compatible with multiple operating systems and web browsers without affecting intended functionality.

2.6 Design Constraints

Below are the constraints imposed by the organisation.

- Environmental constraints: The system must be designed to operate in cold temperatures for the database to work fine. The temperature must be 0-25 Celcius.
- Hardware constraints: The system must be designed to run with enough storage and RAM because the system will run on KADA's on-premises or cloud-based server setup. The processors used are Intel Xeon or AMD EPYC with a minimum of 8 cores. The minimum ram is 16 GB or more for optimal performance. Solid State Drive (SSD) with at least 512 GB for fast data retrieval and secure storage and minimum bandwidth of 1 Gbps.
- Security constraints: The system must be designed to encrypt all the user data with secure protocol using AES-256 for storage and SSL/TLS for transmission. For the communication, the system used HTTPS for secure web-based access to the cooperative system and SMTP for sending and receiving emails within the KADA users.
- Compatibility constraints: The system must be designed to be compatible with many operating systems like Windows and IOS. This will help KADA employees to work with ease.
- Performance constraints: The system must be designed to support a powerful networking environment with a maximum of 1200 concurrent users with low response time which is less than 2 seconds.

3. Architectural Rationale

3.1 Architecture Style and Rationale

The architecture used for the system is the Layered Architecture style. This architecture divides the components of the system into different parts, each of which is in charge of a certain application function. Presentation, Business Logic, and Database are the three main components that make up a typical layered architecture. This architecture only allows the components to interact with its neighbouring layer making it structured and simple.

Scalability is one of the main reasons for using the layered design. New features can be added as the system develops without affecting existing components. Changes on particular components of the system can be made without impacting the whole system by dividing it into separate layers. For example, the Presentation Layer can be modified without affecting the Business Logic or Database layers if improvements on the user experience is needed.

This architecture also has the advantage of being easily maintainable. Because each layer has its own distinct task, troubleshooting and debugging becomes easier. When an issue appears, the error can be quickly determined within a certain layer and apply fixes without causing unwanted interruptions throughout the system. Furthermore, the separation of logic between layers enhances code organization, decreasing complexity and allowing multiple members to work on different areas of the project at the same time.

4. Architectural Views

This section of the SD describes the architectural views of the system namely the scenario view, logical view and physical (deployment) view. These views are representations of the overall architecture of a software system that are meaningful. They enable the architecture to be communicated to and understood further.

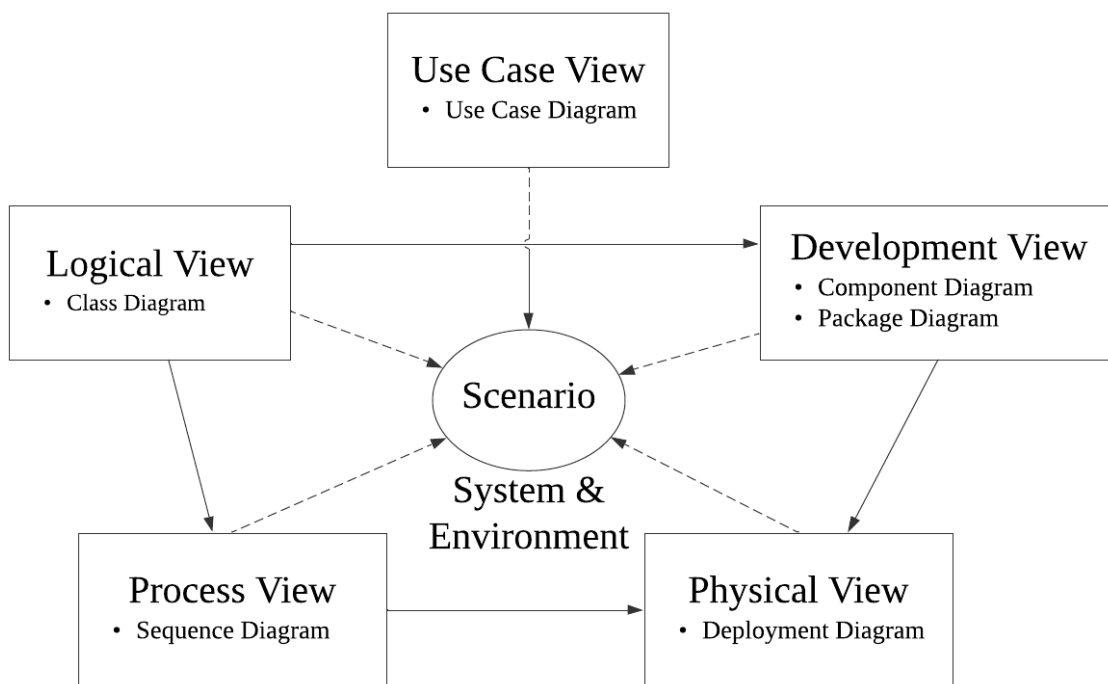


Figure 4.1: Use Case Diagram for KADA's Cooperative Web-Based System

Table 4.1

Architectural Views	Description	Diagram Used
Use Case View	The elements in the four views are shown to work together seamlessly by the use of a small set of important scenarios in a general use cases	Use Case Diagram
Logical View	shows the key abstractions in the system as objects or object classes	Class Diagram

Process View	The process view deals with the dynamic aspects of the system, explains the system processes and how they communicate, and focuses on the run time behavior of the system.	Sequence Diagram
Development (Implementation) View	Shows how the software is decomposed for development	Component Diagram Package Diagram
Physical (Deployment) View	Shows the system hardware and how software components are distributed across the processors in the system	Deployment Diagram

4.1 Use Case View

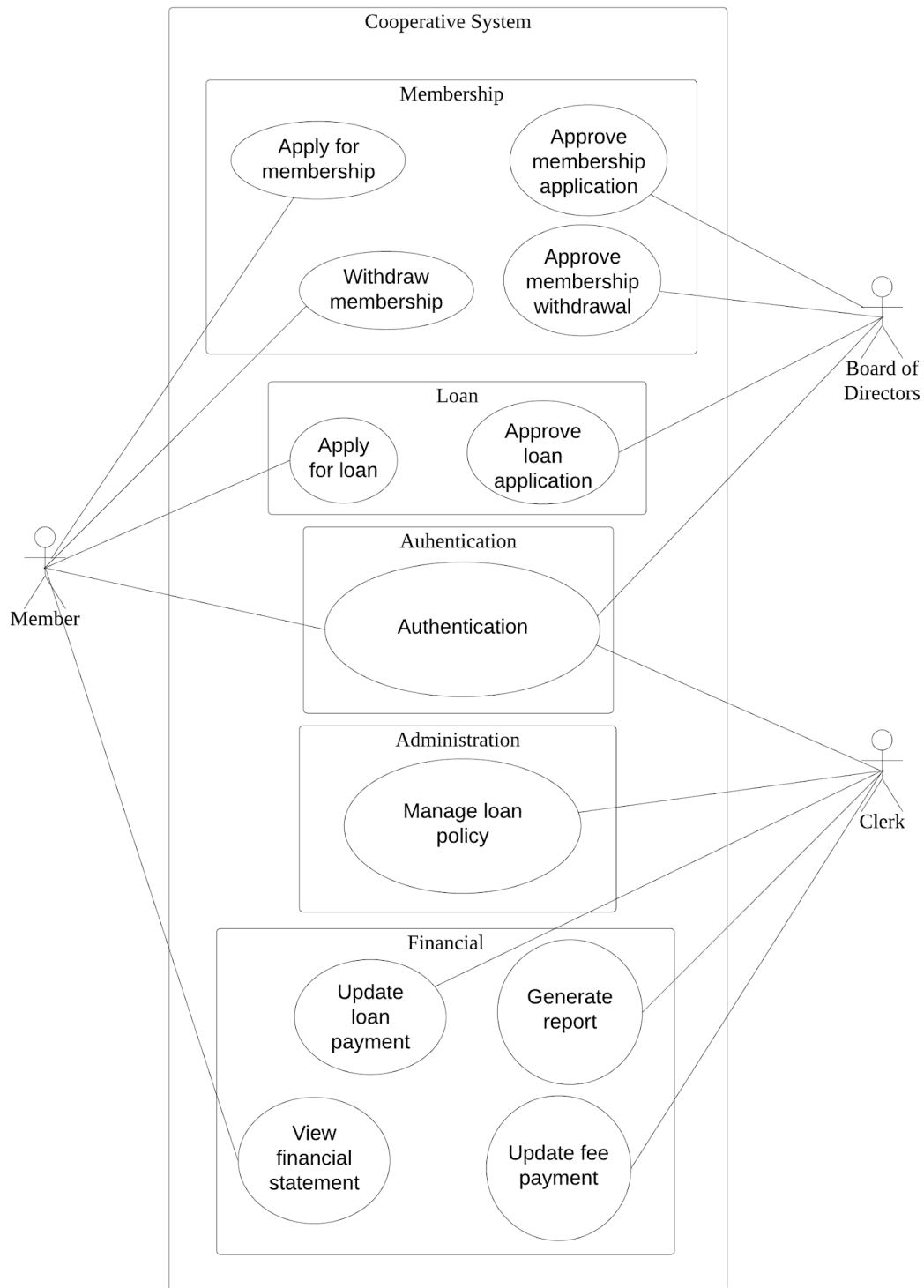


Figure 4.1: Use Case Diagram for KADA's Cooperative Web-Based System

4.2 Implementation View

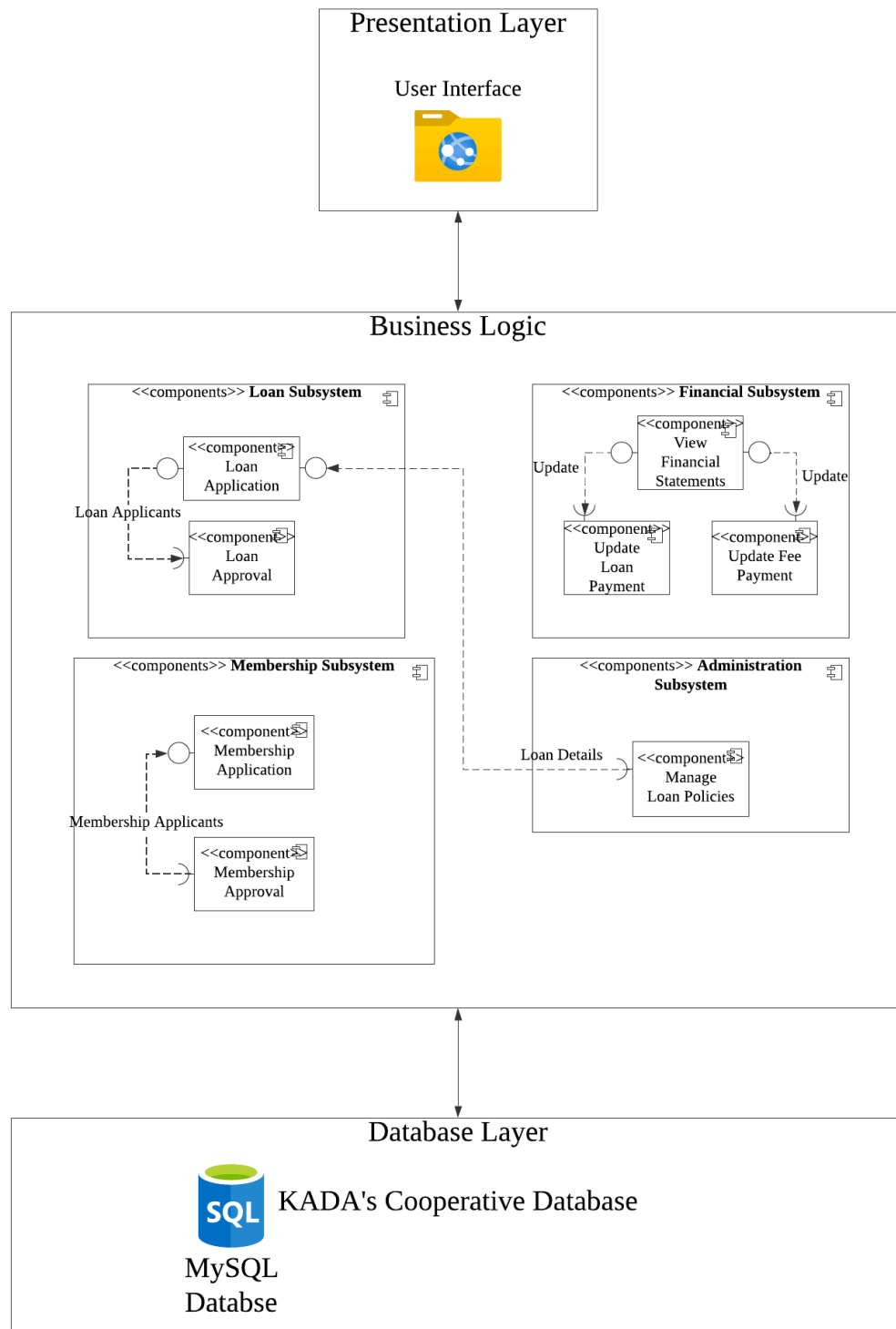


Figure 4.3: Component Diagram of KADA's Cooperative Web-Based System

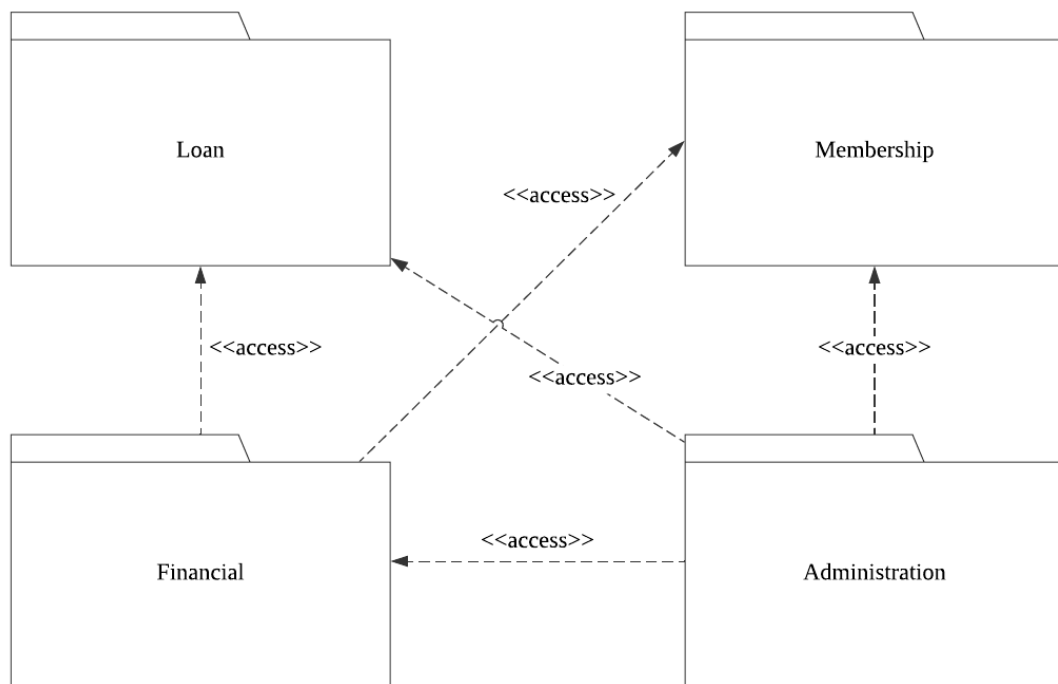


Figure 4.4: Package Diagram for KADA's Cooperative Web-Based System

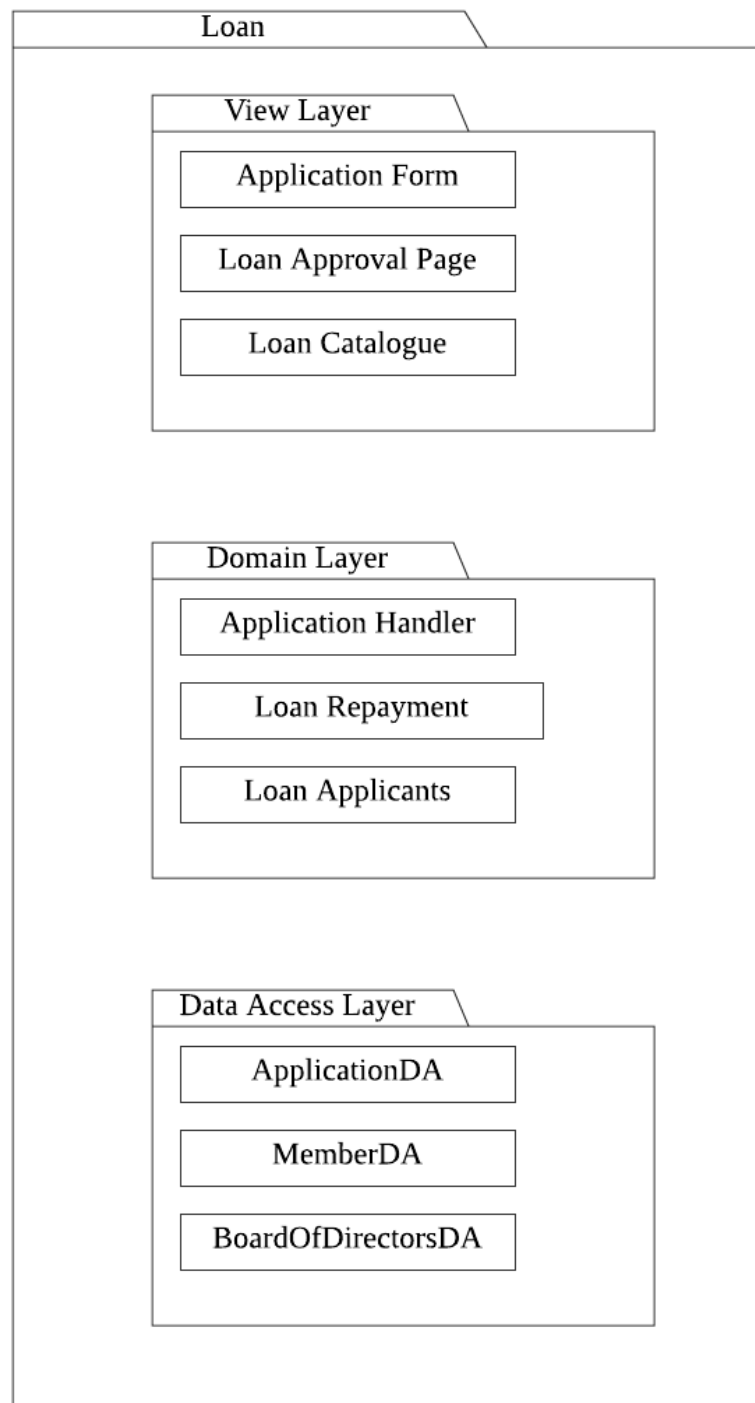


Figure 4.5: Package Diagram for Loan Subsystem

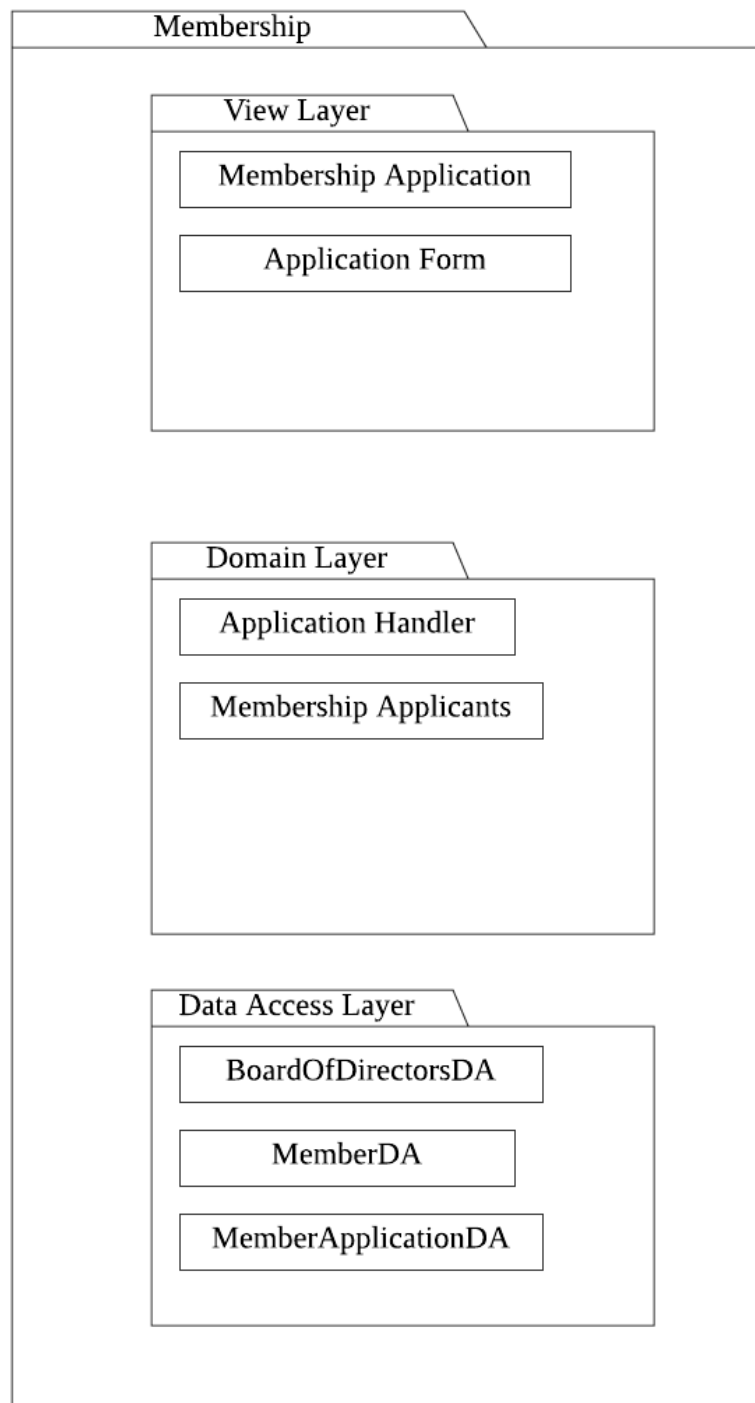


Figure 4.6: Package Diagram for Membership Subsystem

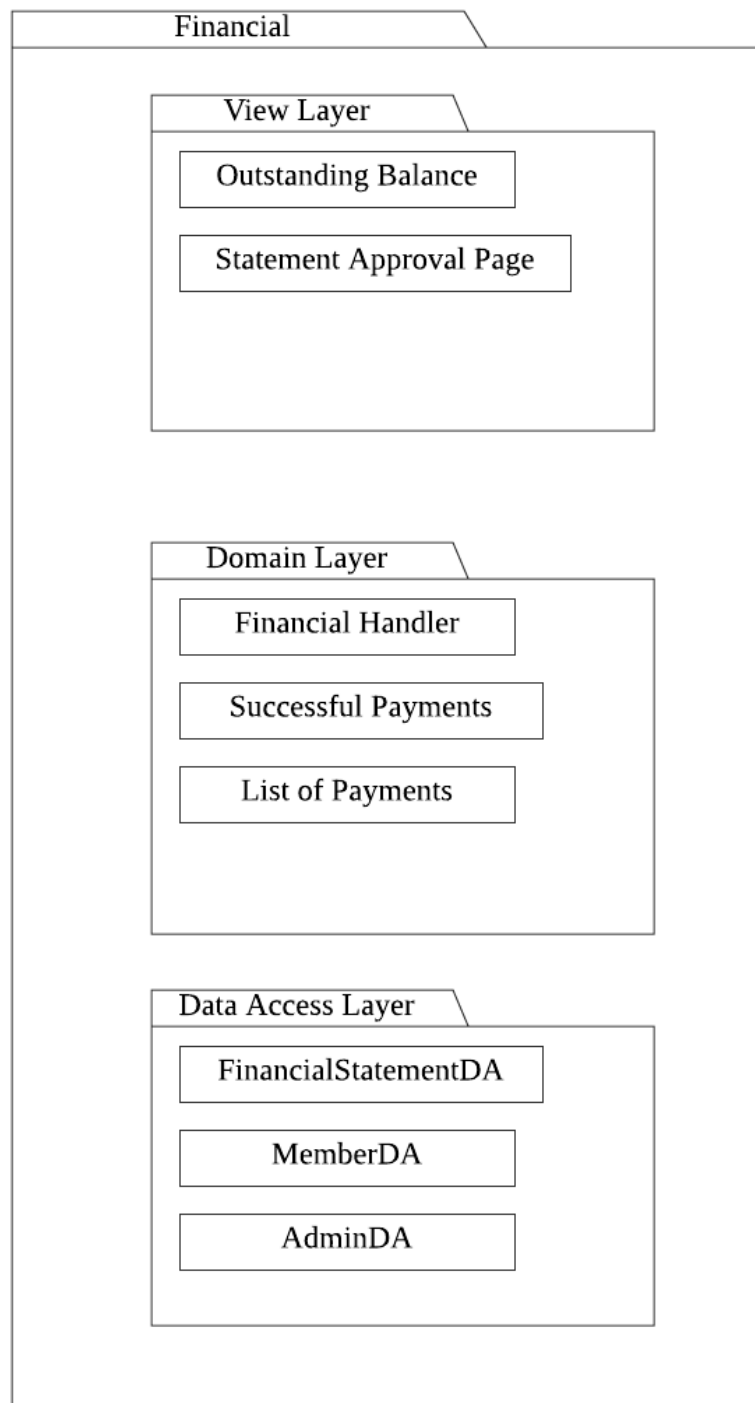


Figure 4.7: Package Diagram for Financial Subsystem

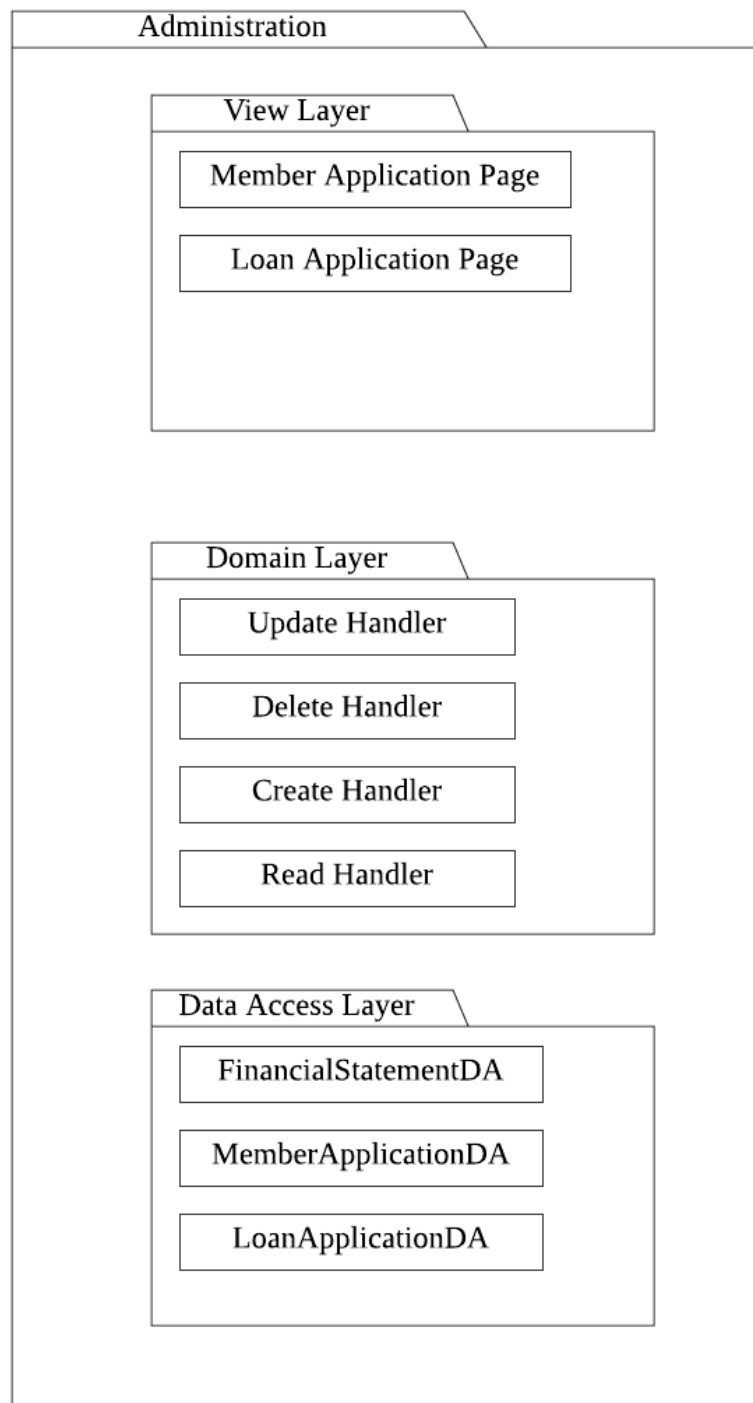


Figure 4.8 Package Diagram of Administration Subsystem

4.3 Logical View

For the logical view, the class diagram is divided by the four modules (subsystems) which are loans, membership, financial and administration.

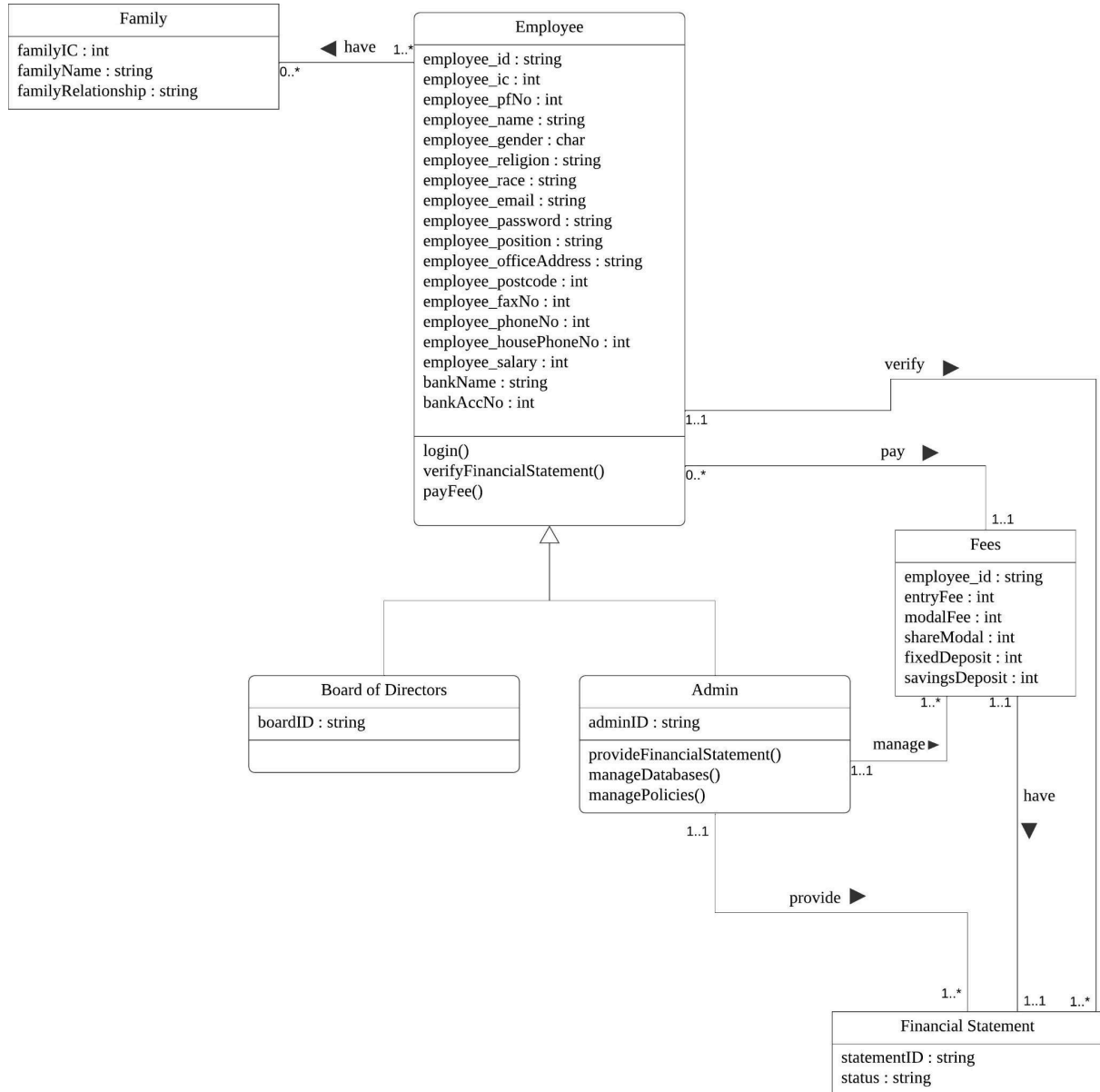


Figure 4.9: Class diagram for Financial Subsystem

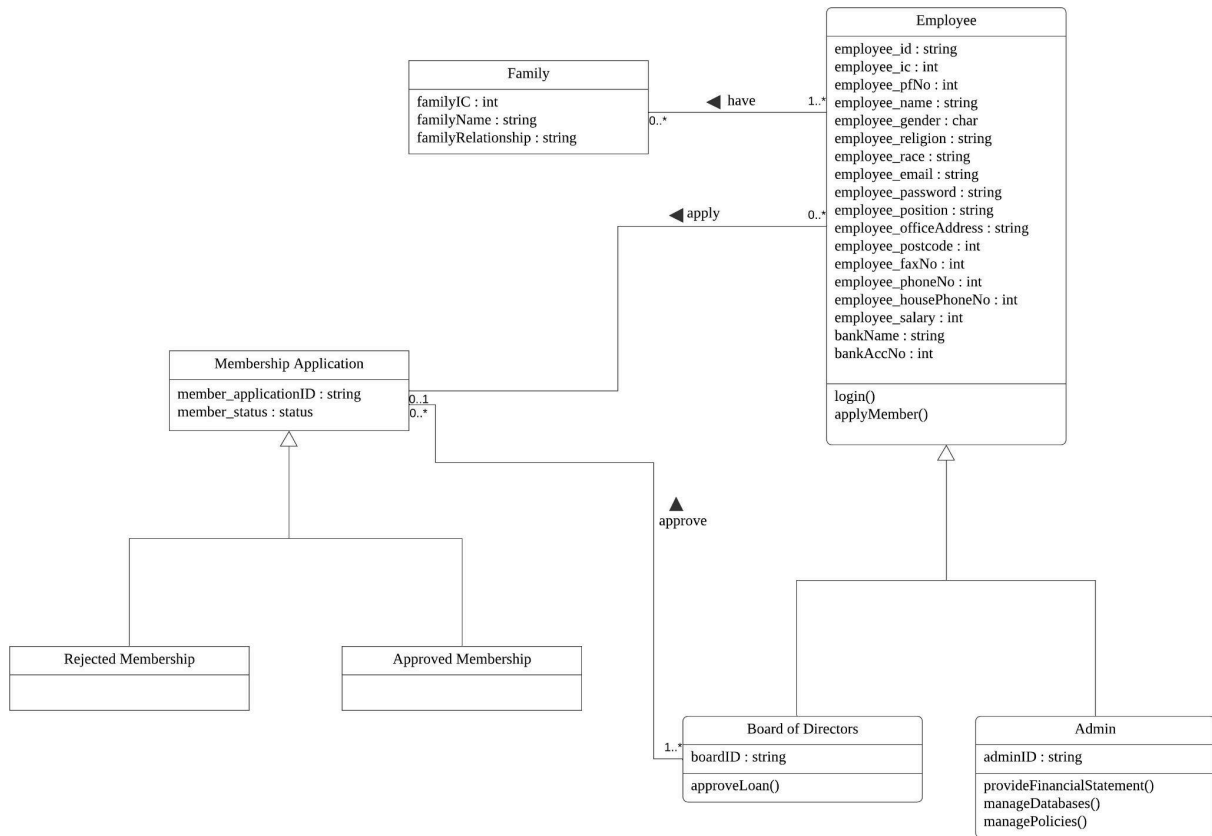


Figure 4.10: Class Diagram for Membership Subsystem

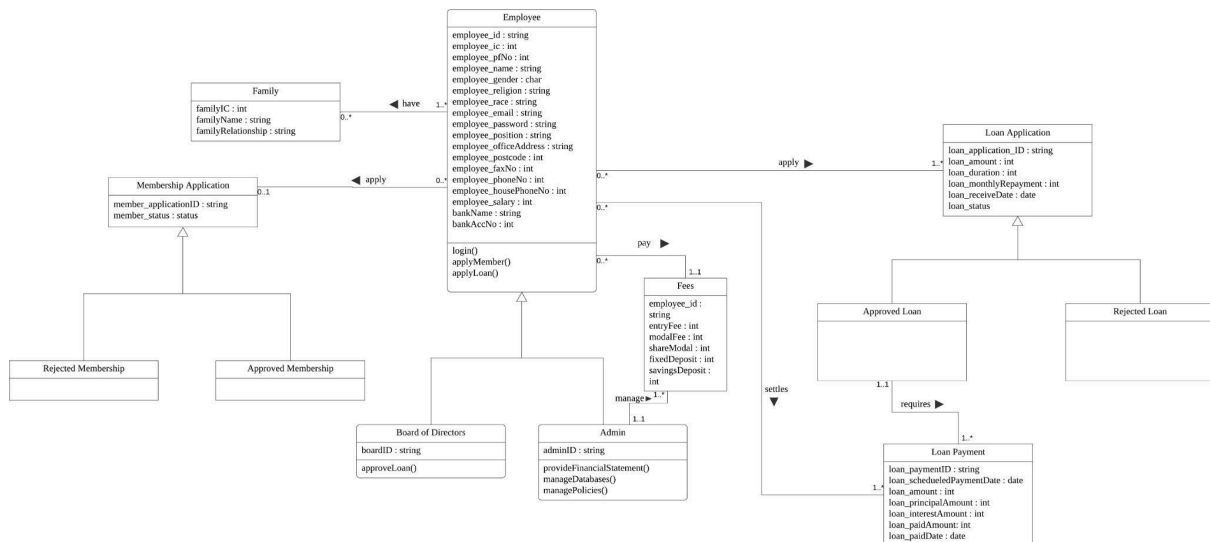


Figure 4.11 Class Diagram for Administration Subsystem

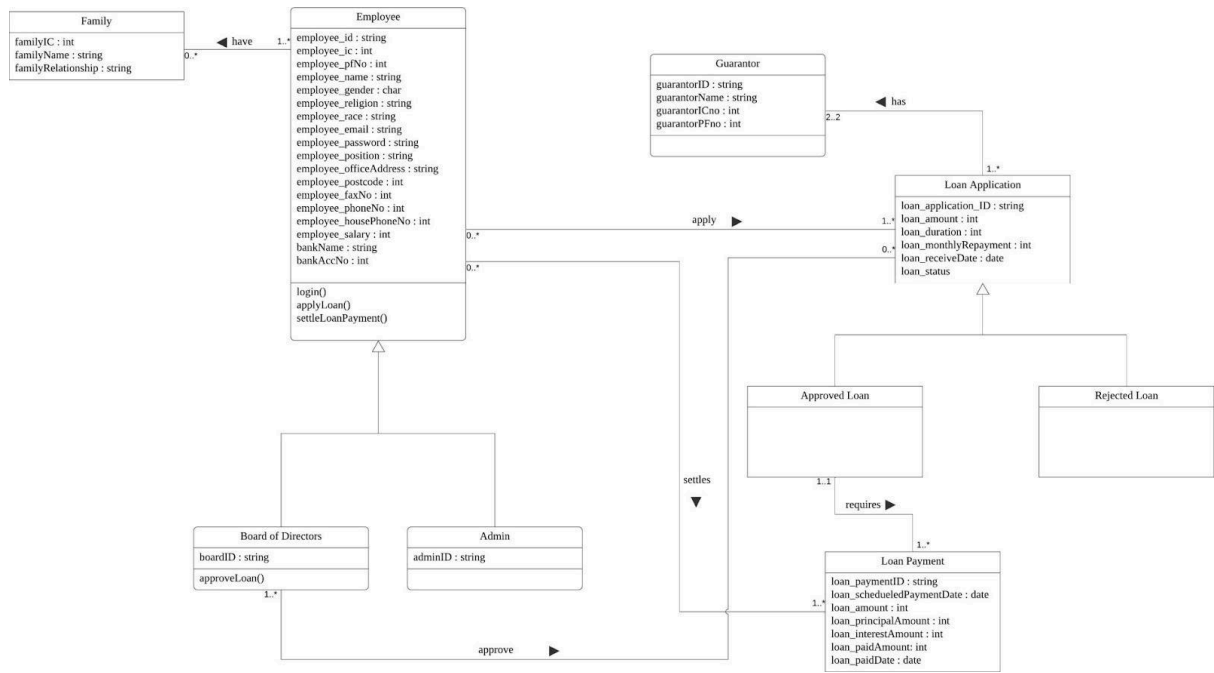


Figure 4.12 Class Diagram for Loan Subsystem

4.4 Process View

The process view is illustrated by using sequence diagrams that detail how operations are carried out.

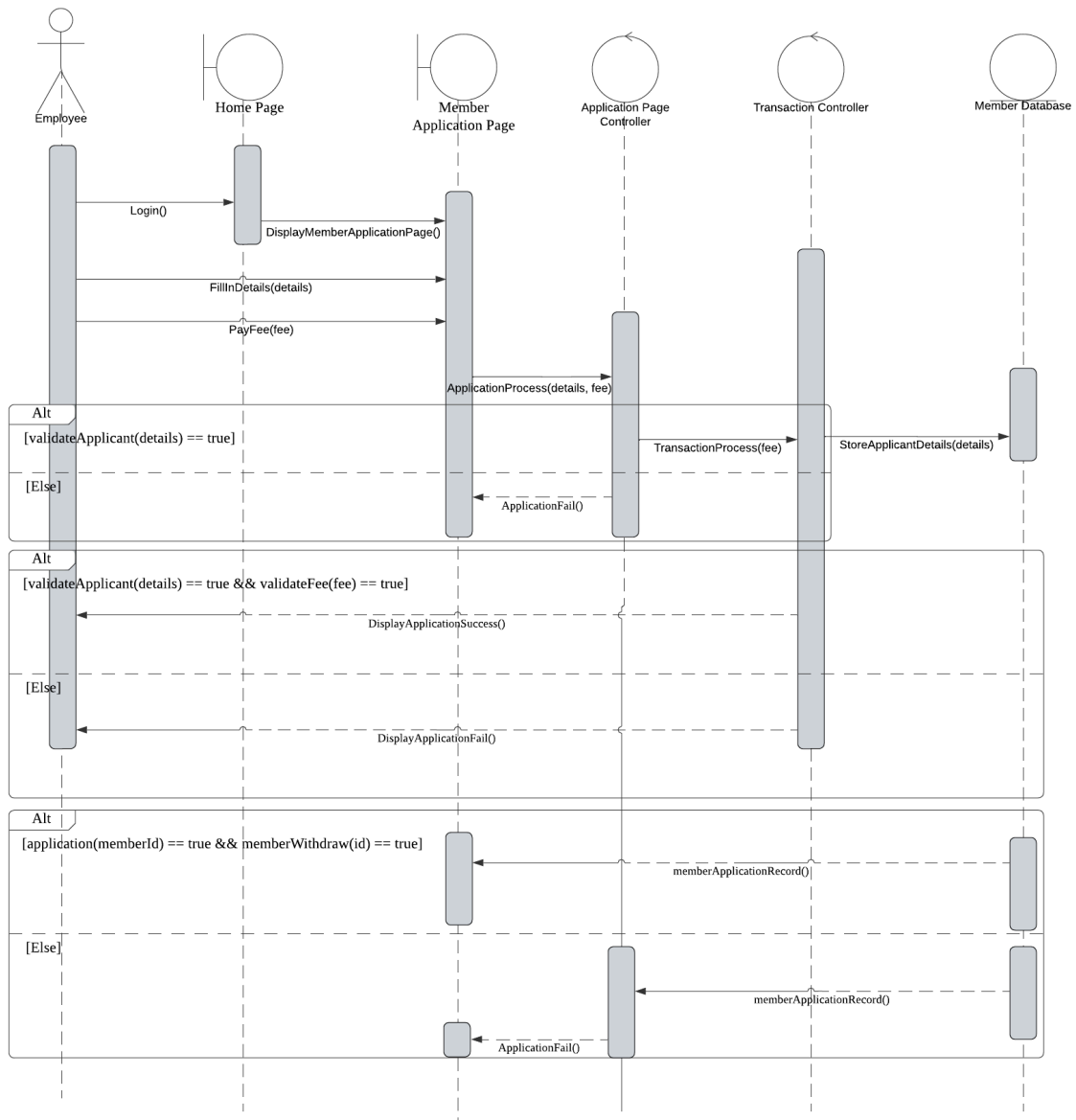


Figure 4.13: Sequence Diagram for Apply for Membership

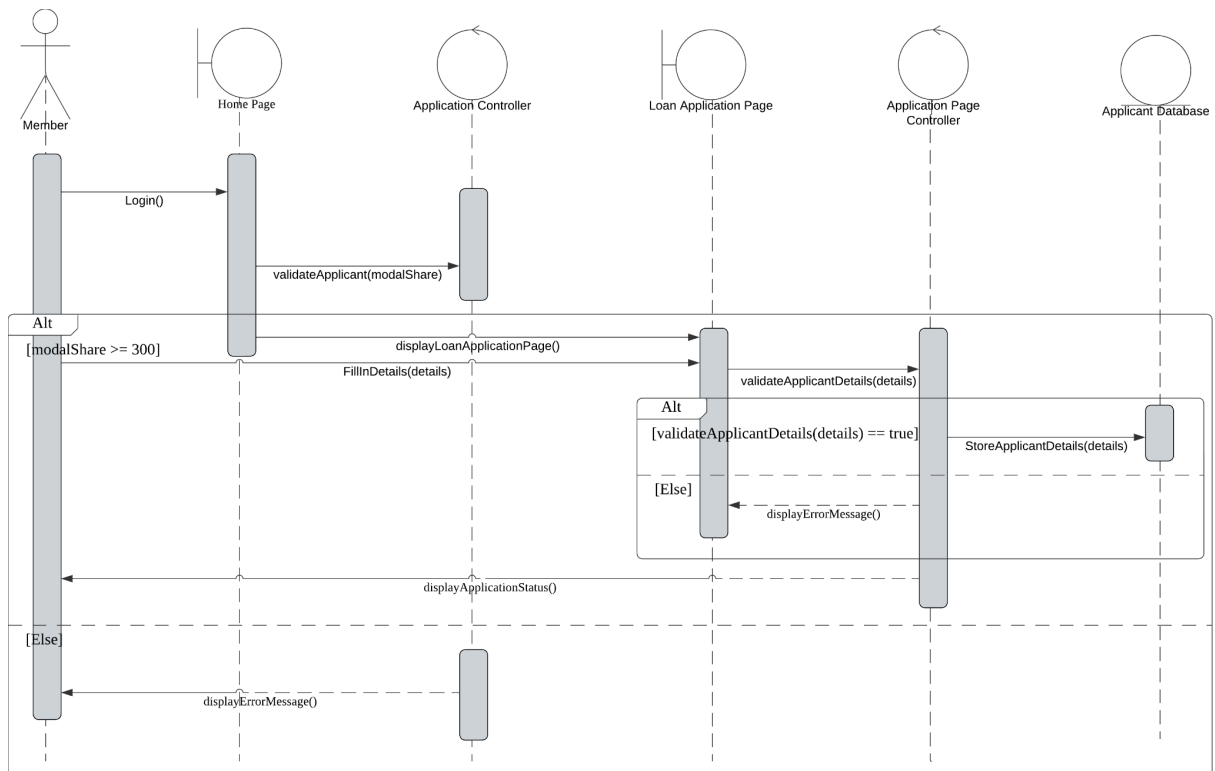


Figure 4.14: Sequence Diagram for Apply for Loan

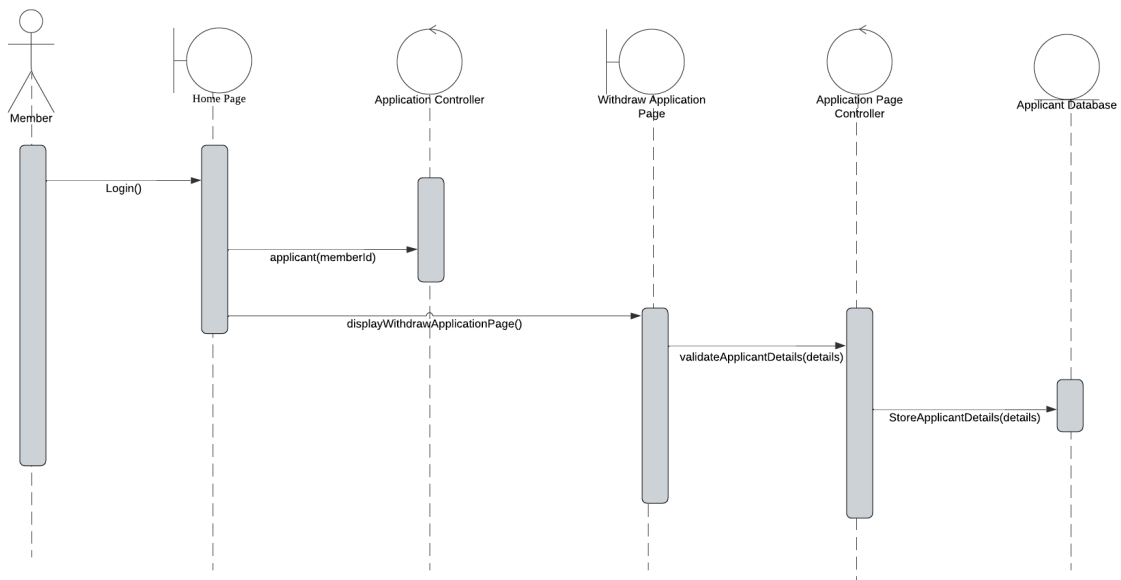


Figure 4.15: Sequence Diagram for Withdraw Membership

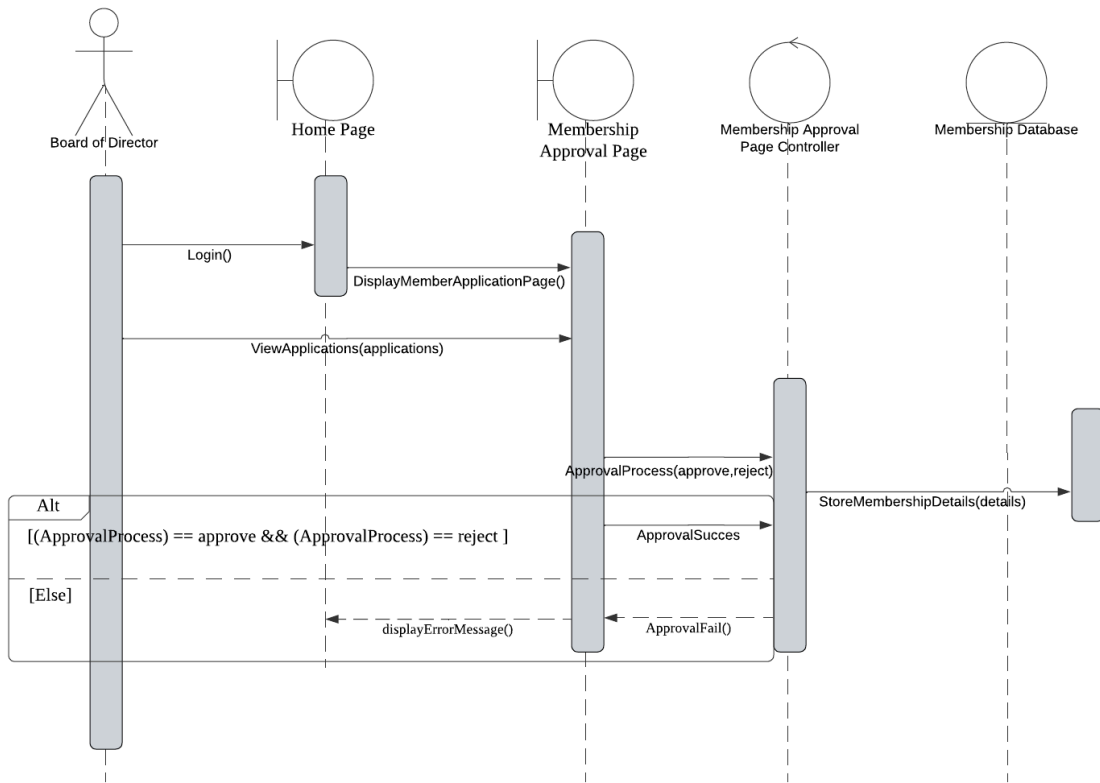


Figure 4.16: Sequence Diagram for Membership Withdrawal Approval

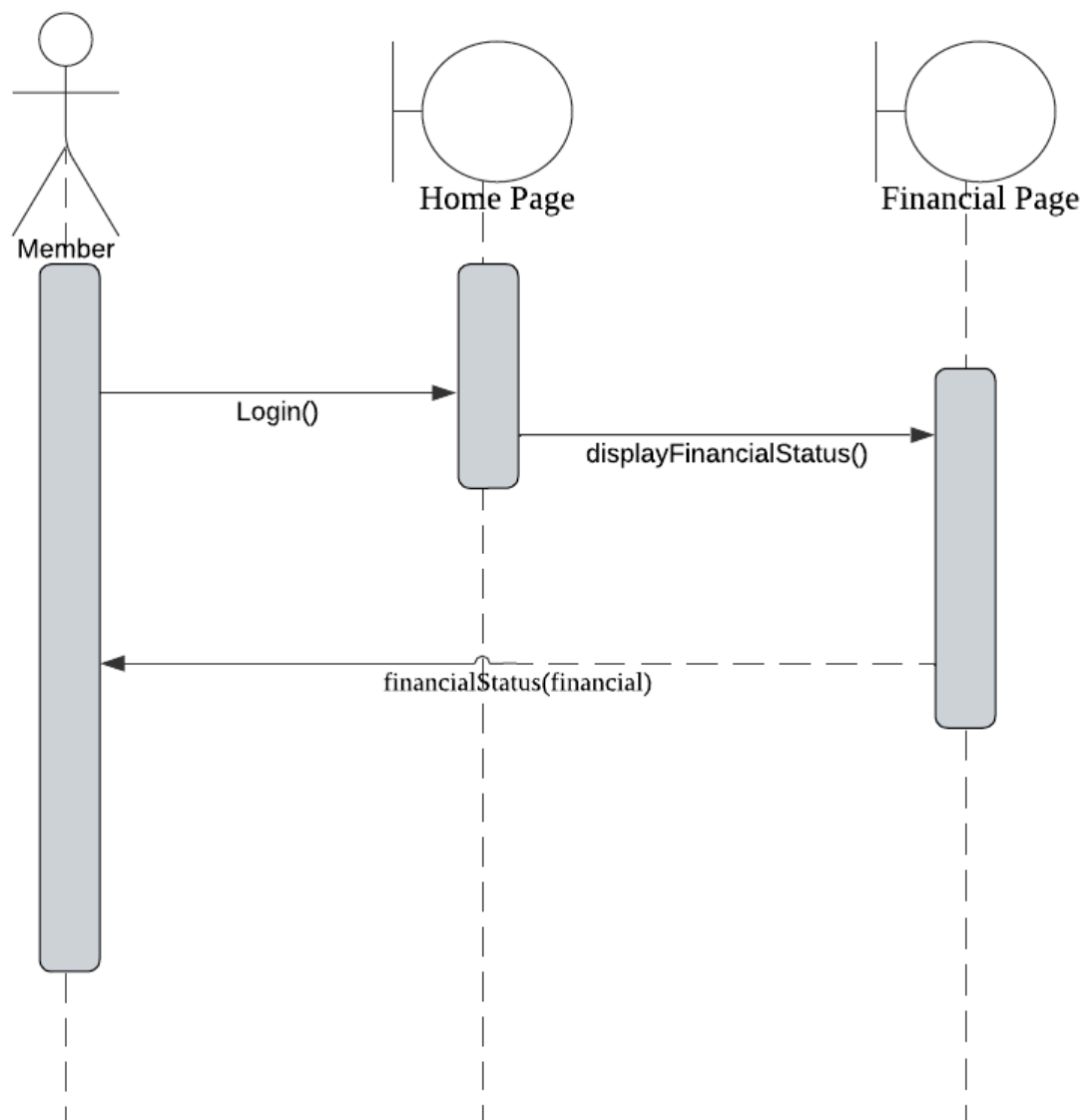


Figure 4.17: Sequence Diagram for View Financial Statement

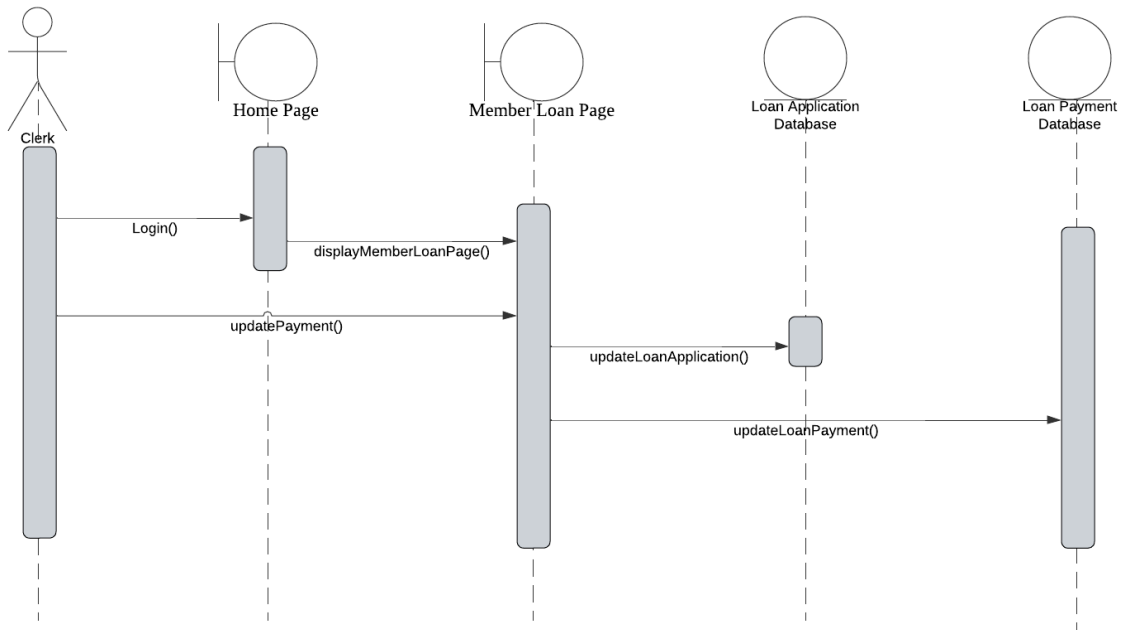


Figure 4.18: Sequence Diagram for Update Loan Payment

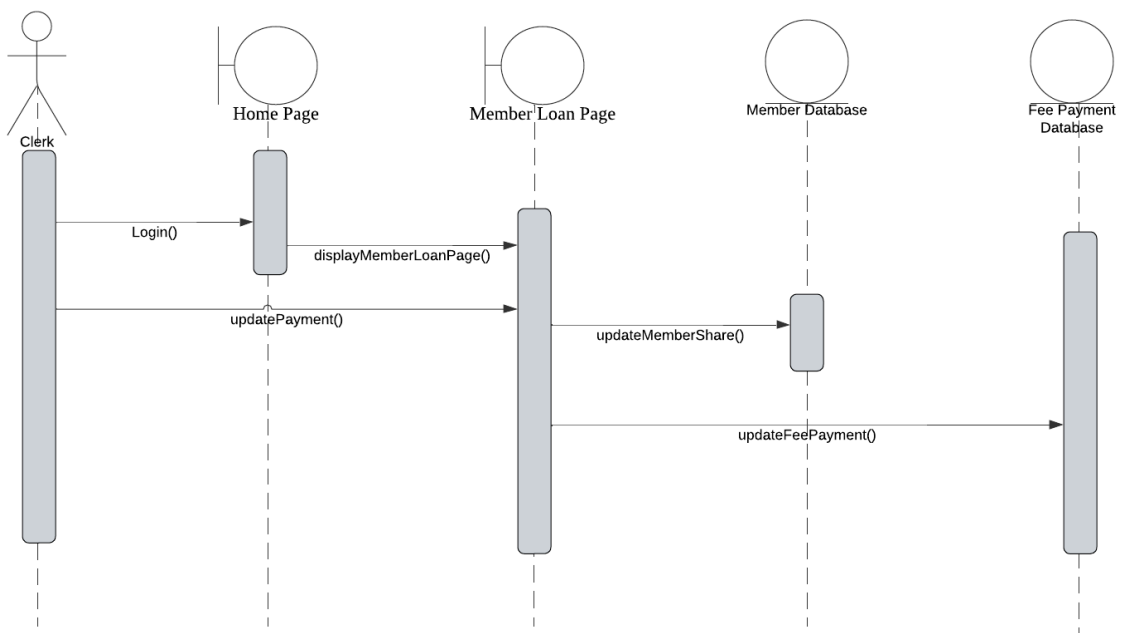


Figure 4.19: Sequence Diagram for Update Fee Payment

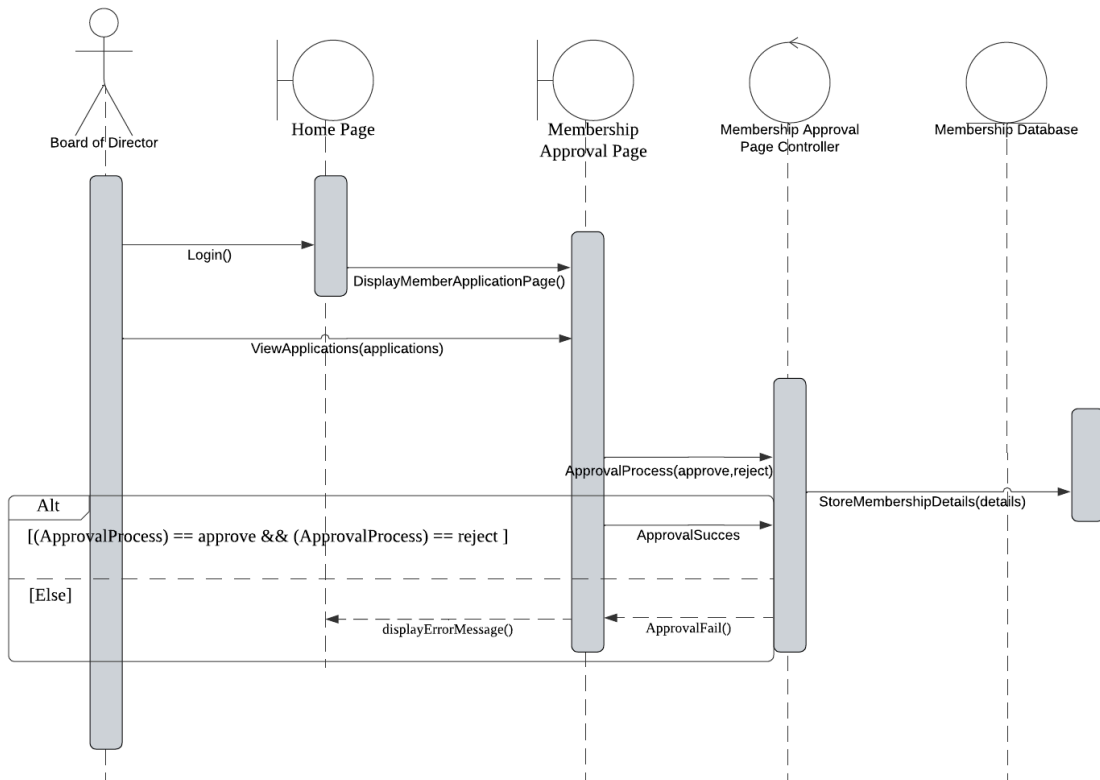


Figure 4.20: Sequence Diagram for Approve Membership Application

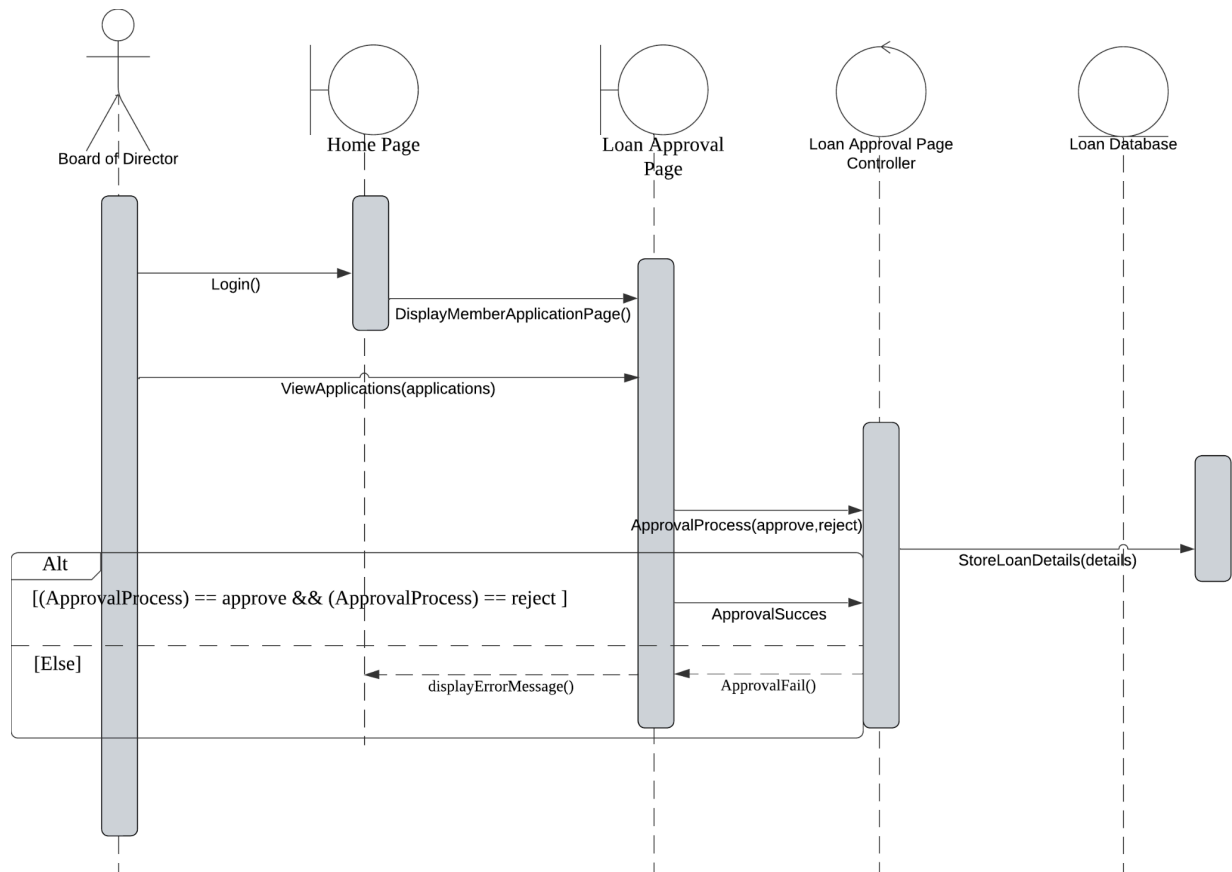


Figure 4.21: Sequence Diagram for Approve Loan Application

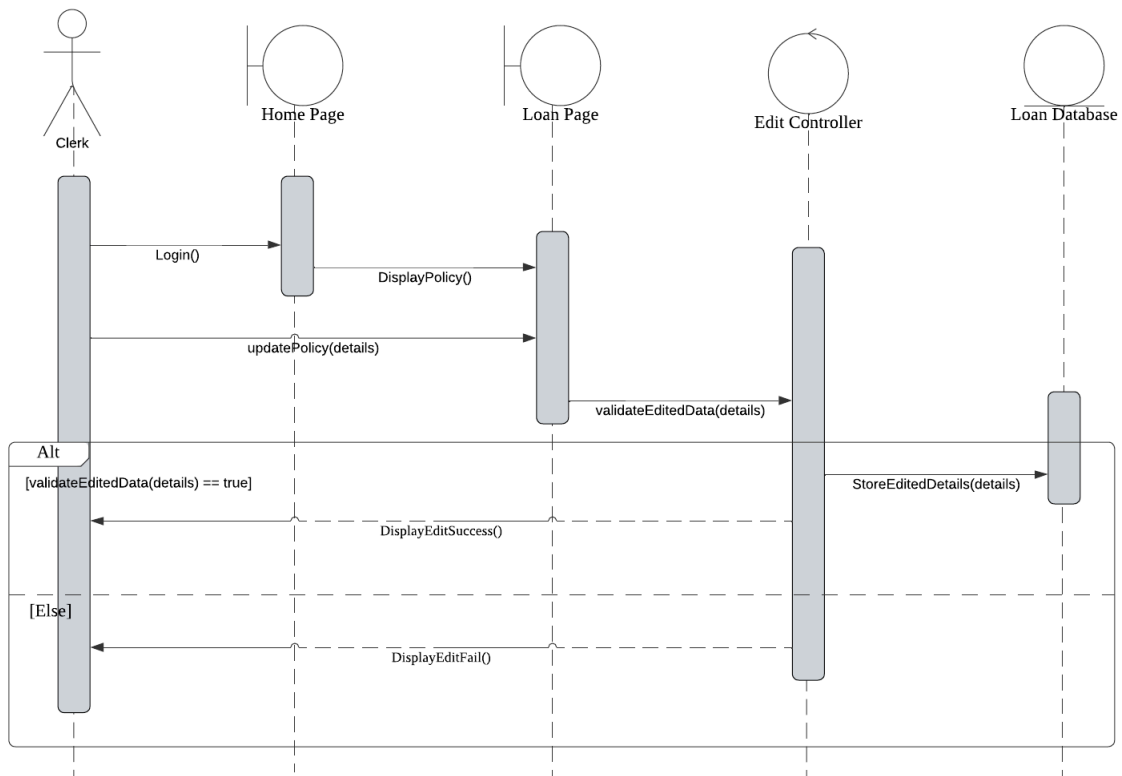


Figure 4.22: Sequence Diagram for Manage Loan Policy

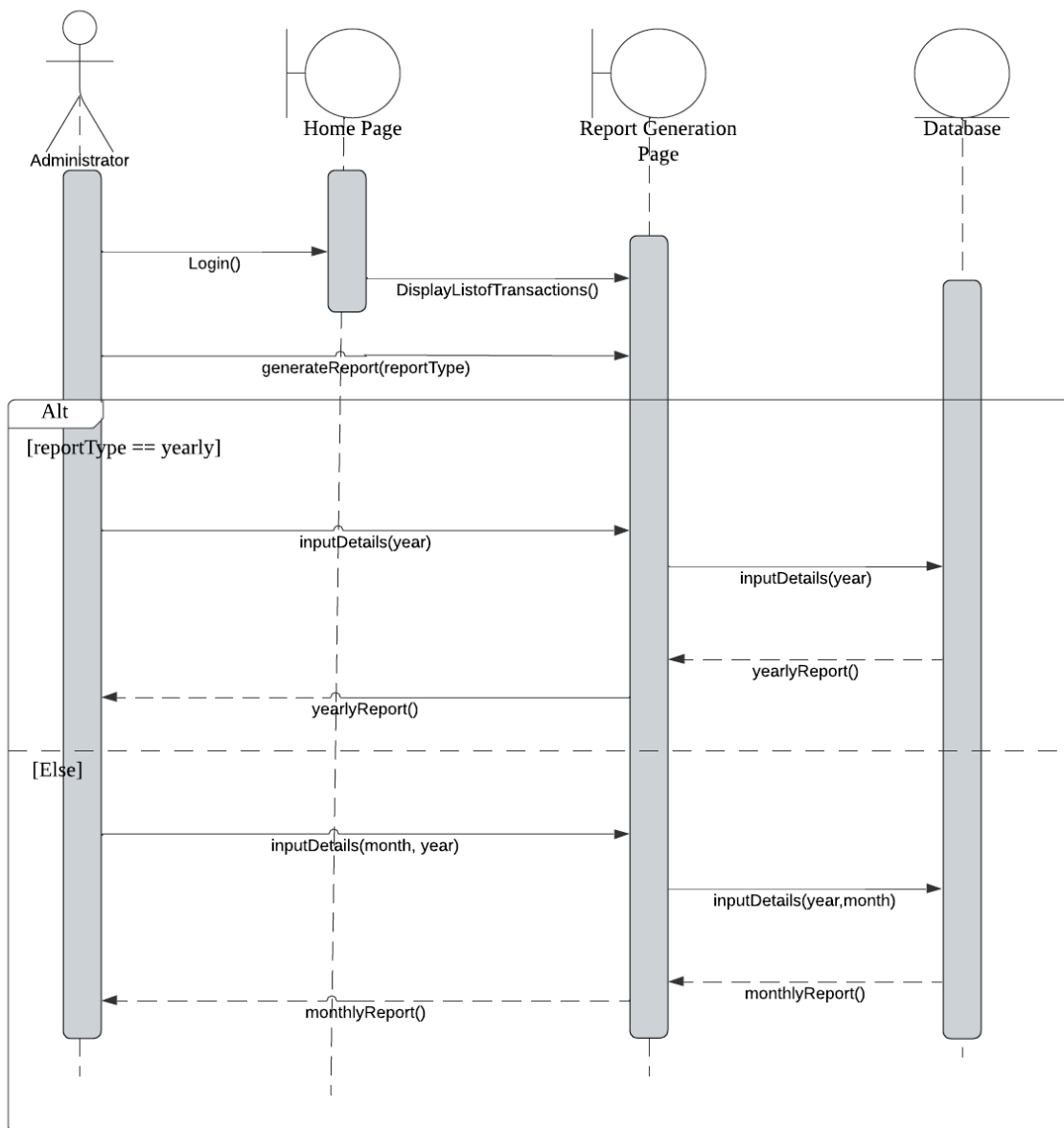


Figure 4.23: Sequence Diagram for Generate Report

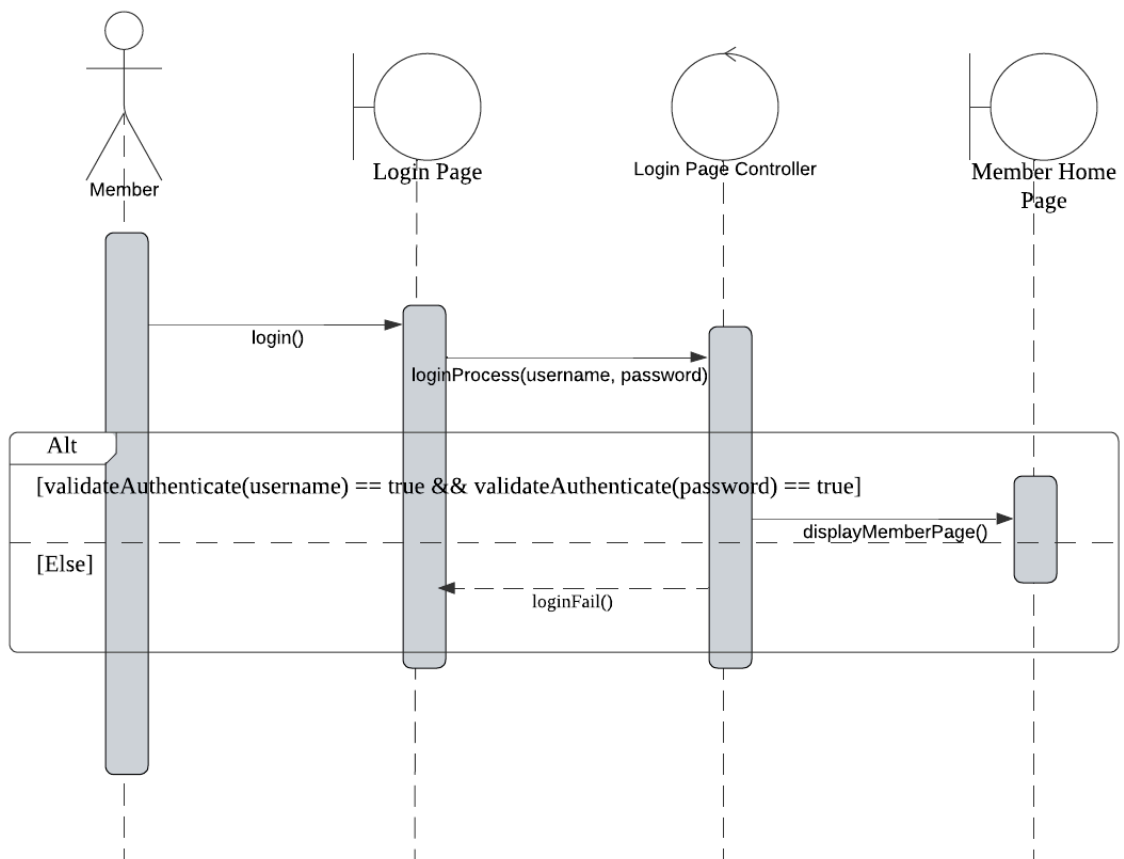


Figure 4.24: Sequence Diagram for Authentication (Member)

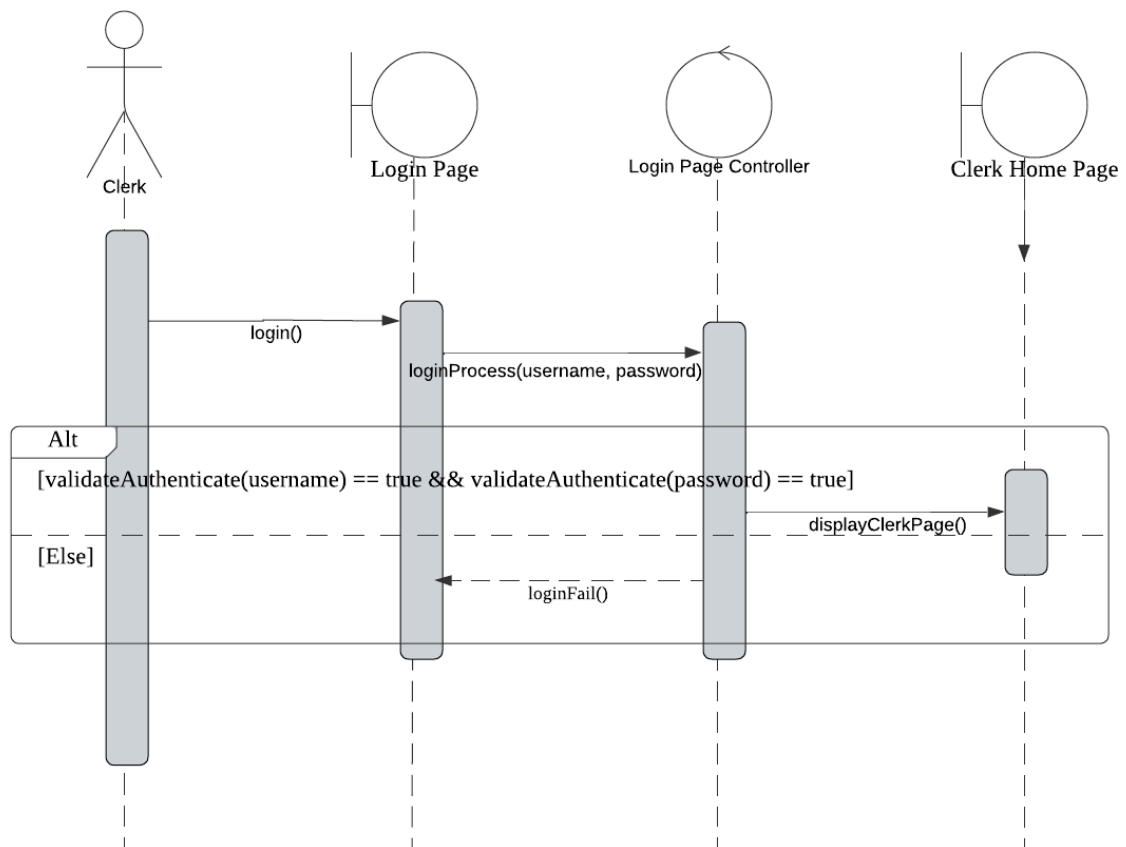


Figure 4.25: Sequence Diagram for Authentication (Clerk)

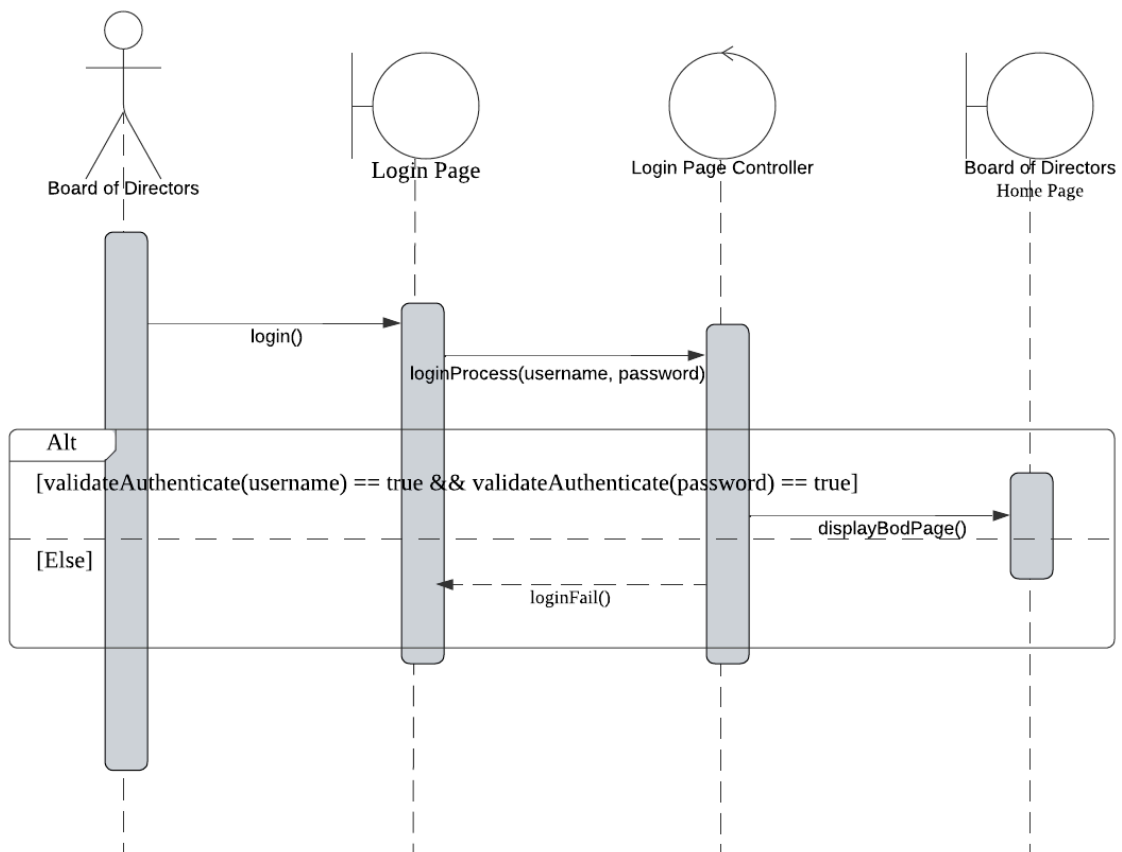


Figure 4.26: Sequence Diagram for Authentication (Board of Directors)

4.5 Physical/ Deployment View

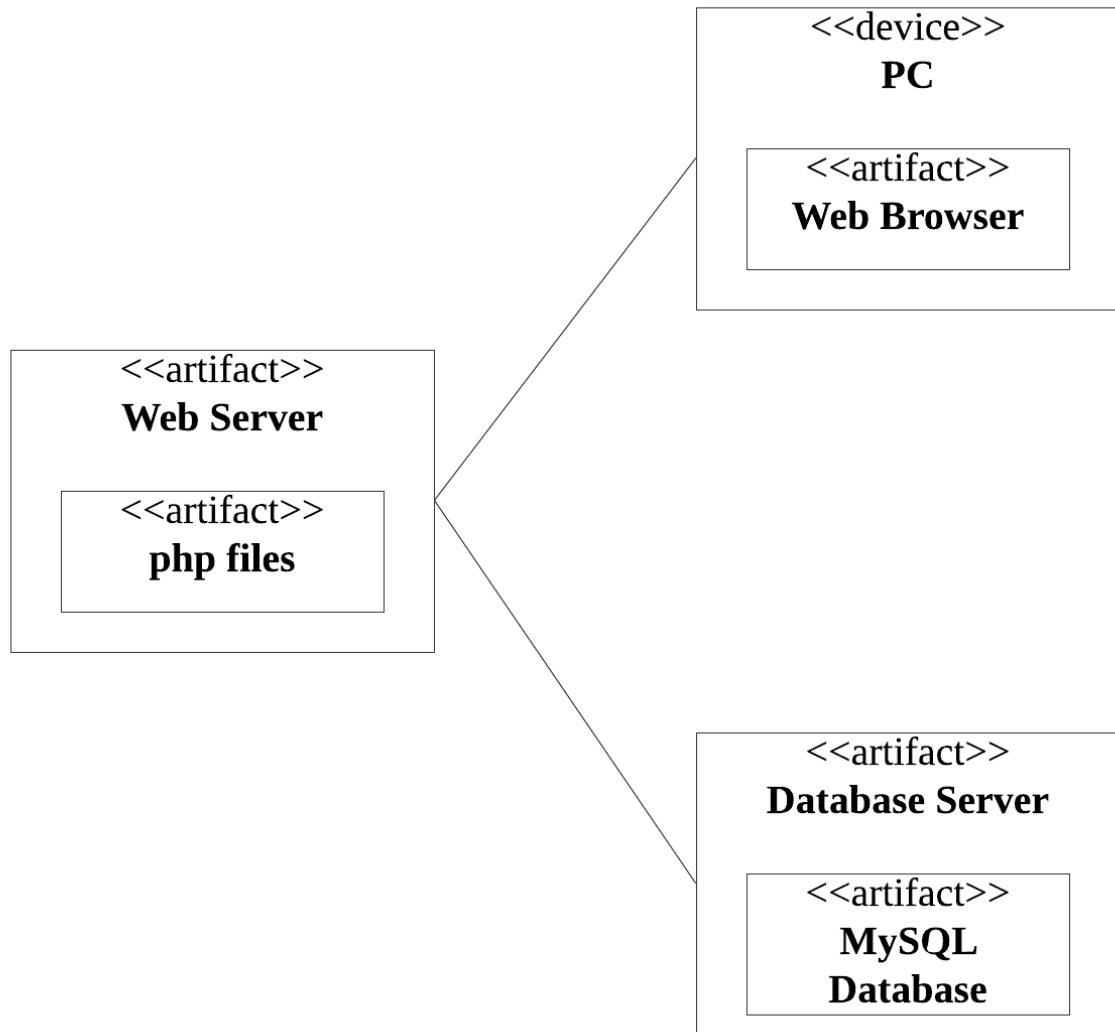


Figure 4.27: Deployment Diagram for KADA's Cooperative Web-Based System

5. Data Design

The major data or systems entities are stored into a relational database named as db_kada, processed and organized into 12 entities as listed in Table 5.1.

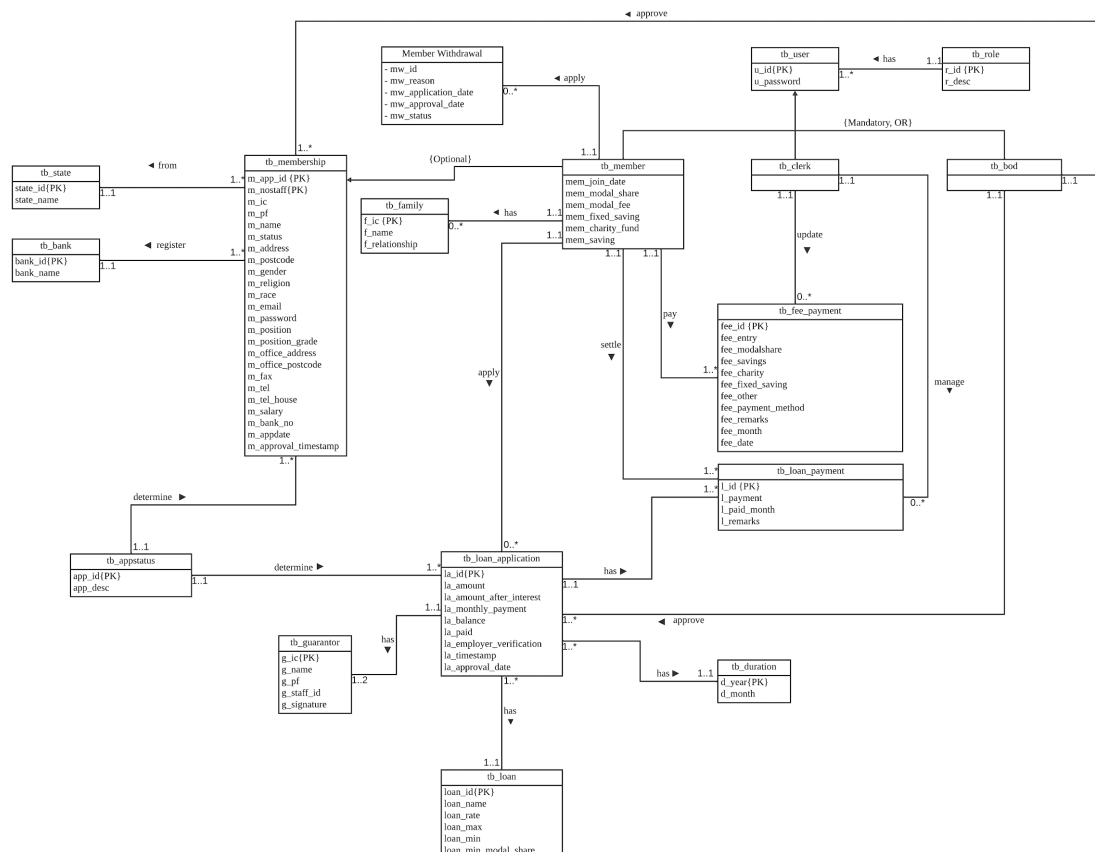


Table 5.1: Description of Entities in the Database

No.	Entity Name	Description
1	User	A general term describing system user
2	Member	A general term describing KADA workers that registered the membership
3	Clerk	A general term describing staff of KADA Cooperative that manages cooperative databases and loan policies
4	Board of Directors	A general term describing manager of KADA Cooperative

5	Membership Application	A general term describing the process of employee want to apply for membership
6	Family	A general term describing member's family members
7	Loan Application	A general term describing the process of employee applying for loan
8	Guarantor	A general term describing employee's guarantors
9	Loan	A general term describing the loan details
10	Fee Payment	A general term describing the process of the member paying the fee
11	Loan Payment	A general term describing the process of the member paying the loan
12	Member Withdrawal	A general term describing the process of the member quitting the membership

5.1 Data Dictionary

The data dictionary shows the attributes name, data type of the attributes and the description.

5.1.1 Entity: User

Table 5.2: Data Dictionary for Entity User

Attribute Name	Type	Description
u_id	int	Unique identifier for a user
u_password	varchar (128)	User's password
u_role	varchar (10)	Role description

5.1.2 Entity: Membership Application

Table 5.3: Data Dictionary for Membership Application

Attribute Name	Type	Description
m_app_id	int	Uniquely identifies each Membership Application
m_nostaff	int	Uniquely identifies each Staff Number
m_ic	varchar(12)	Membership Identity Card number
m_pf	int	Membership 'Personal File' number
m_name	varchar(150)	Membership full name
m_status	varchar(20)	Marital status
m_address	varchar(100)	Home address
m_postcode	int(5)	Home postcode
m_gender	char(1)	Membership gender
m_religion	varchar(20)	Membership religion
m_race	varchar(20)	Membership race
m_email	varchar(50)	Membership email address
m_password	varchar(128)	Membership account password for login

m_position	varchar(20)	Employee position in the company
m_position_grade	varchar(4)	Current position grade in company
m_office_address	varchar(100)	Office address
m_office_postcode	int(5)	Office postcode
m_fax	varchar(11)	Fax number
m_tel	varchar(12)	telephone number
m_tel_house	varchar(11)	house telephone number
m_salary	float	current salary
m_bank_no	varchar(20)	account bank number
m_appdate	timestamp	Membership application timestamp
m_approval_timestamp	timestamp	Approval timestamp

5.1.3 Entity: Family

Table 5.4: Data Dictionary for Family

Attribute Name	Type	Description
f_ic	varchar(12)	Family identity card number
f_name	varchar(150)	Family name
f_relationship	varchar(50)	Family relationship to member

5.1.4 Entity: Loan Application

Table 5.5: Data Dictionary for Loan Application

Attribute Name	Type	Description
la_id	int	Loan application unique ID
la_amount	float	Loan amount
la_amount_after_interest	float	Loan amount including interest
la_monthly_payment	float	Monthly payment
la_balance	float	Loan balance

la_paid	float	Amount of loan already paid
la_employer_verification	mediumblob	Employer verification document
la_timestamp	timestamp	Loan application timestamp
la_approval_date	timestamp	Loan approval timestamp

5.1.5 Entity: Loan

Table 5.6: Data Dictionary for Loan

Attribute Name	Type	Description
loan_name	varchar(30)	Loan type name
loan_rate	float	Loan interest rate
loan_max	float	Maximum loan amount
loan_min	float	Minimum loan amount
loan_min_modal_share	float	Minimum modal share needed to apply loan
loan_name	varchar(30)	Loan type name

5.1.6 Entity: Guarantor

Table 5.7: Data Dictionary for Guarantor

Attribute Name	Type	Description
g_ic	varchar(12)	Guarantor's identity card number
g_name	varchar(150)	Guarantor's full name
g_pf	int	Guarantor's personal file number
g_staff_id	int	Guarantor's staff ID
g_signature	mediumblob	Guarantor's signature document

5.1.7 Entity: Fee Payment

Table 5.8: Data Dictionary for Fee Payment

Attribute Name	Type	Description
fee_id	int	Fee payment unique ID
fee_entry	float	Entry fee
fee_modalshare	float	Modal share fee
fee_savings	float	Savings fee
fee_charity	float	Charity fee
fee_fixed_saving	float	Fixed saving fee
fee_other	float	Other fee
fee_payment_method	varchar(20)	Payment method
fee_remarks	text	Fee payment remarks
fee_month	date	Fee payment month
fee_date	date	Fee payment date

5.1.8 Entity: Loan Payment

Table 5.9: Data Dictionary for Loan Payment

Attribute Name	Type	Description
l_id	int	Loan payment unique ID
l_payment	float	Loan payment amount
l_paid_month	date	Loan payment month
l_remarks	text	Loan payment remarks

5.1.9 Entity: Member Withdrawal

Table 5.10: Data Dictionary for Member Withdrawal

Attribute Name	Type	Description
mw_id	int	Membership withdrawal unique ID
mw_reason	text	Membership withdrawal reason for quitting the membership
mw_application_date	timestamp	Membership withdrawal application date
mw_approval_date	timestamp	Membership withdrawal approval date
mw_status	varchar (20)	Membership withdrawal status

5.1.10 Entity: Board of Director

Table 5.11: Data Dictionary for Board of Director

Attribute Name	Type	Description
b_id	int	BOD unique ID

5.1.11 Entity: Clerk

Table 5.12: Data Dictionary for Clerk

Attribute Name	Type	Description
c_id	int	Clerk unique ID

5.1.12 Entity: Member

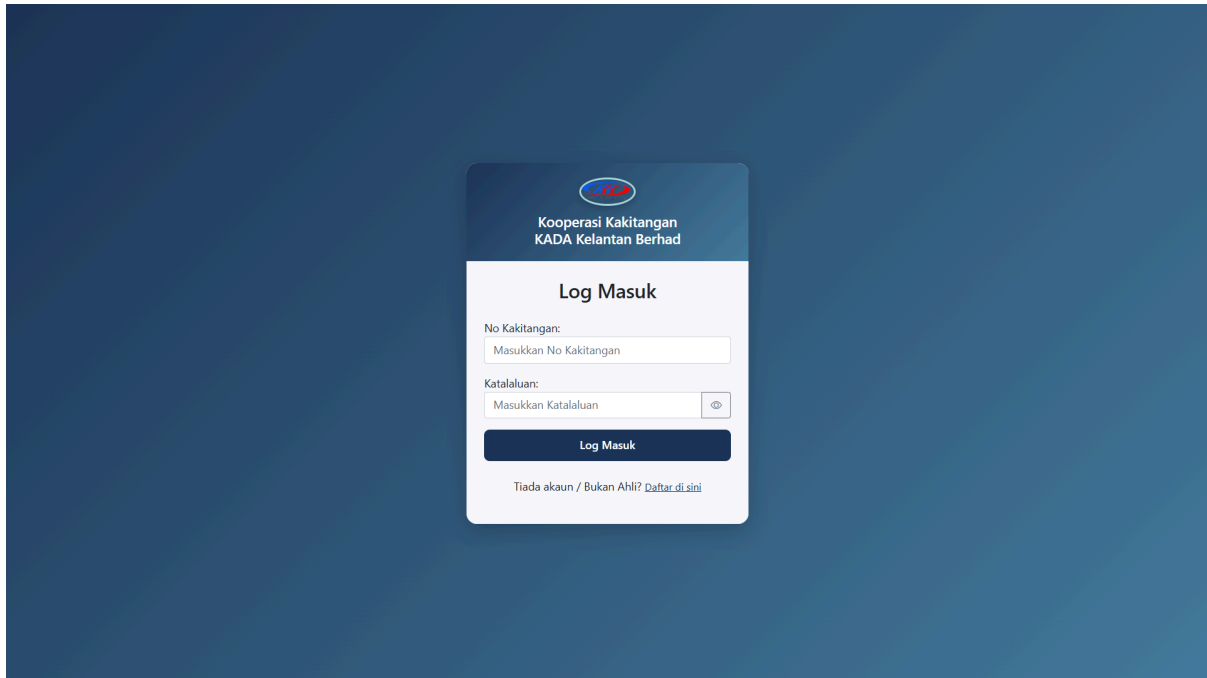
Table 5.13: Data Dictionary for Member

Attribute Name	Type	Description
mem_join_date	timestamp	Member join membership date
mem_modal_share	float	Member's total modal share
mem_modal_fee	float	Member's total modal fee
mem_fixed_saving	float	Member's total fixed saving
mem_charity_fund	float	Member's total charity fund
mem_saving	float	Member's total saving

6. User Interface Design

6.1 Screen Images

This screen image will show a different perspective for a different type of user which are Applicant, Member, Board of Directors and Clerk from start until the end.



The screenshot displays a login form for KADA Kelantan Berhad. The form is centered on a dark blue background. At the top of the form is the KADA logo, followed by the text "Kooperasi Kakitangan KADA Kelantan Berhad". Below this is the title "Log Masuk". The form contains two input fields: "No Kakitangan:" with a placeholder "Masukkan No Kakitangan" and "Katalaluan:" with a placeholder "Masukkan Katalaluan" and a toggle icon. A dark blue "Log Masuk" button is positioned below the input fields. At the bottom of the form, there is a link: "Tiada akaun / Bukan Ahli? [Daftar di sini](#)".

Figure 6.1: Interface for Authentication Login

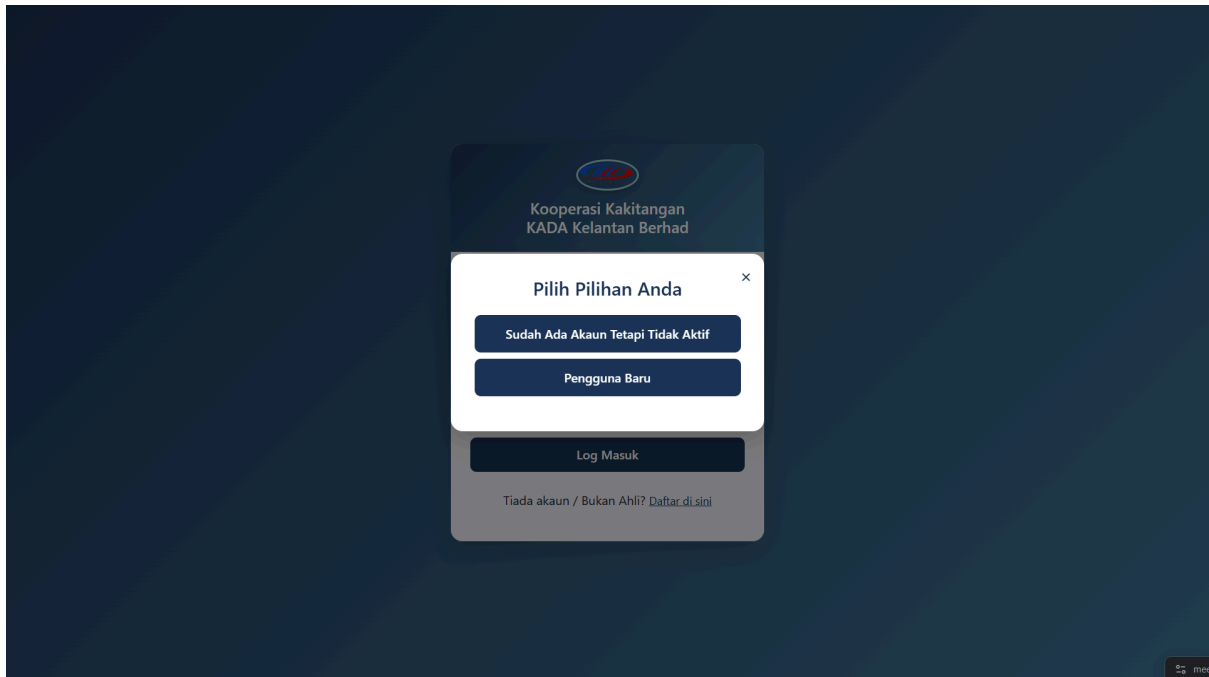


Figure 6.2: Interface for Applying for the first time or not


Koperasi Kakitangan KADA Kelantan Berhad

Permohonan Keahlian

Nota: Semua Bahagian Yang Mempunyai Tanda * Adalah Wajib

Maklumat Peribadi

Nombor Kakitangan *

Nombor Kad Pengenalan *

Nombor pf *

Nama *

Taraf Perkahwinan *

Alamat Rumah

Poskod Rumah *

Negeri (Rumah) *

Jantina *
☐ Lelaki
☐ Perempuan

Agama *
☐ Islam
☐ Christian
☐ Buddha
☐ Lain - lain

Bangsa *
☐ Melayu
☐ Cina
☐ India
☐ Lain - lain

Alamat Emel *

Katalaluan *


Koperasi Kakitangan KADA Kelantan Berhad

Alamat Emel *

Contoh: nama@gmail.com

Katalaluan *

Gunakan kombinasi huruf, nombor, dan simbol untuk katalaluan yang kuat.

Jawatan *

Gred Jawatan *

Alamat Pejabat *

Poskod Pejabat *

Negeri (Pejabat) *

Nombor Fax

Letak '-' jika tiada

Nombor Telefon *

Nombor Telefon Rumah

Letak '-' jika tiada

Gaji Bulanan *

RM

Nama Bank *

Nombor Bank Akaun *

← Ke Login

Reset

Seterusnya →

Figure 6.3: Interface for Member Application

Koperasi Kakitangan KADA Kelantan Berhad

Permohonan Keahlian - Maklumat Keluarga

Ahli Keluarga 1

Nama: No. Kad Pengenalan:

Hubungan:

Ahli Keluarga 2

Nama: No. Kad Pengenalan:

Hubungan:

Figure 6.4: Interface for Family Member Application

) and 'Modal Syer' (RM) in the first row. The second row has 'Fee Masuk' (RM) and 'Wang Deposit Anggota' (RM) in the first row. The third row has 'Sumbangan Tabung Kebajikan (AI - Abrar)' (RM) and 'Simpanan Tetap' (RM) in the first row. Below these fields is a checkbox labeled 'Saya Pasti Bahawa Segala Maklumat Yang Saya Masukkan Adalah Benar Dan Akan Memikul Tanggungjawab Jika Maklumat Tersebut Adalah Salah.' At the bottom, there is a 'Hantar' (Submit) button."/>

Koperasi Kakitangan KADA Kelantan Berhad

Permohonan Keahlian

Yuran & Sumbangan

Modal Yuran (Min: RM35): Modal Syer:

Fee Masuk: Wang Deposit Anggota:

Sumbangan Tabung Kebajikan (AI - Abrar): Simpanan Tetap:

☐ Saya Pasti Bahawa Segala Maklumat Yang Saya Masukkan Adalah Benar Dan Akan Memikul Tanggungjawab Jika Maklumat Tersebut Adalah Salah.

Figure 6.5: Interface for Charity Member Application

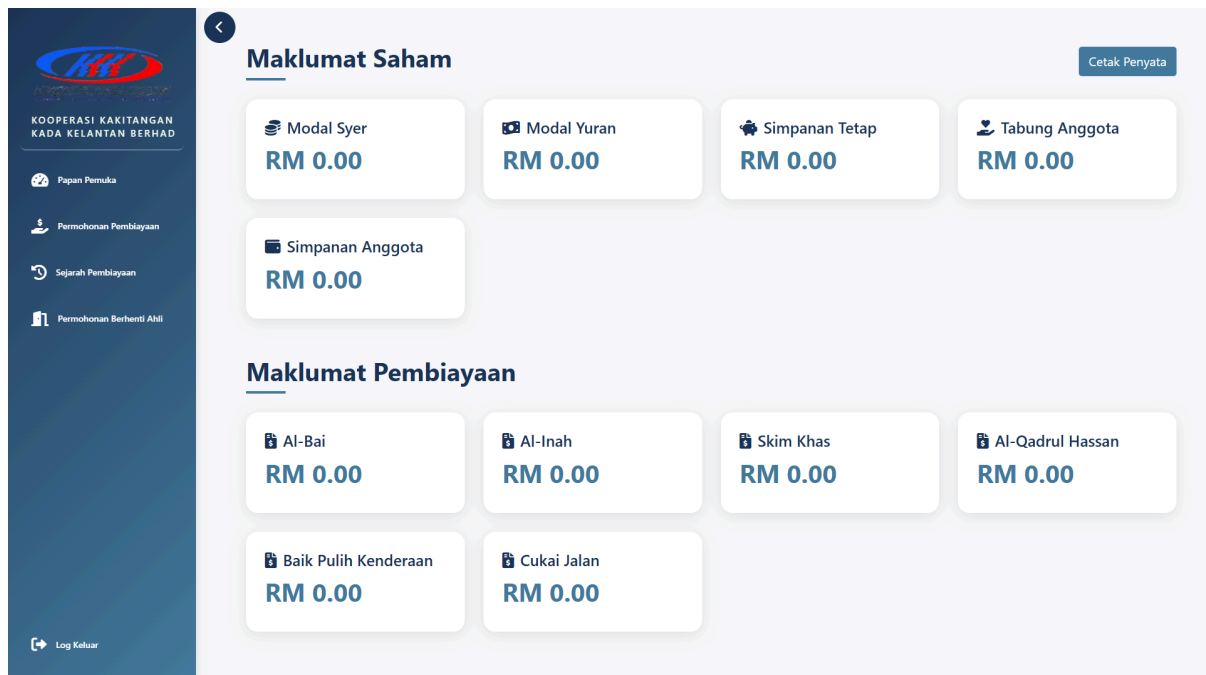


Figure 6.6: Interface for Member Dashboard



Figure 6.7: Interface for Member Loan Application

Butir-Butir Penjamin

Penjamin 1	Penjamin 2
No. KP 041231080417	No. KP 041234080419
Nama test1	Nama test2
No. PF 12	No. PF 23
No. Anggota 1	No. Anggota 2
Borang Tandatangan Penjamin (PDF) Choose File UTM - Course Registration Slip Sem 1.pdf Maksimum Saiz Fail: 1 MB.	Borang Tandatangan Penjamin (PDF) Choose File UTM - Course Registration Slip Sem 2.pdf Maksimum Saiz Fail: 1 MB.

Hantar Permohonan

Figure 6.8: Interface for Member Loan Application Guarantor

Sejarah Pembiayaan

ID Pembiayaan	Nama Pembiayaan	Jumlah Diminta	Status
---------------	-----------------	----------------	--------

Figure 6.9: Interface for Member Loan History



KOPERASI KAKITANGAN
KADA KELANTAN BERHAD

Papan Pemuka

Permohonan Pembiayaan

Sejarah Pembiayaan

Permohonan Berhenti Ahli

Log Keluar



Permohonan Berhenti Menjadi Anggota

Sebab Berhenti Menjadi Anggota

Sila nyatakan sebab-sebab mahu berhenti

Hantar Permohonan

Figure 6.10: Interface for Member Withdraw Application

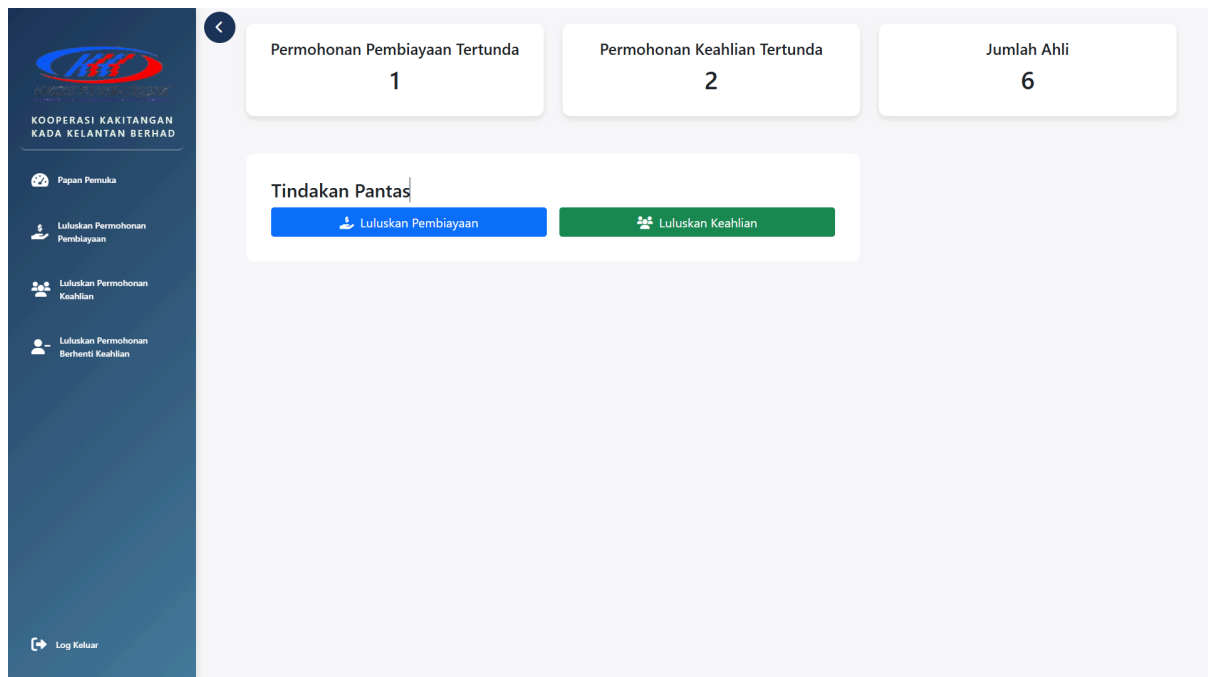


Figure 6.11: Interface for Board of Directors (BoD) Dashboard

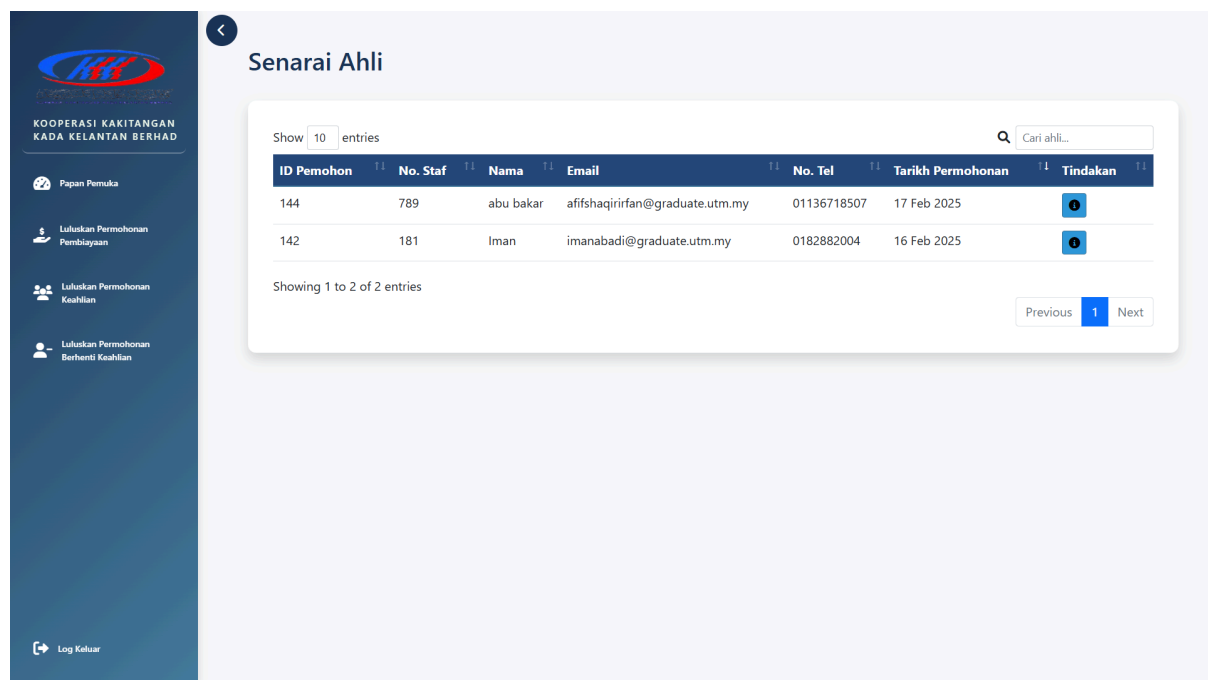
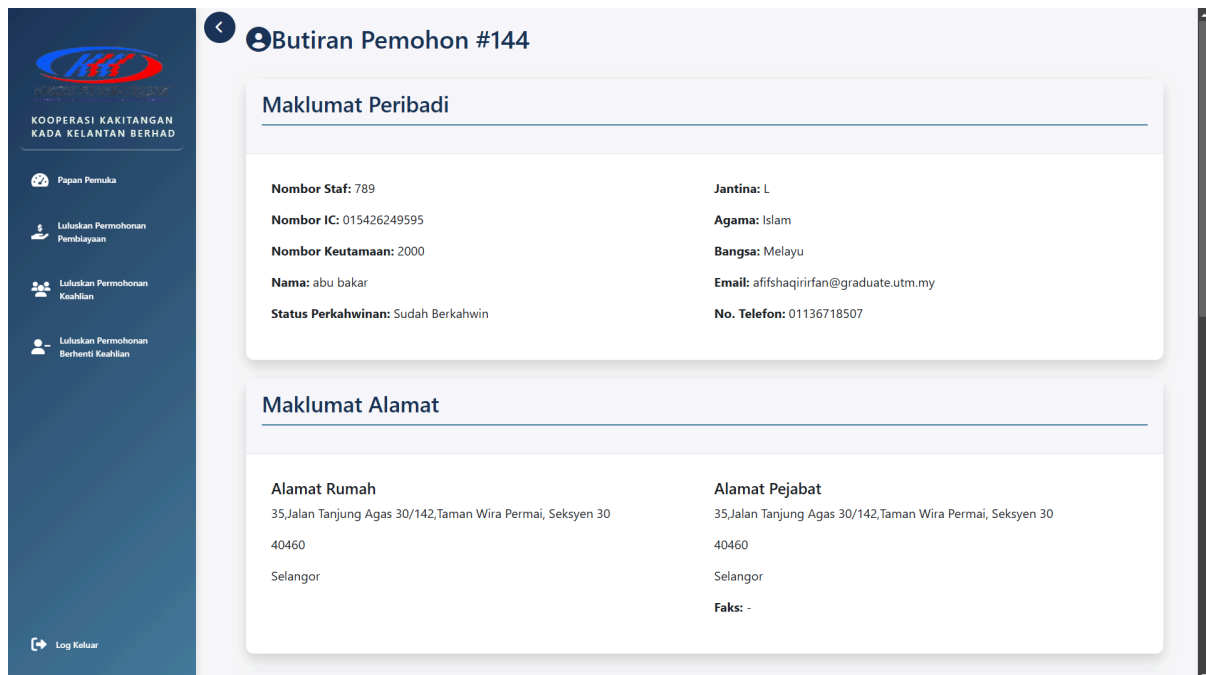


Figure 6.14: Interface for BoD List of Membership Application





KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Luluskan Permohonan Pembiayaan
- Luluskan Permohonan Keahlian
- Luluskan Permohonan Berhenti Keahlian
- Log Keluar

Butiran Pemohon #144

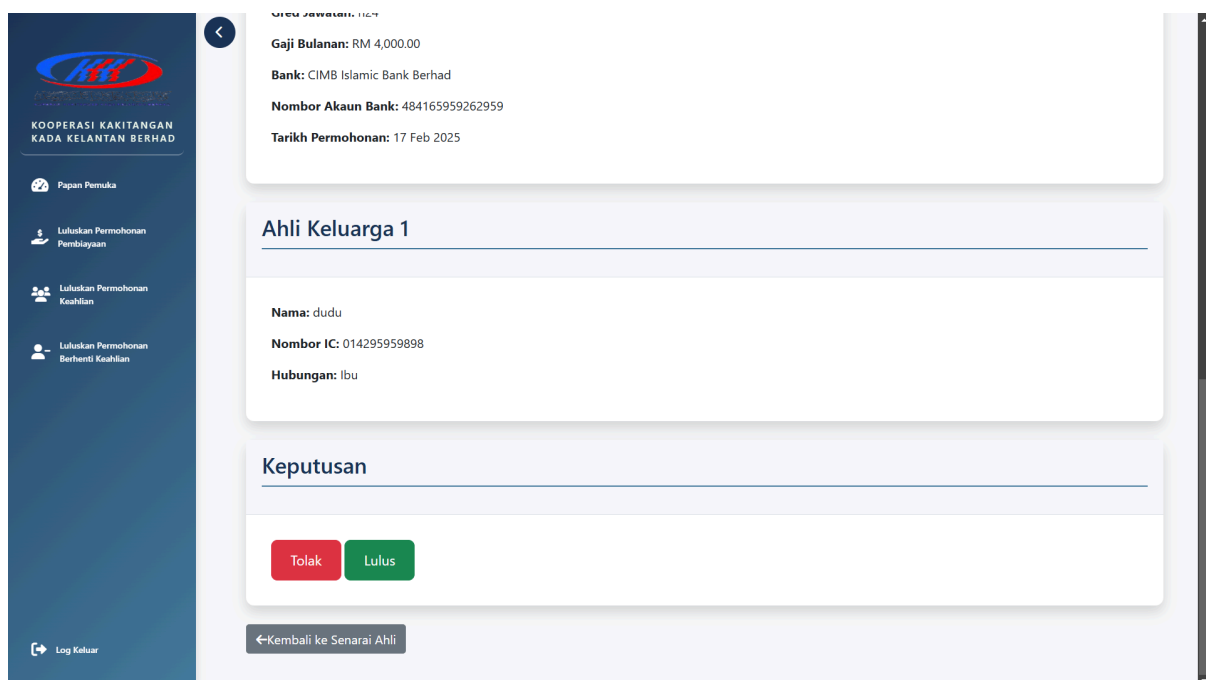
Maklumat Peribadi

Nombor Staf: 789	Jantina: L
Nombor IC: 015426249595	Agama: Islam
Nombor Keutamaan: 2000	Bangsa: Melayu
Nama: abu bakar	Email: afifshaqirirfan@graduate.utm.my
Status Perkahwinan: Sudah Berkahwin	No. Telefon: 01136718507

Maklumat Alamat

Alamat Rumah 35,Jalan Tanjung Agas 30/142,Taman Wira Permai, Seksyen 30 40460 Selangor	Alamat Pejabat 35,Jalan Tanjung Agas 30/142,Taman Wira Permai, Seksyen 30 40460 Selangor Faks: -
--	--

Figure 6.15: Interface for BoD Detail of Membership Application List





KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Luluskan Permohonan Pembiayaan
- Luluskan Permohonan Keahlian
- Luluskan Permohonan Berhenti Keahlian
- Log Keluar

Butiran Pemohon #144

Gaji Bulanan: RM 4,000.00

Bank: CIMB Islamic Bank Berhad

Nombor Akaun Bank: 484165959262959

Tarikh Permohonan: 17 Feb 2025

Ahli Keluarga 1

Nama: dudu
Nombor IC: 014295959898
Hubungan: Ibu

Keputusan

Tolak

Lulus

[←Kembali ke Senarai Ahli](#)

Figure 6.16: Interface for BoD Decision for approving Membership Application



**KOPERASI KAKITANGAN
KADA KELANTAN BERHAD**

- Papan Pemuka
- Luluskan Permohonan Pembiayaan
- Luluskan Permohonan Keahlian
- Luluskan Permohonan Berhenti Keahlian
- Log Keluar

Senarai Permohonan Pinjaman

Show 10 entries

Cari permohonan...

ID Permohonan	ID Ahli	Nama Ahli	Jenis Pinjaman	Amaun Pembiayaan (RM)	Jumlah Pembiayaan (RM)	Bayaran Bulanan (RM)	Tempoh (Tahun / Bulan)	Tarikh Permohonan	Tindakan
101	100	Mohamed Alif Fathi	Al-Bai	1,000.00	1,042.00	86.83	1 / 12	16 Feb 2025	

Showing 1 to 1 of 1 entries

Previous

1

Next

Figure 6.17: Interface for BoD Loan Application List

97



**KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD**

- Papan Pemuka
- Luluskan Permohonan
Pembayaran
- Luluskan Permohonan
Keahlian
- Luluskan Permohonan
Berhenti Keahlian

Log Keluar

Butiran Permohonan #101

Butiran Pinjaman

ID Ahli: 100	Jenis Pinjaman: Al-Bai
Nama Ahli: Mohamed Alif Fathi	Jumlah Pinjaman: RM1,000.00
Nombor IC: 041231050417	Jumlah Pinjaman Selepas Faedah: RM1,042.00
Email: alifptn1234@gmail.com	Tempoh Pinjaman: 1 tahun / 12 bulan
	Bayaran Bulanan: RM86.83

Tandatangan Majikan: [Muat Turun Tandatangan](#)

Butiran Penjamin

Nama Penjamin: Haiqal

Nombor IC: 1234

Nombor PF: 5432

Nombor Staf: 101

Tandatangan Penjamin: [Muat Turun Tandatangan](#)

Nama Penjamin: Iman

Nombor IC: 3241

Bayaran Bulanan: RM86.83

Tandatangan Majikan: [Muat Turun Tandatangan](#)

Butiran Penjamin

Nama Penjamin: Haiqal

Nombor IC: 1234

Nombor PF: 5432

Nombor Staf: 101

Tandatangan Penjamin: [Muat Turun Tandatangan](#)

Nama Penjamin: Iman

Nombor IC: 3241

Nombor PF: 1234

Nombor Staf: 102

Tandatangan Penjamin: [Muat Turun Tandatangan](#)

[← Kembali](#)

Figure 6.18: Interface for BoD Loan Application Details

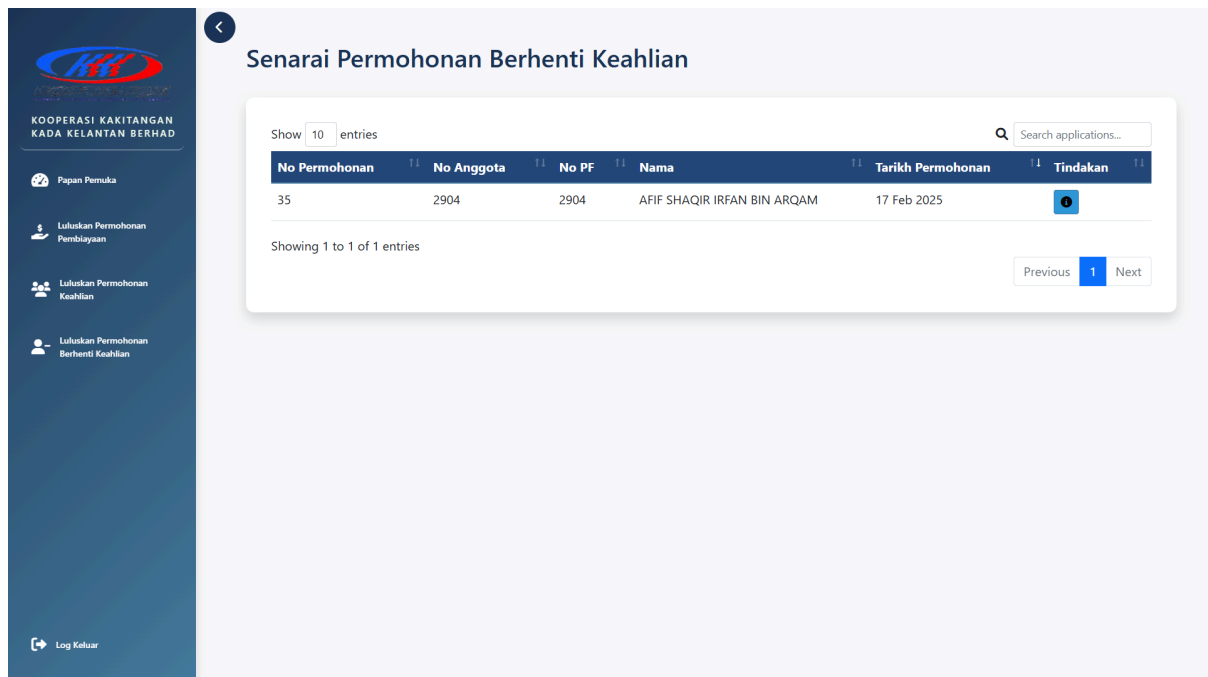


Figure 6.19: Interface for BoD Member Withdraw Application

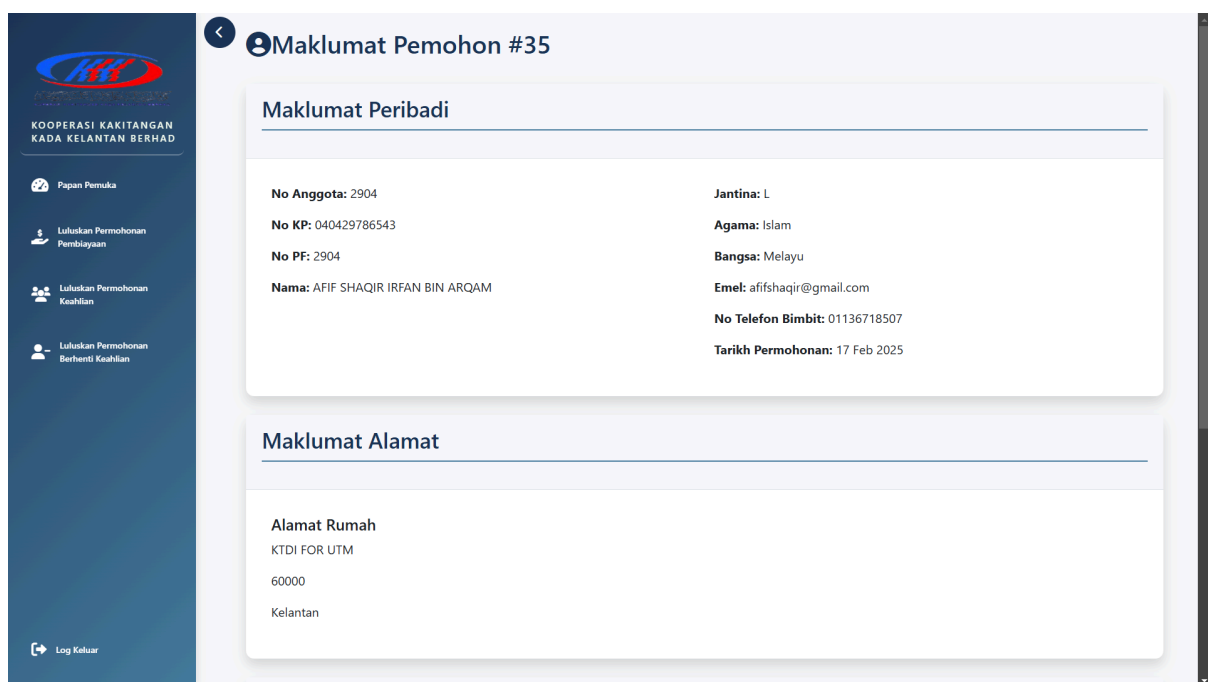


Figure 6.20: Interface for BoD Member Withdraw Application Details



KOPERASI KAKITANGAN
KADA KELANTAN BERHAD

Papan Pemuka

Luluskan Permohonan Pembiayaan

Luluskan Permohonan Keahlian

Luluskan Permohonan Berhenti Keahlian

Log Keluar

Maklumat Alamat

Alamat Rumah

KTDI FOR UTM

60000

Kelantan

Sebab

banyak komitmen

Keputusan

Tidak Diluluskan

Luluskan

←Kembali ke Senarai Berhenti

Figure 6.21: Interface for BoD Member Withdraw Application Decision

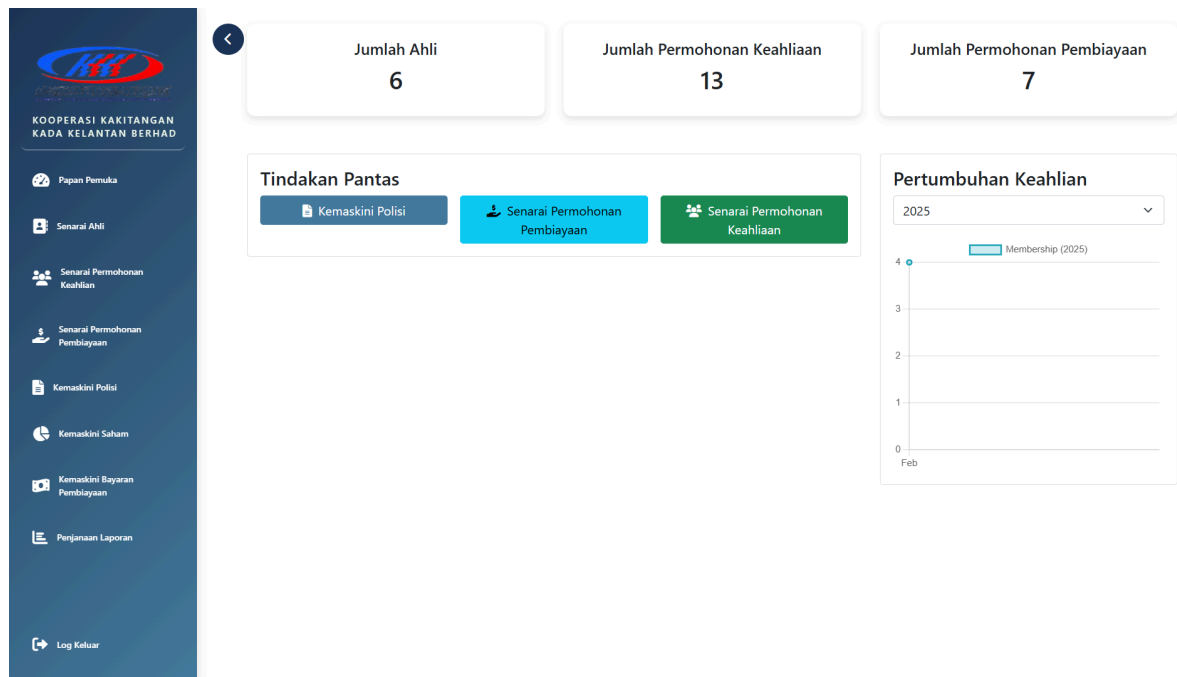


Figure 6.22: Interface for Clerk Dashboard



KOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Kaahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjanaan Laporan
- Log Keluar

Senarai Ahli

ID Staf	Nombor IC	Nombor PF	Nama	Jawatan	Tarikh Sertai	Tindakan
100	041231050417	123	Mohamed Alif Fathi	biasa	2025-02-15 15:52:19	Lihat Butiran
149	041219060149	149	Syahmi Faris	Manager Cawangan	2025-02-15 17:21:44	Lihat Butiran
150	041231050417	345	test	CEO	2025-02-15 10:16:01	Lihat Butiran
181	040510010353	1234	Iman	kerani		Lihat Butiran
789	015426249595	2000	abu bakar	kerani		Lihat Butiran
2904	040429786543	2904	AFIF SHAQIR IRFAN BIN ARQAM	kerani	2025-02-17 07:58:15	Lihat Butiran

Figure 6.23: Interface for Clerk Member List



KOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Kaahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjanaan Laporan
- Log Keluar

Butiran Pemohon, Mohamed Alif Fathi !!

Maklumat Peribadi

Nombor Staf: 100
Nombor IC: 041231050417
Nombor Keutamaan: 123
Nama: Mohamed Alif Fathi
Tarikh Lahir: 2004-12-31
Status Perkahwinan: Belum Berkahwin

Jantina: L
Agama: Islam
Bangsa: Melayu
Emel: alifp1234@gmail.com
No. Telefon: 0189561219

Maklumat Alamat

Alamat Rumah
 F12 Parit Tok Ngah Bawah
 34250
 Perak

Alamat Pejabat
 F12,Parit Tok Ngah Bawah
 34250
 Johor
No. Faks: -

Maklumat Pekerjaan

Jawatan: biasa

Figure 6.24: Interface for Clerk Member List Details



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Keahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjana Laporan
- Log Keluar

Senarai Pemohonan Keahlian

ID Permohonan	Nombor IC	Nombor PF	Nama	Tarikh Permohonan	Status
130	041231050417	123	Mohamed Alif Fathi	2025-02-13 01:31:19	Diluluskan
131	041231050417	345	test	2025-02-15 08:38:01	Diluluskan
132	041231050417	345	test	2025-02-15 08:55:30	Diluluskan
133	041231050417	345	test	2025-02-15 10:15:48	Diluluskan
134	041231080417	124	tester	2025-02-15 15:28:07	Diluluskan
137	041231080417	124	tester	2025-02-15 15:46:53	Ditolak
138	041231050417	123	Mohamed Alif Fathi	2025-02-15 15:50:23	Ditolak
139	041231050417	123	Mohamed Alif Fathi	2025-02-15 15:51:36	Diluluskan
140	041219060149	149	Syahmi Faris	2025-02-15 17:21:19	Diluluskan
141	040429786543	2904	AFIF SHAQIR IRFAN BIN ARQAM	2025-02-15 21:09:22	Diluluskan
142	040510010353	1234	Iman	2025-02-16 01:04:03	Dihantar
143	040429786543	2904	AFIF SHAQIR IRFAN BIN ARQAM	2025-02-17 07:57:23	Diluluskan
144	015426249595	2000	abu bakar	2025-02-17 08:06:25	Dihantar

Figure 6.25: Interface for Clerk List for all the membership applicant



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Keahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjana Laporan
- Log Keluar

Senarai Pemohon Pinjaman

ID	Jenis Pinjaman	Jumlah	Tempoh (Tahun)	Bayaran Bulanan (RM)	Baki (RM)	Dibayar (RM)	Jumlah Keseluruhan (RM)	Tarikh Permohonan	Status	ID BOD Pengesah	ID Ahli
95	Al-Inah	50000	2	2258.33	45166.8	9033.32	54200	2025-02-13 01:56:47	Sedang Berjalan	200	100
96	Al-Bai	30000	6	521.67	37560	0	37560	2025-02-15 08:45:22	Sedang Berjalan	201	100
97	Al-Bai	1233	1	107.07	0	0	1284.79	2025-02-15 09:22:39	Ditolak	201	100
98	Al-Qadrul Hassan	50000	3	1563.89	56300	0	56300	2025-02-15 14:50:34	Sedang Berjalan	201	100
99	Skim Khas	8000	7	123.24	0	0	10352	2025-02-15 15:16:01	Ditolak	201	100
100	Al-Bai	50000	1	4341.67	52100	0	52100	2025-02-15 17:45:28	Sedang Berjalan	200	100
101	Al-Bai	1000	1	86.83	0	0	1042	2025-02-16 00:58:20	Dihantar	200	100

Figure 6.26: Interface for Clerk List for all the loan applicant



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Kaahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjana Laporan
- Log Keluar

Kemaskini Polisi Pinjaman

Pilih Jenis Pinjaman:

Pilih Jenis Pinjaman

Faedah (%):

Maksimum Amaun Pinjaman (RM):

Min Amaun Pinjaman (RM):

Modah Syer (RM):

Kemaskini Polisi Pinjaman

Jenis Pinjaman	Faedah (%)	Maks. Amaun (RM)	Min. Amaun (RM)	Modah Syer (RM)
Al-Bai	4.20%	50,000	1,000	300
Al-Inah	4.20%	50,000	1,000	300
Skim Khas	4.20%	100,000	1,000	300
Al-Qadrul Hassan	4.20%	50,000	1,000	300
Baik Pulih Kenderaan	4.20%	50,000	1,000	300
Cukai Jalan	4.20%	10,000	1,000	300

Figure 6.27: Interface for Clerk Updating Loan Policy



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Kaahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjana Laporan
- Log Keluar

Senarai Ahli

No Staff	Modah Syer	Modal Yuran	Simpanan Tetap	Tabung Anggota	Simpanan Anggota	
100	0	0	0	0	0	Kemaskini
149	0	0	0	0	0	Kemaskini
150	0	0	0	0	0	Kemaskini
181	0	0	0	0	0	Kemaskini
789	0	0	0	0	0	Kemaskini
2904	0	0	0	0	0	Kemaskini

Figure 6.28: Interface for Clerk List of Member Share



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Keahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjanaan Laporan
- Log Keluar

Senarai Ahli

Tarikh Pembayaran:

Nombor Staf: 100

Kaedah Pembayaran

Pembayaran Kaunter

Yuran Masuk: 0

Modal Saham Semasa: 0

50

Yuran Modal Semasa: 0

322

Simpanan Tetap Semasa: 0

5

Dana Kebajikan Semasa: 0

5

Simpanan Semasa: 0

5

Catatan

Figure 6.29: Interface for Clerk Updating Member Share



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli
- Senarai Permohonan Keahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjanaan Laporan
- Log Keluar

Senarai Ahli

No Staf	Nama	Pembiayaan Aktif
100	Mohamed Alif Fathi	4

Lihat

Figure 6.30: Interface for Clerk List member of Updating Loan Payment



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli**
- Senarai Permohonan Keahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjana Laporan
- Log Keluar

Senarai Pembiayaan Ahli

ID Pembiayaan	Nama Pembiayaan	Jumlah Diminta	Jumlah Selepas Faedah	Baki Jumlah	Jumlah Dibayar	Bayaran Bulanan	
95	Al-Inah	50000	54200	45166.8	9033.32	2258.33	Lihat
96	Al-Bai	30000	37560	37560	0	521.67	Lihat
98	Al-Qadrul Hassan	50000	56300	56300	0	1563.89	Lihat
100	Al-Bai	50000	52100	52100	0	4341.67	Lihat

Kembali

Figure 6.31: Interface for Clerk List Loan for one Member



KOOPERASI KAKITANGAN
KADA KELANTAN BERHAD

- Papan Pemuka
- Senarai Ahli**
- Senarai Permohonan Keahlian
- Senarai Permohonan Pembiayaan
- Kemaskini Polisi
- Kemaskini Saham
- Kemaskini Bayaran Pembiayaan
- Penjana Laporan
- Log Keluar

Kemaskini

Tarikh Bayaran:

Jumlah Dibayar: 9033.32

Baki Pinjaman: 45166.8

Jumlah Dibayar Selepas Kemaskini: 11291.65

Baki Pinjaman Selepas Kemaskini: 42908.47

Kaedah Pembayaran

Bayaran Kaunter

Catatan

Kembali Kemaskini

ID Pembayaran Pinjaman	Bayaran(RM)	Bulan Dibayar
34	2258.33	2025-04-01
35	2258.33	2025-05-01

Figure 6.32: Interface for Clerk Updating Payment Loan

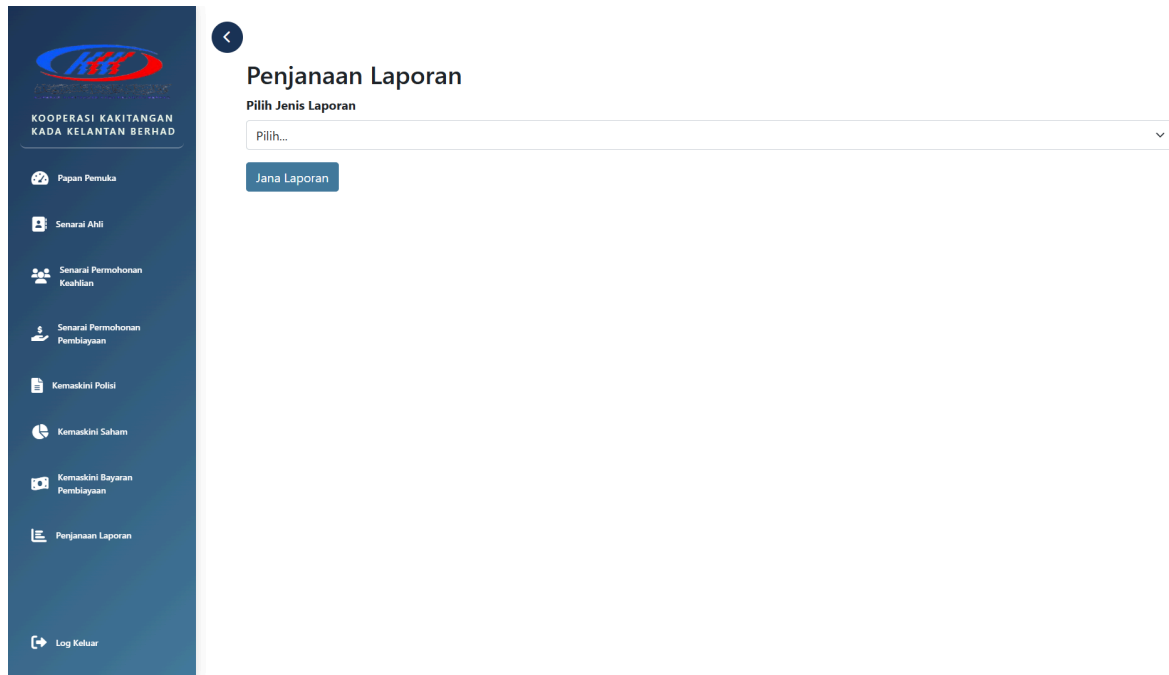


Figure 6.33: Interface for Clerk Generate Monthly or Yearly Report

7. Traceability

The traceability matrix below provides a structured mapping of user stories from the product backlog to their respective system components. This table ensures that each user story is linked to its corresponding Agile artifact, specifically the subsystem or package it belongs to. By using unique user story IDs, the table efficiently tracks and validates development progress across different sprints.

Table 7.1: Traceability

Sprint #	Subsystem/Package	User Story ID
1	Package Membership	US001
	Package Loan	US002
	Package Financial	US006
		US007
2	Package Administration	US010
	Package Financial	US005
	Package Membership	US008
	Package Loan	US009
3	Package Authentication	US012
	Package Financial	US011
4	Package Membership	US003
		US004

8. Test Cases

8.1 TC001: Test Membership Subsystem: Apply for Membership (US001)

This test contains the following test cases:

(a) TC001_01: Test Apply for membership (SD001)

DESCRIPTION	VERSION	TEST DATE
Test the Apply for membership	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Afiq	

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to landing page	kadakoperasi.fun	The landing page should be open	As Expected	PASS		
2	Click "Daftar Sini"		Display options which are "Daftar Baru" and "Sudah Ada Akaun Tapi Tidak Aktif"	As Expected	PASS		
3	Click "Pengguna Baru"		Display membership application form	As Expected	PASS		
4	Fill in all the form	No. Kakitangan = 1234 No.KP = 040510010353 No.PF = 1234 Nama = Iman Alamat = No 12, Taman Larkin Jaya, Johor Bahru Poskod = 80000	Be able to continue to the next page and display the family information page	As Expected	PASS		if the user still have the active account then the form application cannot be displayed and cannot continue to the next page

		Emel = imanabadi@graduate.utm.my Katalaluan = admin123 Jawatan = Kerani Gred Jawatan = H13 Alamat Pejabat = UTM, Johor Bahru Poskod Pejabat = 80350 Nombor Fax = - Nombor Telefon = 0182882004 Nombor Telefon Rumah = - Gaji Bulanan = 5000 Nombor Akaun Bank = 12345678					
5	Fill the the family information form	Nama = Azman Zamri No. KP = 890123300101 Hubungan = Bapa	Be able to continue and display Fee and Charity form	As Expected	PASS		
6	Fill in the charity or fee form	Modal share = 50	Button "Submit" will display	As Expected	PASS		
7	Click button "Hantar"		Able to submit form and wait for approval	As expected	PASS		

8.2 TC002: Test Loan Subsystem: Apply for Loan (UC002)

This test contains the following test cases:

(a) TC002_01: Test Apply for Loan (SD002)

DESCRIPTION	VERSION	TEST DATE
Test the Apply for Loan	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Alif	

Test Case ID (TC002_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to loan application page		The landing page should be open	As Expected	PASS		
2	Choose "Jenis Pembiayaan"		Display the loan that is offered from the database	As Expected	PASS		If the modal share of the member is less than RM 300, the member cannot apply for loan
3	Fill in loan form	Amaun Pembiayaan = 10000 Tempoh Bayaran (Bulan) = 12	Displays the monthly payment and the total repayment after interest.	As Expected	PASS		If the amount requested is less than the minimum amount or more than the maximum amount, the monthly payment will not be shown
4	Fill in all guarantors form	No. KP Penjamin 1 = 041231080514 Nama Penjamin 1 = Encik Ahmad No. PF Penjamin 1 = 1234 Tandatangan Penjamin 1 = signature1.pdf	Able to submit form and wait for approval	As Expected	PASS		If the IC number is the same for two guarantors, it will display an error and the form reset.

		No. KP Penjamin 2 = 041231080517 Nama Penjamin 2 = Encik Ahmed No.PF Penjamin 2 = 4321 Tandatangan Penjamin 2 = signature2.pdf					
5.	Submit the application		The system shows a pop-up of successful application	As Expected	PASSD		

8.3 TC003: Test Membership Subsystem: Membership Withdrawal (UC003)

This test contains the following test cases:

(a) TC003_01: Test Membership Withdrawal (SD003)

DESCRIPTION	VERSION	TEST DATE
Test the Membership Withdrawal	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Afiq	

Test Case ID (TC003_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to withdraw application		The landing page should be open	As Expected	PASS		
2	Fill in "Sebab berhenti menjadi anggota"	Sebab = Alasan peribadi	Able to submit the form	As Expected	PASS		
3	Click "Hantar"		Submit and wait for approval	As Expected	PASS		

8.4 TC004: Test Membership Subsystem: Approve Membership Withdrawal (UC004)

This test contains the following test cases:

(a) TC004_01: Test Approve Membership Withdrawal (SD004)

DESCRIPTION	VERSION	TEST DATE
Test the Approve Membership Withdrawal	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Alif	

Test Case ID (TC004_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to approve membership withdrawal application		The landing page should be open and display the list of application	As Expected	PASS		
2	Click the "Tindakan" button on chosen member		Display the personal information of the member and reason for withdrawal	As Expected	PASS		
3	Click the "Tidak Luluskan" or "Luluskan"		The system alerts the user of successful approval.	As Expected	PASS		

8.5 TC005: Test Financial Subsystem: View Financial Statement (UC005)

This test contains the following test cases:

(a) TC005_01: Test View Financial Statement (SD005)

DESCRIPTION	VERSION	TEST DATE
Test the Test View Financial Statement	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Iman	

Test Case ID (TC005_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to member dashboard		The landing page should be open	As Expected	PASS		
2	Click "Cetak Penyata"		Redirect to the statement in PDF	As Expected	PASS		
3	Display the statement		Show the detail of member's loan details and fee details	As Expected	PASS		

8.6 TC006: Test Financial Subsystem: Update Loan Payment (UC006)

This test contains the following test cases:

(a) TC006_01: Test View Financial Statement (SD006)

DESCRIPTION	VERSION	TEST DATE
Test the Update Loan Payment	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Iman	

Test Case ID (TC006_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to update loan payment		The landing page should be open	As Expected	PASS		
2	Display the list of member that apply loan		Shows all the member list that applied loan	As Expected	PASS		
3	Click the "Lihat" button		Show the detail of member's loan list	As Expected	PASS		
4	Click the "Lihat" button in the loan list		Redirect to page for clerk to update loan payment	As Expected	PASS		
5	Fill the date of payment	Tarikh bayaran = 2025-03-12	The date successfully key in	As Expected	PASS		
6	Choose the "Kaedah Pembayaran"		Display the chosen data	As Expected	PASS		
7	Fill the "Catatan"	Catatan = bayaran bulan 2	The remarks successfully key in	As Expected	PASS		
8	Click "Kemaskini"		The Loan Payment successfully updated and added to loan payment database	As Expected	PASS		

8.7 TC007: Test Financial Subsystem: Update Fee Payment (UC007)

This test contains the following test cases:

(a) TC009_01: Test Update Fee Payment (SD007)

DESCRIPTION	VERSION	TEST DATE
Test the Update Fee Payment	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Iman	

Test Case ID (TC007_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to "Kemaskini Saham"		The page should display all members and their respective shares	As Expected	PASS		
2	Click on "Kemaskini" on any one of the members		Display the update form page.	As Expected	PASS		
3	Pick the month of payment.		Display the month of payment	As Expected	PASS		
4	Fill in details	Fee Masuk = 50 Catatan = Bayaran untuk bulan ini.	Details successfully key in	As Expected	PASS		
5	Click "Kemaskini"		The payment is successfully updated	As Expected	PASS		

8.8 TC008: Test Membership Subsystem: Approve Membership Application (UC008)

This test contains the following test cases:

(a) TC008_01: Test Approve Membership Application (SD008)

DESCRIPTION	VERSION	TEST DATE
Test the Approve Membership Application	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Afiq	

Test Case ID (TC008_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to "Luluskan Permohonan Keahlian"		The landing page should be open	As Expected	PASS		
2	Click on details		Display a member application details	As Expected	PASS		
3	Click the "Lulus" or "Tolak"		The system alerts the user of successful approval.	As Expected	PASS		

8.9 TC009: Test Loan Subsystem: Approve Loan Application (UC009)

This test contains the following test cases:

(a) TC009_01: Test Approve Loan Application (SD009)

DESCRIPTION	VERSION	TEST DATE
Test the Approve Loan Application	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Alif	

Test Case ID (TC009_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to "Luluskan Permohonan Pembiayaan"		The landing page should be open	As Expected	PASS		
2	Click on details		Display a loan application details	As Expected	PASS		
3	Click the "Tolak" or "Lulus"		The system alerts the user of successful approval.	As Expected	PASS		

8.10 TC010: Test Manage Loan Policies Subsystem: Manage Loan Policies (UC010)

This test contains the following test cases:

(a) TC010_01: Manage Loan Policies (SD010)

DESCRIPTION	VERSION	TEST DATE
Test the Manage Loan Policies	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Alif	

Test Case ID (TC010_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to "Kemaskini Polisi"		The landing page should be open	As Expected	PASS		
2	Choose "Jenis Pinjaman"		Loan type chosen is displayed in page	As Expected	PASS		
3	Fill in the details to update the policies	Faedah(%) = 2.5 Maksimum Amaun Pinjaman (RM) = 100000 Min Amaun Pinjaman (RM) = 5000 Modah Syer (RM) = 300	Details successfully key in	As Expected	PASS		
4	Click "Kemaskini Polisi Pinjaman"		The data successfully update the loan database	As Expected	PASS		

8.11 TC011: Test Administration Subsystem: Generate Report (UC011)

This test contains the following test cases:

(a) TC011_01: Test Generate Report (SD011)

DESCRIPTION	VERSION	TEST DATE
Test the Generate Report	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Iman	

Test Case ID (TC011_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to "Penjanaan Laporan"		The landing page should be open	As Expected	PASS		
2	Choose the type report	Report Type = Bulanan	Display a user input date month	As Expected	PASS		
3	Choose a month	Month = January	The system generate and display the monthly report	As Expected	PASS		
4	Choose the type report	Report Type = Tahunan	Display a user input year	As Expected	PASS		
5	Choose a year	Year = 2024	The system generate and display the yearly report	As Expected	PASS		

8.12 TC012: Test Authentication Subsystem: Authentication (UC012)

This test contains the following test cases:

- (a) TC012_01: Test Authentication (SD012)

8.12.1 TC012_01: Test Authentication (SD012)

- (a) TC012_01_01: Test Authentication (Member) (SD012)
(b) TC012_01_02: Test Authentication (Board of Directors) (SD013)
(c) TC012_01_03: Test Authentication (Clerk) (SD014)

DESCRIPTION	VERSION	TEST DATE
Test the Authentication (Member)	2.0	15-Feb-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
	Iman	

Test Case ID (TC012_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to landing page	kadakoperasi.fun	The landing page should be open	As Expected	PASS		
2	Enter "No Kakitangan" and "Katalaluan"	No Kakitangan = 100 Password = 123	Credential can be entered	As Expected	PASS		
3	Click "Log Masuk"		Logged in as Member	As Expected	PASS		

Test Case ID (TC012_01_02)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to landing page	kadakoperasi.fu n	The landing page should be open	As Expected	PASS		
2	Enter "No Kakitangan" and "Katalaluan"	No Kakitangan = 200 Password = admin123	Credential can be entered	As Expected	PASS		
3	Click "Log Masuk"		Logged in as Board of Directors	As Expected	PASS		

Test Case ID (TC012_01_03)	Test Steps	Input Data	Expected Results	Actual Results	PASS/FAIL	Tested By	Tester Comments
1	Navigate to landing page	kadakoperasi.fu n	The landing page should be open	As Expected	PASS		
2	Enter "No Kakitangan" and "Katalaluan"	No Kakitangan = 300 Password = admin123	Credential can be entered	As Expected	PASS		
3	Click "Log Masuk"		Logged in as Clerk	As Expected	PASS		

9. Traceability Matrix

Table 9.1: Example of Traceability Matrix for KADA's Cooperative Web-Based System

Sprint No	Package Item	User Story ID	User Story Description	Sequence Diagram ID	Sequence Diagram Description	Test Case ID
#1	Package Membership	US_001	Apply for Membership	SD_001	Illustrates the steps for member registration.	TC_001
#1	Package Loan	US_002	Apply for Loan	SD_002	Illustrates the steps for loan registration.	TC_002
#4	Package Membership	US_003	Withdrawal as Membership	SD_003	Illustrates the steps for withdrawing registration.	TC_003
#4	Package Membership	US_004	Approve Membership Withdrawal	SD_004	Shows the process for approve membership withdrawal	TC_004
#2	Package Financial	US_005	View Financial Statement	SD_005	Shows the process for member to view the financial statement for loan and fee details to the member.	TC_005
#1	Package Administration	US_006	Update Loan Payment	SD_006	Illustrates the steps for updating loan payment.	TC_006
#1	Package Administration	US_007	Update Fee Payment	SD_007	Show the process of updating fee payment for each members	TC_007

#2	Package Membership	US_008	Approve Membership Application	SD_008	Shows the process for updating a user's profile.	TC_008
#2	Package Loan	US_009	Approve Loan Application	SD_009	Shows process for loan application approval	TC_009
#1	Package administration	US_010	Manage Loan Policies	SD_010	Shows the process of searching for a post	TC_010
#3	Package Administration	US_011	Generate Report	SD_011	Shows the process of report generation	TC_011
#2	Package Authentication	US_012	Authentication	SD_012	Shows the process of logging in as a member	TC_012_01_01
				SD_013	Shows the process of logging in as a board of director	TC_012_01_02
				SD_014	Shows the process of logging in as a clerk	TC_012_01_03